

BIOLOGY FOR ENGINEERS

MODULE 1

BIOMOLECULES AND THEIR APPLICATIONS

INTRODUCTION TO BIOLOGY: The cell: the basic unit of life, Structure and functions of a cell. The Plant Cell and animal cell, Prokaryotic and Eukaryotic cell, Stem cells and their application. Biomolecules: Properties and functions of Carbohydrates, Nucleic acids, proteins, lipids. Importance of special biomolecules; Enzymes (Classification (with one example each), Properties and functions), vitamins and hormones.

CELL

- A cell is the structural and fundamental unit of life.
- The study of cells from its basic structure to the functions of every cell organelle is called Cell Biology.
- Robert Hooke was the first Biologist who discovered cells.
- All organisms are made up of cells. They may be made up of a single cell (unicellular), or many cells (multicellular).

CELL STRUCTURE

1. Cell Wall

- It is made up of cellulose, hemicellulose and pectin.
- The cell wall is present exclusively in plant cells. It protects the plasma membrane and other cellular components.

2. Cell Membrane (plasma membrane)

- The cell membrane is the outer covering of a cell within which all other organelles, such as the cytoplasm and nucleus, are enclosed.
- It separates the cell from the external environment. The cell membrane is present in all the cells.
- It is semi permeable membrane

3. Cytoplasm

- The cytoplasm is a thick, clear, jelly-like substance present inside the cell membrane.
- Most of the chemical reactions within a cell take place in this cytoplasm.
- The cell organelles such as endoplasmic reticulum, vacuoles, mitochondria, ribosomes, are suspended in this cytoplasm.

4. Nucleus

- The nucleus contains the hereditary material of the cell, the DNA.
- It sends signals to the cells to grow, mature, divide and die.
- The nucleus is surrounded by the nuclear envelope that separates the DNA from the rest of the cell.

5. Cell Organelles

- Cells are composed of various cell organelles that perform certain specific functions to carry out life's processes.
- Some of the cell organelles are endoplasmic reticulum, golgi apparatus, mitochondria etc

FUNCTIONS OF CELL

1. Provides Support and Structure

- All the organisms are made up of cells. They form the structural basis of all the organisms. The cell wall and the cell membrane are the main components that function to provide support and structure to the organism.

2. Facilitate Growth Mitosis

- In the process of mitosis, the parent cell divides into the daughter cells. Thus, the cells multiply and facilitate the growth in an organism.

3. Allows Transport of Substances

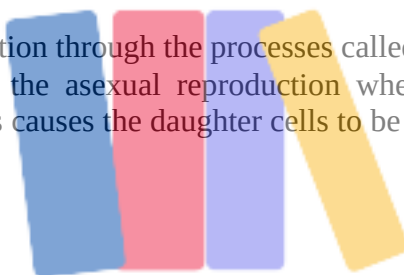
- Various nutrients are imported by the cells to carry out various chemical processes going on inside the cells.
- The waste produced by the chemical processes is eliminated from the cells by active and passive transport.
- Small molecules such as oxygen, carbon dioxide, and ethanol diffuse across the cell membrane through passive transport.

4. Energy Production

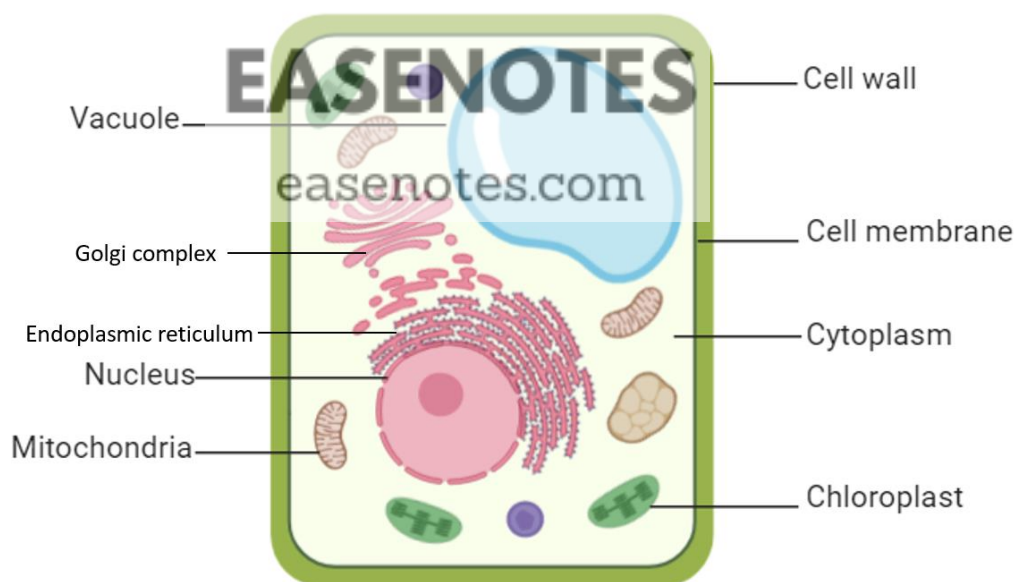
- Cells require energy to carry out various chemical processes. This energy is produced by the cells through a process called photosynthesis in plants and respiration in animals.

5. Helps in Reproduction

- A cell aids in reproduction through the processes called mitosis and meiosis.
- Mitosis is termed as the asexual reproduction where the parent cell divides to form daughter cells. Meiosis causes the daughter cells to be genetically different from the parent cells.



PLANT CELL



1. Cell Wall

- Plant cell wall is a tough layer that is made up of cellulose, lignin, pectin, and hemicellulose located outside the cell membrane.
- The primary function of the cell wall is to protect and provide structural support to the cell as well as protect the cell against mechanical stress and to provide form and structure to the cell.

2. Cell Membrane

- Cell membrane is a semi-permeable membrane present in the plant cell.
- It is one of the essential organelles as it regulates the entry and exit of different substances within the cell.

3. Nucleus

- The nucleus is a membrane-based organelle that is found in eukaryotic cells.
- Nucleus stores DNA, RNA and controls cell division, growth and protein production.

4. Plastids

- Plastids are double-membrane organelles that are found in the cells of plants and algae.
- These organelles play a crucial role in the manufacturing and storage of food. It also contains pigments that contribute to the process of photosynthesis.

5. Central Vacuole

- This is the largest organelle found in plant cells and is surrounded by a membrane.
- It not only stores materials and wastes but also provides proper structure to the plant, which maintains a proper pressure for a growing plant.

6. Mitochondria

- Often referred to as the “powerhouse of the cell,” mitochondria are the double-membraned organelles that provide energy to the cell by breaking down carbs and sugar molecules.

7. Lysosomes

- Lysosomes are membrane-bound cell organelles that contain digestive enzymes.
- Lysosomes are responsible for functions like recycling old cells, digesting materials both inside and outside of the cell, and releasing enzymes.

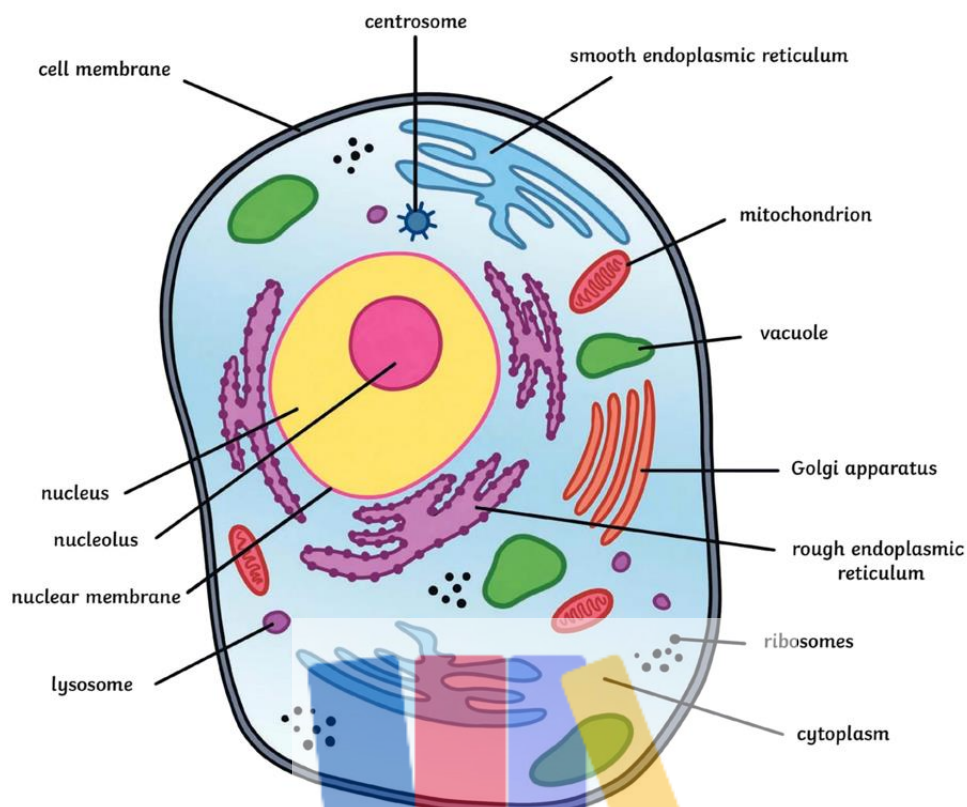
8. Golgi Apparatus

- Golgi apparatus also known as Golgi Complex. It helps in packaging of proteins and lipids and transports molecules to the different parts of the cell.

9. Endoplasmic reticulum

- It helps in protein synthesis and transport

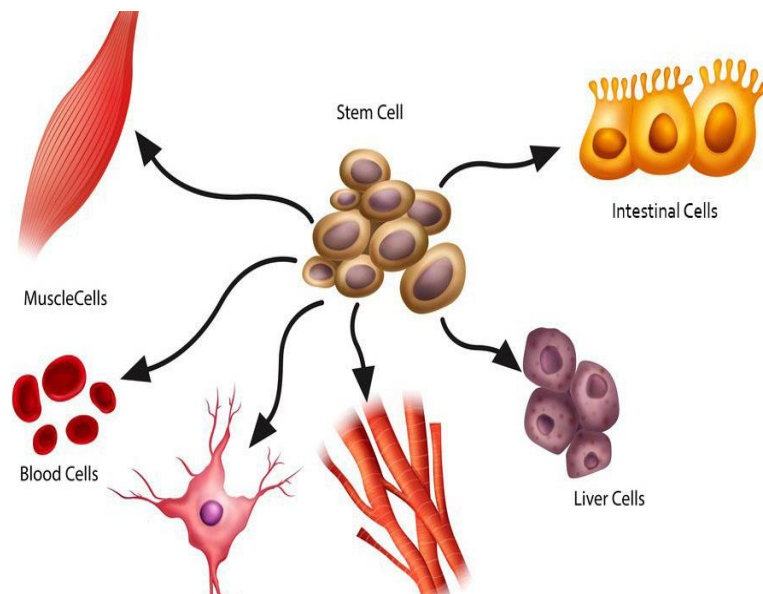
ANIMAL CELL



- The **cell membrane** is a double-layered membrane made up of phospholipids that surrounds the entire cell. The membrane is selectively permeable and allows only certain molecules to pass through.
- **Cytoplasm** is a gel-like substance that contains various cell organelles. It is protected by the cell membrane.
- The **nucleus** contains the genetic material DNA located in the nucleolus region of the nucleus. The nucleus is separated from the rest of the cell by a nuclear membrane. It also regulates the growth and division of cells.
- **Ribosomes** are found freely in the cytoplasm of the cell or attached to the membranes of endoplasmic reticulum. They help in the synthesis of proteins.
- The **endoplasmic reticulum** consists of a network of membranous sacs. It is of two types, rough endoplasmic reticulum and smooth endoplasmic reticulum. It helps in production and transporting proteins and calcium storage.
- **Golgi Apparatus** is a flat, layered, sac-like organelle present close to the nucleus. It packs and transports lipids and proteins.
- **Mitochondria** is also known as the “powerhouse of the cell”. The process of cellular respiration occurs here during which energy is released in the form of ATP
- **Lysosomes** are spherical organelles membrane-bound organelles. The enzymes present in the lysosome break down cellular debris, waste products, and foreign substances.
- **Vacuoles** are membrane-bound sac that stores water, food, ions, nutrients, and waste products. Unlike plant cell, they’re smaller and less prominent in animal cells.

STEM CELLS

- Stem cells also have the ability to repair damaged cells. These cells have strong healing power. They can evolve into any type of cell.
- Research on stem cells is going on, and it is believed that stem cell therapies can cure ailments like paralysis and Alzheimer’s as well.



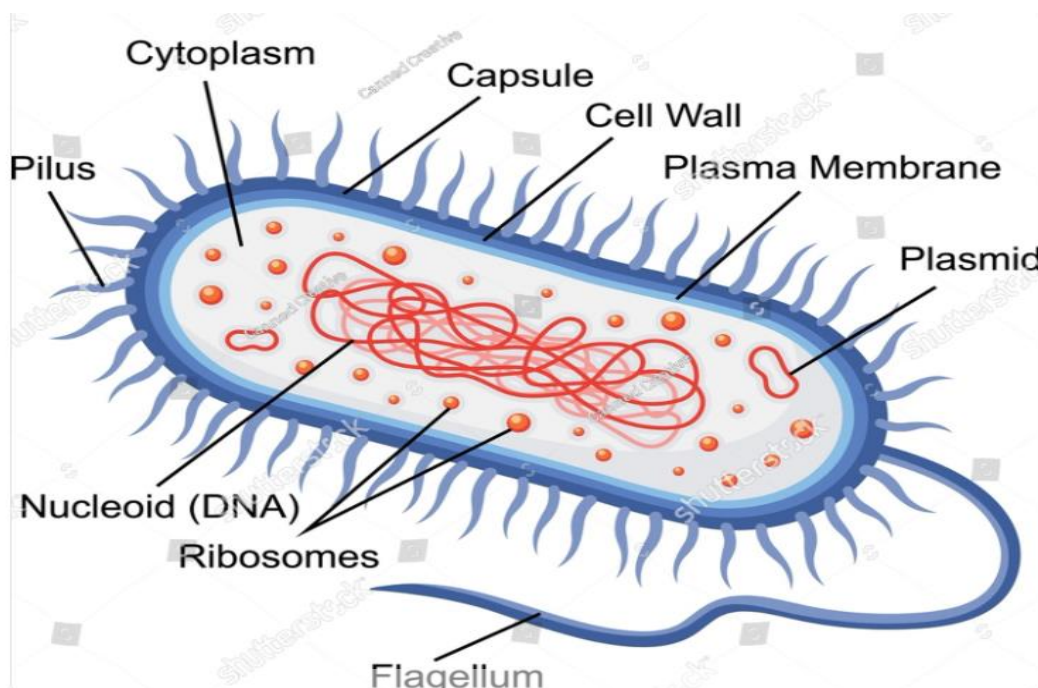
APPLICATIONS OF STEM CELLS

- **Treatment of Cardiovascular Disease-** A team of researchers have developed blood vessels in mice using human stem cells. Within two weeks of implantation, the blood vessels formed their network and were as efficient as the natural vessels.
- **Tissue Regeneration-** This is the most important application of stem cells. The stem cells can be used to grow a specific type of tissue or organ. This can be helpful in kidney and liver transplants.
- **Treatment of Brain Diseases-** Stem cells can also treat diseases such as Parkinson's disease and Alzheimer's. These can help to replenish the damaged brain cells.
- **Blood Disease Treatment-** The adult stem cells are used to treat cancers, sickle cell anaemia, and other immunodeficiency diseases. These stem cells can be used to produce red blood cells and white blood cells in the body.
- **Cosmetic and Anti-aging Treatments-** Stem cell therapies are being explored for their potential in rejuvenating aging skin, promoting hair growth, and improving overall appearance.

PROKARYOTIC CELL

- "Prokaryotic cells are the cells that do not have a true nucleus and membrane-bound organelles."
- Prokaryotic cells are single-celled microorganisms known to be the earliest on earth.
- Prokaryotes include Bacteria and Archaea.
- The photosynthetic prokaryotes include cyanobacteria that perform photosynthesis.

COMPONENTS OF PROKARYOTIC CELLS



The prokaryotic cells have the following components:

- **Plasma Membrane**- It is an outer protective covering of phospholipid molecules which separates the cell from the surrounding environment.
- **Cytoplasm**- It is a jelly-like substance present inside the cell. All the cell organelles are suspended in it.
- **DNA**- It is the genetic material of the cell. All the prokaryotes possess a circular DNA. It directs what proteins the cell creates. It also regulates the actions of the cell.
- **Ribosomes**- Protein synthesis occurs here
- **Plasmid** – carries genes that provide antibiotic resistance to the cell.
- Some prokaryotic cells possess pilus and flagella which helps in locomotion.

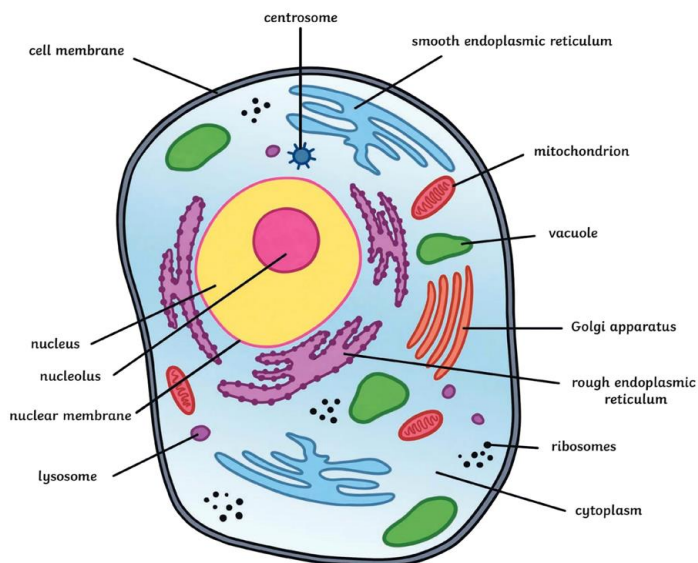
REPRODUCTION IN PROKARYOTES

A prokaryote reproduces in two ways:

- Asexually by binary fission
- Sexually by conjugation

EUKARYOTIC CELL

- “Eukaryotic cells are the cells that contain a membrane bound nucleus and organelles.”
- Protozoa, fungi, plants, and animals all have eukaryotic cells. They are classified under the kingdom Eukaryota.



CHARACTERISTICS OF EUKARYOTIC CELLS

The features of eukaryotic cells are as follows:

1. **Membrane-bound Organelles:** Eukaryotic cells contain membrane-bound organelles such as the nucleus, mitochondria, endoplasmic reticulum, Golgi apparatus, lysosomes, and chloroplasts (in plant cells).
2. **Nucleus:** Eukaryotic cells have a distinct nucleus enclosed within a nuclear envelope, which separates the genetic material (DNA) from the cytoplasm. The nucleus serves as the control center of the cell.
3. **Complex Cytoskeleton:** Eukaryotic cells possess a dynamic cytoskeleton made up of microtubules, microfilaments, and intermediate filaments. The cytoskeleton provides structural support, facilitates cell movement, and plays a role in intracellular transport and cell division.
4. **Endomembrane System:** Eukaryotic cells have an endomembrane system composed of interconnected membranes, including the endoplasmic reticulum (ER), Golgi apparatus, vesicles, and lysosomes.
5. **Mitochondria (and Chloroplasts in Plants):** Eukaryotic cells typically contain mitochondria, which are responsible for cellular respiration and energy production through the process of oxidative phosphorylation. Plant cells additionally contain chloroplasts, which perform photosynthesis, converting light energy into chemical energy in the form of glucose.
6. **Larger Size:** Eukaryotic cells are generally larger and more complex than prokaryotic cells, with a typical size ranging from 10 to 100 micrometers in diameter.
7. **Reproduction:** Eukaryotic cells reproduce through mitosis and meiosis processes that ensure the faithful transmission of genetic material to daughter cells.
8. **Multicellularity:** Many eukaryotic organisms are multicellular, consisting of numerous specialized cells organized into tissues, organs, and organ systems. This cellular specialization allows for division of labour and increased complexity.

CARBOHYDRATES

- Carbohydrates are one of the essential macronutrients needed by the body for energy production.
- They are primarily composed of carbon, hydrogen, and oxygen molecules.
- They can be classified into three main categories: sugars, starches, and fibers.

Properties of Carbohydrates:

- Chemical Structure: Carbohydrates are composed of simple sugars (monosaccharides) or larger molecules formed by the bonding of multiple sugar units (polysaccharides).
- Solubility: Many carbohydrates are soluble in water, making them easily digestible in the body.
- Energy Density: Carbohydrates provide 4 calories per gram, serving as the body's primary source of energy.
- Color and Taste: Carbohydrates can vary in color (e.g., white sugar, brown rice) and taste (e.g., sweet, starchy).

Functions of Carbohydrates:

- Energy Production: The main function of carbohydrates is to provide the body with energy. Glucose, derived from carbohydrates, is the preferred source of fuel for the brain and muscles.
- Energy Storage: Carbohydrates are stored in the form of glycogen in the liver and muscles, serving as a reserve energy source that can be mobilized when needed.
- Structural Support: Carbohydrates, particularly fibers, play a vital role in providing structural support to plant cells and contributing to the bulk and texture of food.
- Regulation of Blood Sugar: Carbohydrates help regulate blood sugar levels by controlling the release of glucose into the bloodstream.
- Digestive Health: Dietary fiber, a type of carbohydrate, aids in digestion by promoting regular bowel movements and maintaining gut health.
- Weight Management: Consuming complex carbohydrates, such as whole grains and vegetables, can help with weight management by providing a feeling of fullness and reducing the risk of overeating.



PROTEINS -

- Proteins are macromolecules made up of amino acids, which are the building blocks of life.
- They play crucial roles in various biological processes and are involved in a wide range of functions in living organisms.

Properties -

- Colour and Taste: Proteins are usually colourless and tasteless, homogeneous and crystalline.
- Shape and Size: They range in shape from simple crystalloid spherical structures to long fibrillar structures.



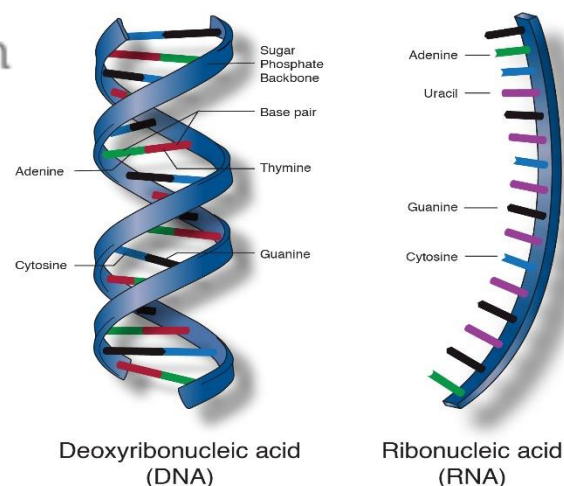
- Colloidal nature: Due to their giant sizes, proteins exhibit many colloidal properties, light-scattering in solution, resulting in visible turbidity (Tyndall effect).
- Denaturation: This is a process that involves the disruption of the secondary and tertiary structures of the protein, leading to the loss of biological activity. Often, denaturation is followed by coagulation, a process in which denatured protein molecules form large aggregates which precipitate from solution.
- Amphoteric nature: Just like amino acids, proteins are amphoteric in nature, i.e. they act as both acids and alkalies.

Functions of proteins -

- Structural role: Proteins provide structural support to cells and tissues. For example, collagen is a fibrous protein that forms the structural framework of connective tissues like skin, tendons, and bones.
- Enzymatic activity: Many proteins act as enzymes, which are biological catalysts that facilitate biochemical reactions. Examples include digestive enzymes like amylase and protease.
- Transportation: Certain proteins function as carriers or transporters, facilitating the movement of molecules across cell membranes or throughout the body. For instance, hemoglobin is a protein in red blood cells that transports oxygen from the lungs to tissues and organs.
- Muscle contraction: Proteins such as actin and myosin are vital for muscle contraction and movement. These proteins interact to generate the force required for muscle contraction, enabling bodily movements and locomotion.
- Hormonal regulation: Proteins serve as hormones. Eg -Hormones such as insulin, growth hormone, and adrenaline are protein-based and play essential roles in controlling metabolism, growth, and stress responses.
- Immune response: Proteins are integral to the immune system, where they act as antibodies to recognize and neutralize pathogens such as bacteria, viruses, and toxins.

NUCLEIC ACIDS

- Nucleic acids are macromolecules that serve as the information storage and transmission molecules in living organisms.
- Nucleotides are the building blocks of nucleic acid.
- These make up all living things genetic material.
- In a live cell, **DNA**- deoxyribonucleic acid and **RNA**-ribonucleic acid are two forms of nucleic acids.
- A nucleotide is made up of three chemically different components-nitrogenous base, a monosaccharide pentose sugar and phosphate group.



Functions -

1. Information storage: Nucleic acids, particularly DNA, store genetic information in the form of nucleotide sequences. This genetic information contains the instructions for the synthesis of proteins and regulates various cellular processes.
2. Transmission of genetic information: During cellular replication, DNA is faithfully copied and transmitted to daughter cells, ensuring the inheritance of genetic traits from one generation to the next.
3. Protein synthesis: Nucleic acids, specifically RNA, play a central role in protein synthesis

through the process of translation.

4. Energy transfer and metabolism: Nucleic acids participate in energy transfer and metabolic processes through molecules like adenosine triphosphate (ATP) which are involved in cellular energy metabolism and redox reactions.
5. Immune response: Nucleic acids can trigger immune responses when recognized as foreign by the immune system.
Eg- viral RNA or double-stranded RNA, can stimulate the production of antiviral proteins as part of the host defense against infections.
6. Diagnostic and therapeutic applications: Nucleic acids are widely used in molecular biology techniques for diagnostic purposes and in gene therapy which aim to modulate gene expression and treat genetic disorders and diseases.

LIPIDS

Lipids are fatty, waxy, or oily compounds that are soluble in organic solvents and insoluble in polar solvents such as water.

Ex- Fats and oils



Properties of Lipids

- Lipids may be either liquids or non-crystalline solids at room temperature.
- Pure fats and oils are colorless, odorless, and tasteless.
- They are energy-rich organic molecules
- Insoluble in water
- Soluble in organic solvents like alcohol, chloroform, acetone, benzene, etc.
- No ionic charges
- Solid triglycerols (Fats) have high proportions of saturated fatty acids.
- Liquid triglycerols (Oils) have high proportions of unsaturated fatty acids.

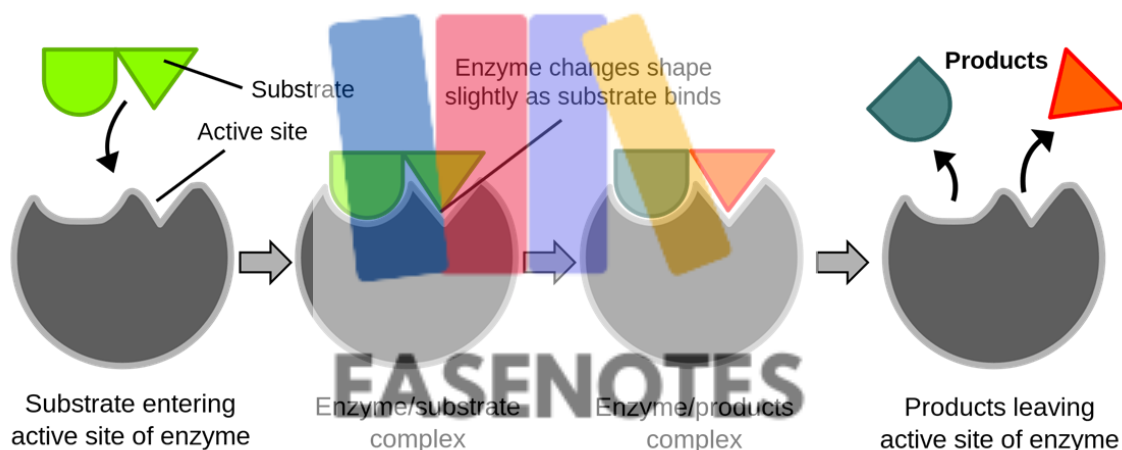
Functions-

1. Hydrophobicity: The hydrophobic nature allows lipids to form membranes and act as barriers between aqueous compartments within cells of organisms.
2. Energy storage: One of the primary functions of lipids is energy storage. Lipids such as **triglycerides** store energy in the form of long-term fuel reserves. When broken down through metabolism, lipids yield more energy per gram than carbohydrates or proteins, making them efficient energy storage molecules.
3. Structural component of membranes: Lipids are essential components of cell membranes,

where they form the lipid bilayer that surrounds and encloses cells. Lipids provide structural integrity, regulate membrane fluidity, and participate in various cellular processes such as signaling and transport.

4. **Insulation and protection:** Lipids, particularly adipose tissue, serve as insulation to help maintain body temperature by reducing heat loss. **Adipose tissue** also acts as a cushion to protect organs from physical shock or injury.
5. **Hormone synthesis:** Lipids are precursors for the synthesis of various hormones and signaling molecules. **Estrogen and testosterone**, are derived from cholesterol. These hormones regulate diverse physiological processes including metabolism, growth, reproduction, and stress responses.
6. **Transport of fat-soluble vitamins:** Lipids facilitate the absorption and transport of fat-soluble vitamins (A, D, E, and K) in the bloodstream to various tissues where they are needed for biological functions.
7. **Waterproofing:** Lipids provide waterproofing properties to the skin, fur, feathers, and other biological structures. Lipid-based substances like **sebum** protect the skin and hair from dehydration.

ENZYMES



Properties -

1. **Specificity:** Enzymes exhibit high specificity for their substrates, meaning they catalyze specific reactions involving particular molecules (substrates).
2. **Catalytic Activity:** Enzymes increase the rate of reactions without being consumed in the process. They facilitate reactions by providing an alternative pathway with lower activation energy.
3. **Efficiency:** Enzymes can accelerate reactions by factors of up to millions or even billions. This efficiency allows cells to carry out essential processes rapidly at physiological conditions.
4. **Reusability:** Enzymes can catalyze multiple reactions without being altered or consumed in the process. They can repeatedly bind to substrates and facilitate the conversion of substrates to products.
5. **Optimal Conditions:** Enzymes have specific optimal conditions for activity, including pH, temperature, and substrate concentration. Deviations from these optimal conditions can affect enzyme activity.

Functions of Enzymes:

- **Metabolism:** Enzymes play a central role in metabolism, the collective set of biochemical reactions that sustain life. They catalyze the breakdown and synthesis of molecules such as carbohydrates, lipids, proteins, and nucleic acids.

- Digestion: Digestive enzymes facilitate the breakdown of complex nutrients (e.g., carbohydrates, proteins, lipids) in food into simpler molecules that can be absorbed and utilized by the body.
- DNA Replication and Repair: Enzymes such as **DNA polymerases and DNA ligases** are involved in DNA replication, ensuring faithful duplication of genetic material. Additionally, DNA repair enzymes correct errors and damage in DNA molecules to maintain genomic integrity.
- Detoxification: Enzymes such as **cytochrome P450** oxidases catalyze the detoxification and elimination of drugs, pollutants, and toxins, from the body.
- Immune Response: Enzymes such as **proteases, nucleases, and lipases** are involved in the immune response by degrading foreign pathogens (e.g., viruses, bacteria) and cellular debris, facilitating their clearance from the body.
- Biotechnology: Enzymes are widely used in biotechnological applications, including food processing, pharmaceutical production, environmental remediation, and DNA technology.

ENZYME CLASSIFICATION

- Oxidoreductases: Catalyze oxidation-reduction reactions by transferring electrons between substrates. Eg – glucose oxidase
- Transferases: Transfer functional groups (e.g., methyl, acyl, phosphoryl) between substrates. Eg -Sulfotransferase
- Hydrolases: Catalyze hydrolysis reactions, breaking bonds by adding water. Eg- Peptidases
- Lyases: Catalyze the addition or removal of groups to form double bonds or the reverse reaction without hydrolysis. Eg -ammonia-lyase
- Isomerases: Catalyze the rearrangement of atoms within a molecule to form isomeric forms. Eg-Glucose Isomerase
- Ligases: Group of enzymes that catalyzes the joining of two molecules, using ATP. Eg- DNA ligases

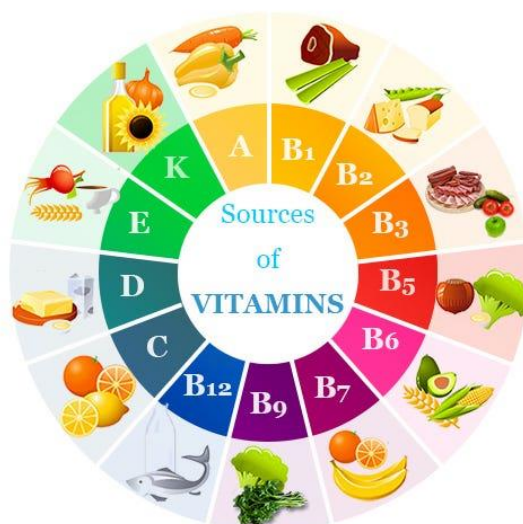
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VITAMINS

Properties of Vitamins:

1. Organic Nature: Vitamins are organic compounds, meaning they contain carbon atoms bonded to hydrogen and other elements such as oxygen, nitrogen, or sulfur.
2. Essentiality: Vitamins are considered essential nutrients because they cannot be synthesized in sufficient quantities by the body and must be obtained from the diet or supplements.
3. Micronutrient Status: Vitamins are classified as micronutrients because they are required in relatively small quantities compared to macronutrients like carbohydrates, proteins, and fats.
4. Diverse Chemical Structures: Vitamins encompass a wide range of chemical structures, including water-soluble vitamins (e.g., B vitamins, vitamin C) and fat-soluble vitamins (e.g., vitamins A, D, E, K).
5. Role as Coenzymes or Precursors: Many vitamins serve as coenzymes facilitating enzymatic reactions involved in metabolism and other biochemical processes.



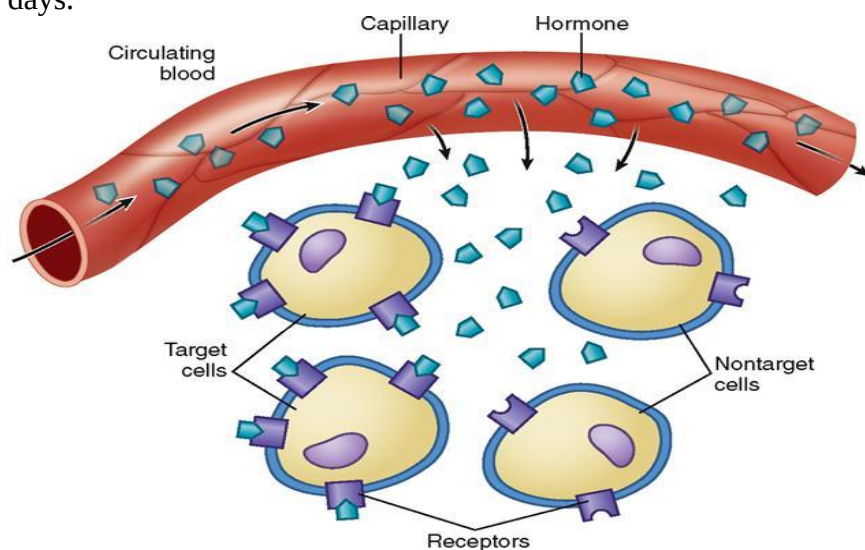
Functions of Vitamins:

- **Antioxidant Activity:** Certain vitamins, such as vitamins C and E, function as antioxidants, neutralizing harmful free radicals and protecting cells from oxidative damage caused by reactive oxygen species.
- **Energy Metabolism:** B vitamins serve as coenzymes in biochemical reactions that convert nutrients into energy (e.g., ATP production).
- **Cellular Function and Growth:** Vitamins are involved in various cellular functions and processes, including DNA synthesis and repair, cell division, and cell differentiation, which are essential for growth, development, and maintenance of tissues and organs.
- **Bone Health:** Vitamin D is crucial for calcium absorption and bone mineralization, playing a vital role in maintaining bone health.
- **Blood Clotting:** Vitamin K is necessary for the synthesis of clotting factors in the blood.
- **Immune Function:** Vitamins A, C, and D, as well as certain B vitamins play roles in supporting immune function by enhancing immune cell function, and promoting antibody production.
- **Vision:** Vitamin A is essential for vision, particularly for the synthesis of visual pigments in the retina, which are necessary for low-light and color vision.
- **Neurological Function:** Several vitamins, including B vitamins play critical roles in neurological function, nerve signaling, neurotransmitter synthesis.
- **Healthy Skin and Hair:** Various vitamins, including vitamins A, C, E, and biotin, contribute to the maintenance of healthy skin, hair, and nails by supporting cell growth, repair, and antioxidant protection.

HORMONES

- **Hormones** are signaling molecules produced by glands in multicellular organisms. They are secreted into the bloodstream or other body fluids, where they travel to target tissues or organs to regulate physiological and behavioral processes.
- **Chemical Nature:** Hormones can be classified based on their chemical structure into several categories, including peptides, steroids, amino acid derivatives, and fatty acid derivatives.
- **Target Specificity:** Hormones typically act on specific target cells or tissues that possess receptors for them. These receptors can be located on the cell surface or within the cell, depending on the chemical nature of the hormone.
- **Transportation:** Hormones are transported through the bloodstream, allowing them to reach distant target tissues. Some hormones can also act locally, without entering the bloodstream, in a process known as paracrine signaling.

- **Duration of Action:** The effects of hormones can vary in duration, ranging from seconds to days.



Functions-

1. **Reproduction and Sexual Development:** Hormones such as testosterone, estrogen, progesterone regulate reproductive functions, including sexual maturation, menstruation, and pregnancy.
2. **Stress Response:** Hormones such as cortisol and adrenaline are involved in the body's response to stress, helping to mobilize energy reserves and increase alertness and physiological readiness for action.
3. **Homeostasis:** Hormones play key roles in maintaining homeostasis by regulating various physiological processes, including blood pressure, electrolyte balance, body temperature, and water balance.
4. **Mood Regulation:** Hormones such as serotonin, dopamine, and oxytocin are involved in regulating mood, emotions, and social behaviors.
5. **Immune Function:** Some hormones, such as cytokines modulate immune function and help to regulate the body's defense mechanisms.

QUESTION BANK

1. Explain the structure of plant and animal cell with a neat labelled diagram.
2. Differentiate between prokaryotic and eukaryotic cells.
3. What are stem cells? Explain their applications.
4. Explain the properties and functions of the following
 - i. Carbohydrates
 - ii. Proteins
 - iii. Nucleic acids
 - iv. Vitamins
 - v. Hormones
 - vi. Lipids
 - vii. Enzymes
5. Explain the classification of enzymes with one example each.



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BIOLOGY FOR ENGINEERS

MODULE 2

BIOMOLECULES AND THEIR APPLICATIONS

Contents- Carbohydrates (cellulose-based water filters, PHA and PLA as bioplastics), Nucleic acids (DNA Vaccine for Rabies and RNA vaccines for Covid19, Forensics – DNA fingerprinting), Proteins (Proteins as food – whey protein and meat analogs, Plant based proteins), lipids (biodiesel, cleaning agents/detergents), Enzymes (glucose-oxidase in biosensors, lignolytic enzyme in bio-bleaching).

BIOMOLECULES AND THEIR APPLICATIONS:

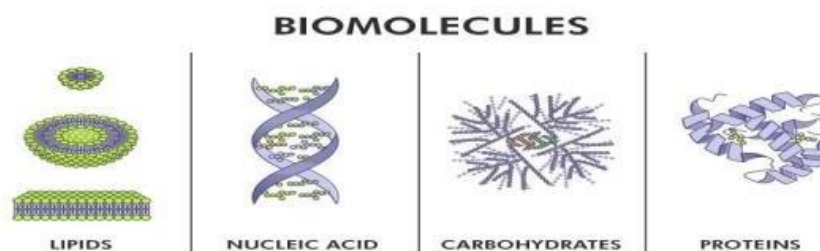
Biomolecule, also called biological molecule, any of numerous substances that are produced by cells and living organisms.

Biomolecules have a wide range of sizes and structures and perform a vast array of functions. The four major types of biomolecules are carbohydrates, lipids, nucleic acids, and proteins. Among biomolecules, nucleic acids, namely DNA and RNA, have the unique function of storing an organism's genetic code—the sequence of nucleotides that determines the amino acid sequence of proteins, which are of critical importance to life on Earth.

Carbohydrates, which are made up primarily of molecules containing atoms of carbon, hydrogen, and oxygen, are essential energy sources and structural components of all life, and they are among the most abundant biomolecules on Earth. They are built from four types of sugar units— monosaccharides, disaccharides, oligosaccharides, and polysaccharides.

Lipids, another key biomolecule of living organisms, fulfil a variety of roles, including serving as a source of stored energy and acting as chemical messengers. They also form membranes, which separate cells from their environments and compartmentalize the cell interior, giving rise to organelles, such as the nucleus and the mitochondrion, in higher (more complex) organisms.

The main applications of biomolecules are: The biomolecules may involve in several processes such as energy storage (carbohydrates), catalyzing the biochemical reactions (hormones), storing/transmitting the genetic codes (RNA/DNA), or altering biological and neurological activities (neurotransmitter/hormones).



CARBOHYDRATES

Carbohydrates are sugar molecules. Along with proteins and fats, Carbohydrates are one of three main nutrients found in foods and drinks. Body breaks down carbohydrates into Glucose. Glucose or blood sugar, is the main source of energy for your body's cells, tissues and organs. Glucose can be used immediately or stored in the liver and muscles for later use.



(carbohydrates)

There are three main types of Carbohydrates

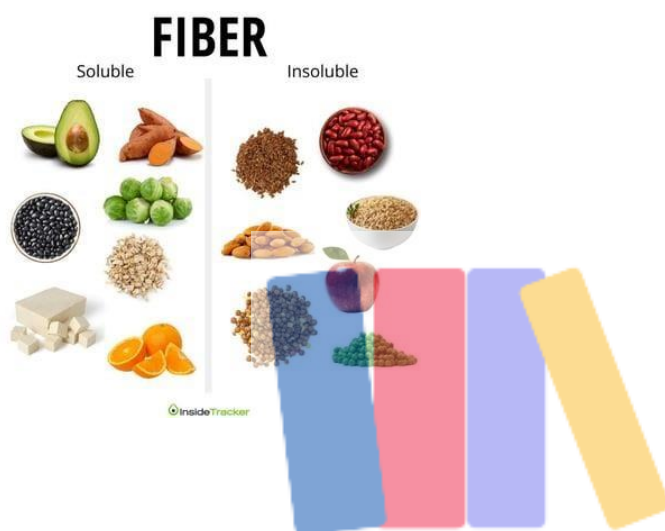
1. **Sugars:** They are also called as simple- carbohydrates because they are in the most basic form. They can be added to foods, such as the sugar in candy, desserts, processed foods, and regular soda. They also include the kinds of sugar that are found naturally in fruits, vegetables and milk.



2. **Starches:** They are complex carbohydrates, which are made up of lots of simple sugars strung together. Your body needs to break starches down into sugars to use them for energy. Starches include bread, cereal and pasta. They also include certain vegetables like potatoes, peas and corn.

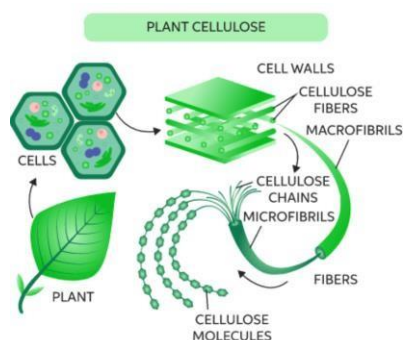


3. **Fiber:** It is also complex Carbohydrate. Our body cannot breakdown most fibers, so eating foods with fiber can help you feel full and make you less likely to overeat. Diets high in fiber have other health benefits. They may help prevent stomach or intestinal problems, such as constipation. They may also help lower cholesterol and blood sugar. Fiber is found in many foods such as fruits and vegetables.



CELULOSE : It is a complex carbohydrate , or polysaccharide, consisting of 3,000 or more glucose units.

It is extremely abundant, easily renewable and biodegradable. Due to inter- and intramolecular hydrogen bonding between the hydroxyl groups of the neighbouring cellulose chains, cellulose is insoluble in water, despite being hydrophilic and is difficult to dissolve with common organic solvents.



Taking benefit of these advantages of cellulose, we have a best application of cellulose that is Cellulose based water filters.

The interest in the use of biobased filters for water purification has increased in recent years, as such have the potential to be affordable, lightweight and biodegradable. Research has been focused on creating biobased membranes for micro and ultrafiltration from cellulose nanofibrils (CNFs).

Filters based on cellulose pulp filters do usually have large pores that facilitate water percolation but they do not sufficiently remove bacteria through size exclusion; other techniques are therefore needed to achieve a bacteria-reducing effect. Several groups have addressed this issue by incorporating antibacterial metal nanoparticles into cellulose-based water filters, both silver nanoparticles (AgNPs), and copper nanoparticles (CuNPs) are known to have good antibacterial effects.

An alternate method to physically remove bacteria from water, while keeping the filter pore size larger than bacteria, is to use positively charged filters that adsorb negatively charged bacteria onto the surfaces of the filters.

This allows negatively charged particles much smaller than the filter pore size to be efficiently removed from water and this is an interesting approach for removing bacteria from water without adding any toxic chemicals or reducing the flow by reducing pore size. Both Gram-positive and gram-negative bacteria have a negative net surface charge on the cell envelope, due to peptidoglycans, liposaccharides and proteins in the cell wall, and this makes their removal non-selective and efficient for most types of bacteria.



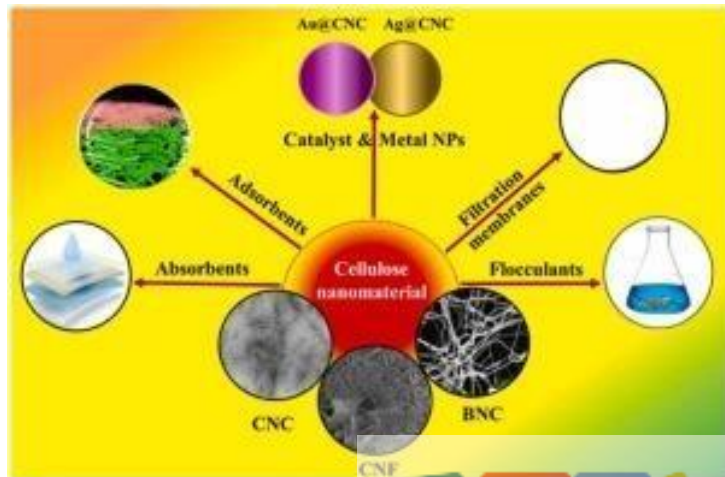
(cellulose based water filter)

Methods used for the same are:

- LBL (Layer By Layer) MODIFICATION
- NITROGEN ANALYSIS
- SEM (Scanning Electron Microscope)
- FLOW RATE FOR FREE FLOW FILTRATION
- BACTERIAL REMOVAL EFFICIENCY FOR FILTRATION
- FILTRATION OF NATURAL WATER SAMPLES

- FLOURESCENCE MICROSCOPY

Cellulose filter papers are versatile and diverse tools for micro filtration, that work by trapping particulates within a random matrix of cellulose fibers. Cellulose filter papers can be categorized as quantitative or qualitative, depending on their application.



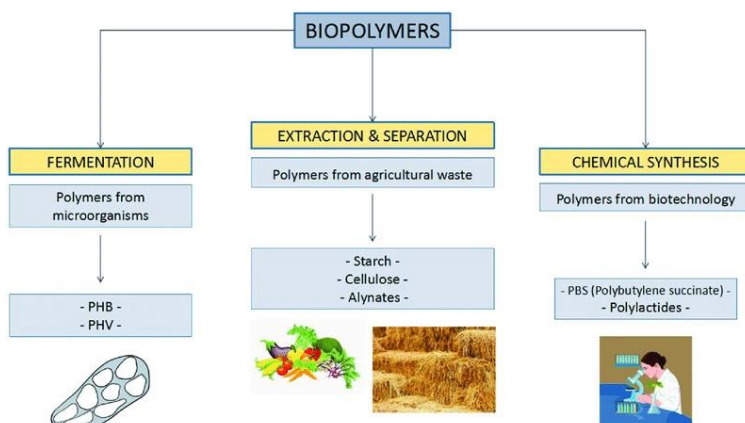
(cellulose based nanomaterials for water treatment)

POLY LACTIC ACID [PLA] AND POLY HYDROXY ALKANOATES [PHA] :

Just like, how we introduced biodegradable water treatment plant from cellulose, we also have a replacement for toxic, non-biodegradable plastics which we are using in our daily life. That is bioplastics.

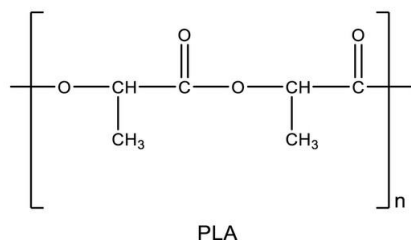
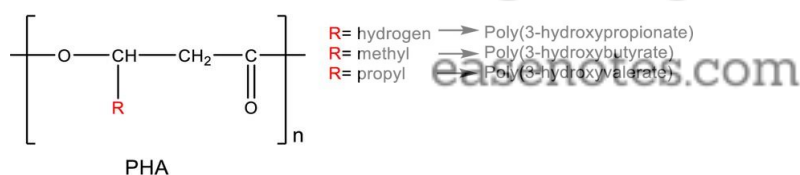
Bioplastics are one type of plastic which can be generated from natural resources such as starches and vegetable oils. Bioplastics are basically classified as bio based and/or biodegradable. Not all bio-based plastics are biodegradable and similarly not all biodegradable plastics are bio based. Bioplastics are referred to as bio based when the focus of the material is on the origin of the carbon building block and not by where it ends up at the end of its cycle life. Bio plastics are said to be biodegradable if they are broken down with the effect of the right environment conditions and microbes which in turn use them as a food source. The bioplastics are considered compostable if within 180days, a complete microbial assimilation of the fragmented food sources takes place in a compost environment.

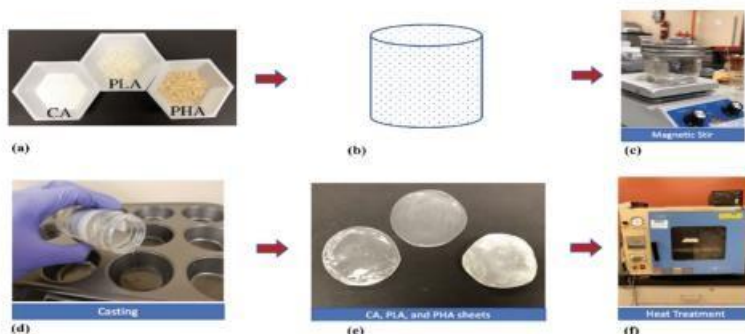
Based upon this, we have **PHA** and **PLA**.



PLA: It is both biobased and biodegradable under industrial composting conditions (at high temperature around 58⁰c). Because of its good mechanical properties, processability, renewability, and non-toxicity, PLA is considered today as one of the most commercially promising bioplastics. When compared with most other biodegradable polymers, PLA has better durability, transparency and mechanical strength.

PHAs: These are a significant polymer family that are 100% bio-based and bio-degradable. PHAs are microbiologically produced polyesters that have tunable physical and mechanical properties. This is accompanied by low environmental impact due to their biodegradability and non-toxicity nature. Therefore, they are promising candidates for a sustainable future manufacturing. Ranging from brittle thermoplastics to gummy elastomers, PHAs properties can be altered by the selection of bacteria, fermentation conditions and substrate. Due to their flexible properties, PHAs can eventually substitute PP, Polyethylene (PE), and polystyrene(PS), which are the main polymers of today's global polymer market.





(Steps in producing bioplastic from PHA and PLA.)

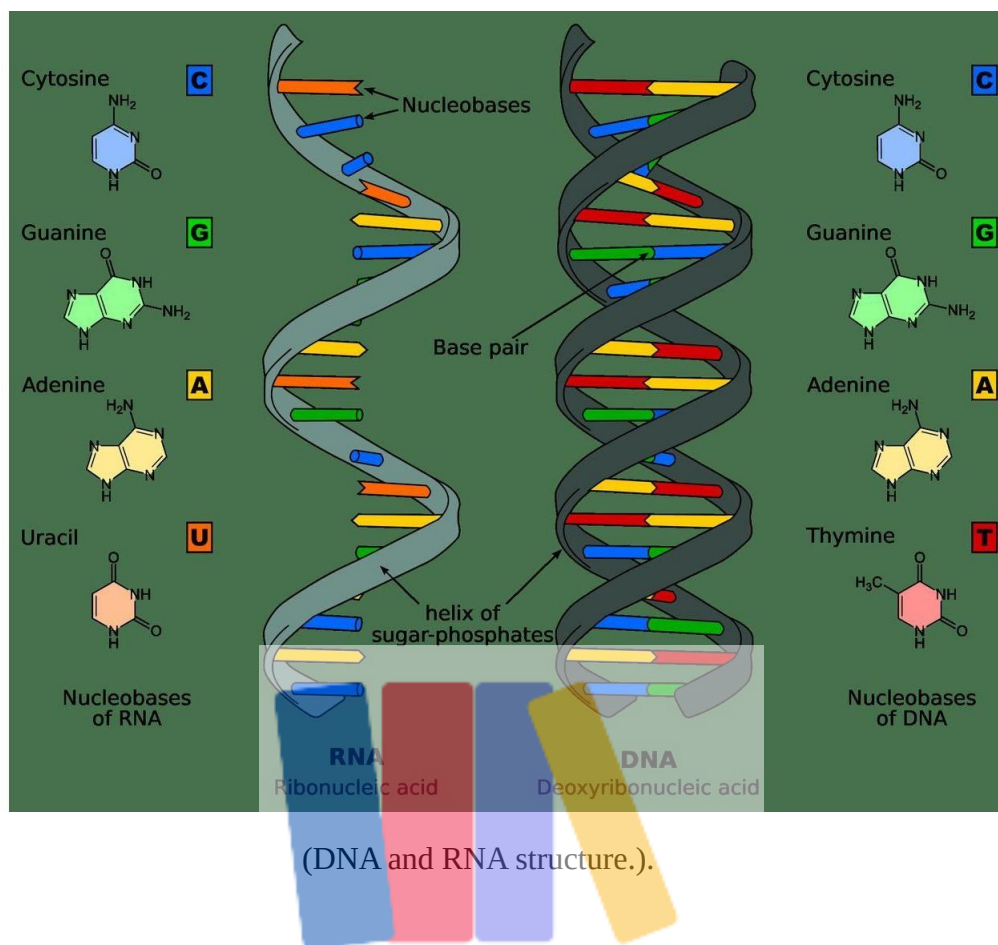
NUCLEIC ACID

Nucleic acids are biopolymers, macromolecules, essential to all known forms of life. They are composed of nucleotides, which are the monomers made of three components: a 5-carbon sugar, a phosphate group and a nitrogenous base. The two main classes of nucleic acids are Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). If the sugar is ribose, the polymer is RNA. If the sugar is the ribose derivative deoxyribose the polymer is DNA.

Nucleic acids are naturally occurring chemical compounds that serve as the primary information-carrying molecules in cells and make up the genetic material. Nucleic acids are found in abundance in all living things, where they create, encode, and then store information of every living cell of every life-form on earth. In turn, they function to transmit and express that information inside and outside the cell nucleus to the interior operations of the cell and ultimately to the next generation of each living organism. The encoded information is contained and conveyed via the nucleic acid sequence, which provides the "ladder-step" ordering of nucleotides within the molecules of RNA and DNA. They play an especially important role in directing protein synthesis.

Strings of nucleotides are bonded to form helical backbones- typically, one for RNA, two for DNA- and assembled into chains of base-pairs selected from the five primary, or canonical, nucleobases, which are: adenine, cytosine, guanine, thymine and uracil. Thymine occurs only in DNA and uracil only in RNA.

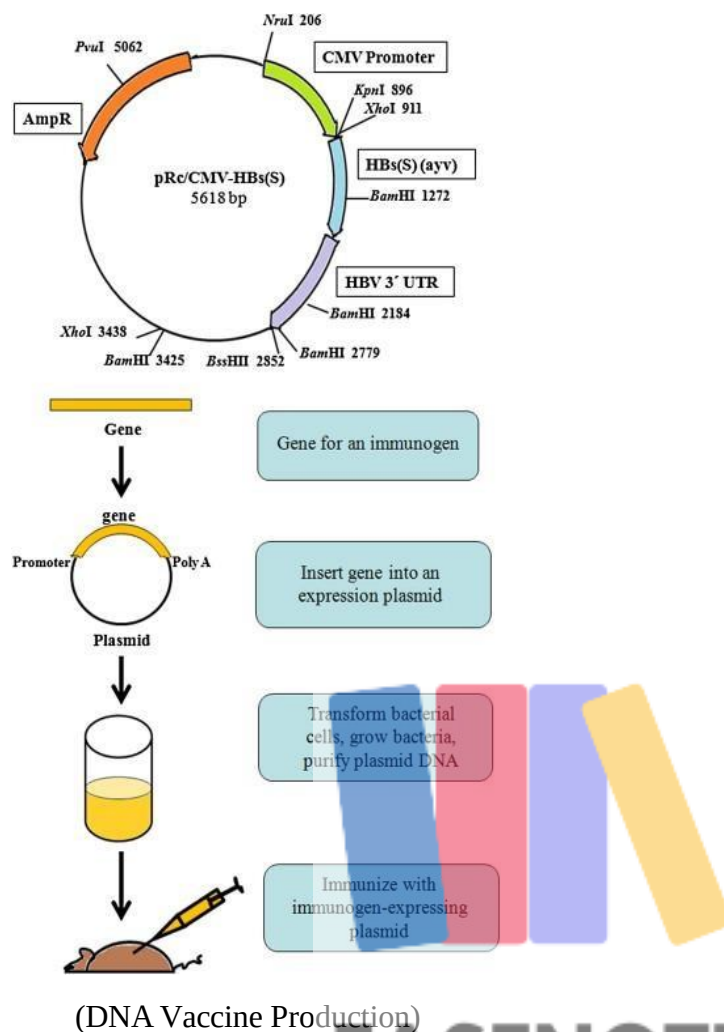
The two main acids are DNA and RNA, which is the fundamental unit of any living organisms. Based on these factors, there are many applications for the same of which are explained below:



1. DNA VACCINE FOR RABIES

Rabies is a preventable viral disease most often transmitted through the bite of a rabid animal. The rabies virus infects the central nervous system of mammals, ultimately causing disease in the brain and death. Most rabies cases reported to the Centers for Disease Control and Prevention (CDC) each year occur in wild animals like bats, raccoons, skunks and foxes, although any mammal can get rabies.

A DNA Vaccine, using a pCI-nco plasmid encoding the glycoprotein gene of a Mexican isolate of rabies virus, was developed to induce long-lasting protective immunity against rabies virus in dogs. The worldwide incidence of rabies and high rates of therapy failure, despite availability of effective vaccines indicate the need of timely and improved prophylactic approaches. DNA vaccination based on optimized formulation of lysosome-targeted glycoprotein of the rabies virus provides potential platform for preventing and controlling rabies. A range of parameters including physical, physiological, clinical, immunological, hematological along with histopathology profiles of target organs was monitored to assess the impact of vaccination. There were no observational adverse effects despite high dose administration of the DNA vaccine formulation. Thus, this study indicates the safety of next generation of vaccines as well as highlights their potential application.



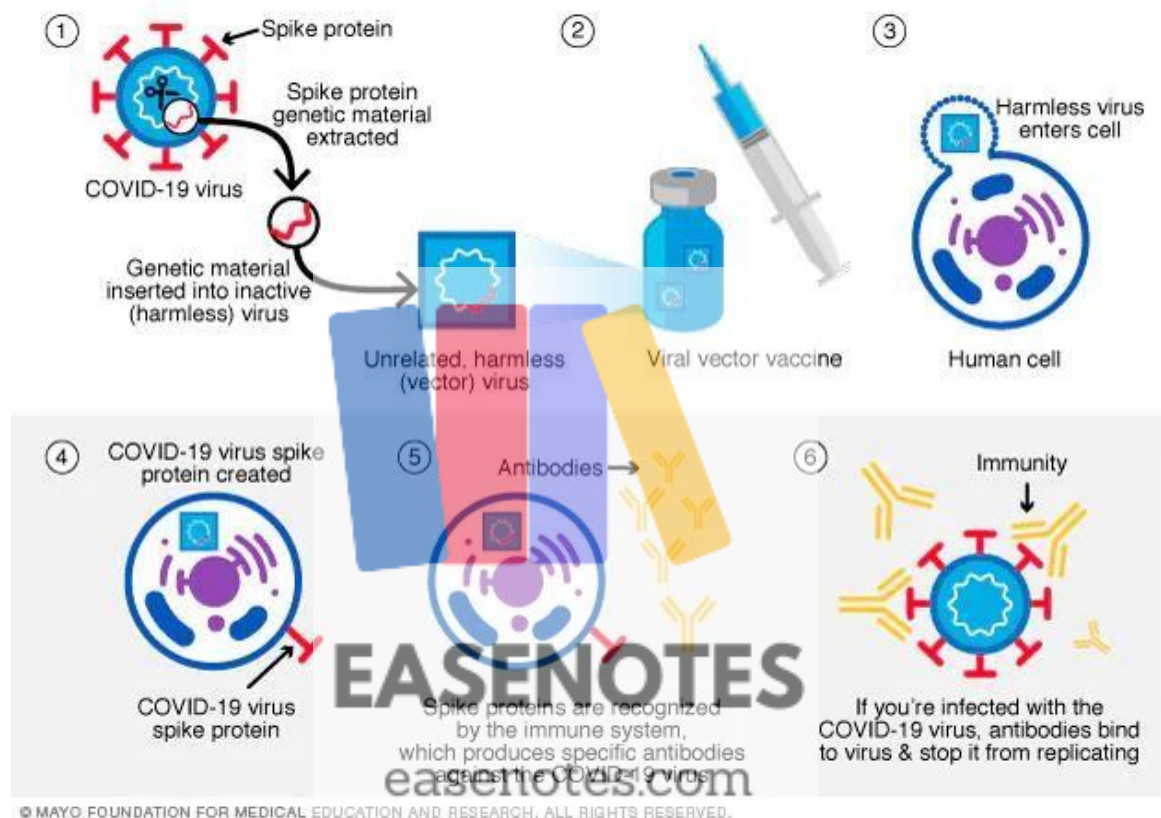
2. RNA VACCINE FOR COVID:

Coronavirus disease (Covid-19) is an infectious disease caused by the SARS-CoV-2 virus. Messenger RNA, or mRNA technology, instructs cells to make a protein that generates an immune response in the body, thus producing the antibodies that provide protection against a disease. It is the basis for the Pfizer/BioNTech and Moderna covid 19 vaccines being used by governments worldwide, and in the UN-supported COVAX global vaccine solidarity initiative. Messenger ribonucleic acid (mRNA) is a molecule that provides cells with instructions for making proteins. mRNA vaccines contain the instructions for making the SARS-CoV-2 spike protein. This protein is found on the surface of the virus that causes COVID-19.

The mRNA molecule is essentially a recipe, telling the cells of the body how to make the spike protein. COVID-19 mRNA vaccines are given by injection, usually into the muscle of the upper arm. After the protein piece is made, the cell breaks down the instructions and

gets rid of them. The mRNA never enters the central part(nucleus) of the cell, which is where our DNA (genetic material) is found. Your DNA can't be altered by mRNA vaccines.

The cell then displays the protein piece on its surface. Our immune system recognizes that the protein doesn't belong there and begins building an immune response and making antibodies.



(RNA Vaccine for covid 19)

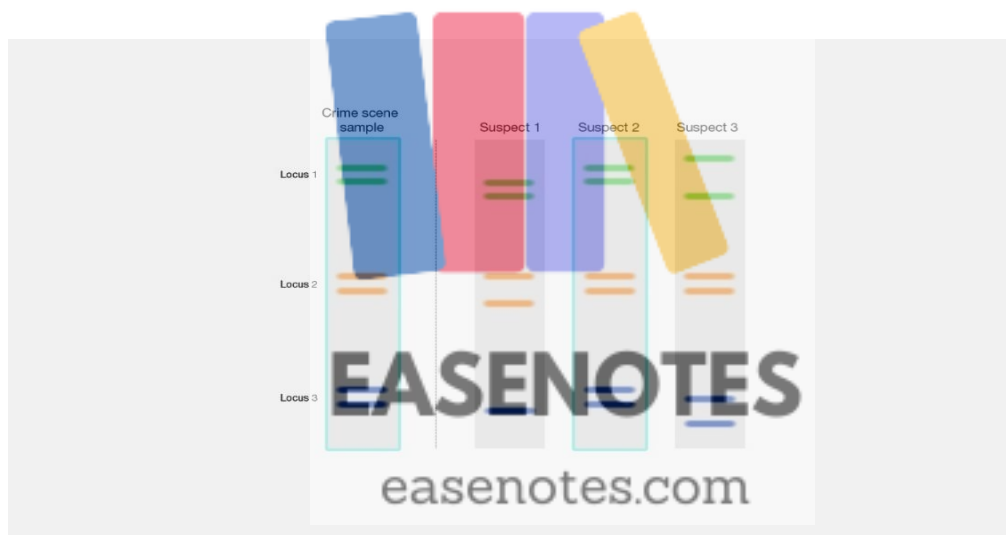
3. FORENSIC-DNA FINGERPRINTING:

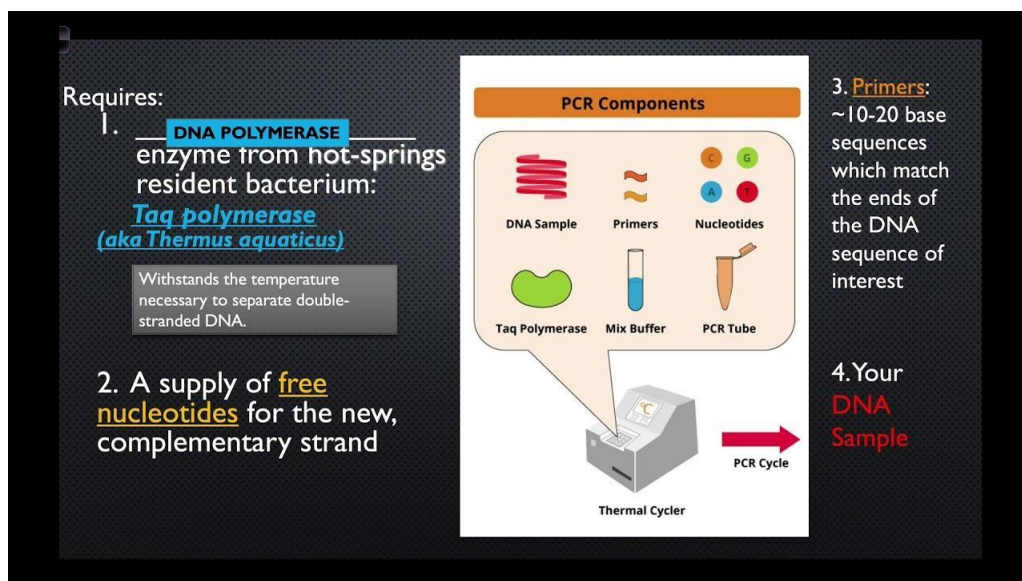
DNA fingerprinting is also called DNA typing, DNA profiling, genetic fingerprinting, genotyping, or identify testing, in genetics, method of isolating and identifying variable elements within the base-pair sequence of DNA(deoxyribonucleic acid).

The procedure for creating a DNA fingerprint consists of first obtaining a sample of cells, such as skin, hair, or blood cells, which contains DNA. The DNA is extracted from the cells and purified, the DNA was then cut at specific points along the strand with proteins known as restriction enzymes. The enzymes produced fragments of varying lengths that were sorted by placing them on a gel and then subjecting the gel to an electric current (electrophoresis): the shorter the fragment, the more quickly it moved toward the positive pole(anode).

The sorted double-stranded DNA fragments were then subjected to a blotting technique in which they were split into single strands and transferred to a nylon sheet. The fragments underwent autoradiography in which they were exposed to DNA probes- pieces of synthetic DNA that were made radioactive and that bound to the minisatellites. A piece of X-ray film was then exposed to the fragments, and dark mark was produced at any point where radioactive probe had become attached. The resultant pattern of marks could then be analyzed. The DNA testing process is comprised of four main steps, including **extraction, quantitation, amplification and capillary electrophoresis**.

DNA fingerprinting is a laboratory technique used to determine the probable identify of a person based on the nucleotide sequences of certain regions of human DNA that are unique to individuals. Forensic genetic fingerprinting can be defined as the comparison of the DNA in a person's nucleated cells with that identified in biological matter found at the scene of a crime or with the DNA of another for the purpose of identification or exclusion. The application of these techniques introduces new factual evidence to criminal investigations and court cases.





DNA FINGERPRINTING

So, this is all about nucleic acids, considering the two main nucleic acids, that is, DNA and RNA, discussing about some of the most important applications of these nucleic acids.

PROTEINS

Protein is found throughout the body in muscle, bone, skin, hair, and virtually every other body part or tissue. It makes up the enzymes that power many chemical reactions and the hemoglobin that carries oxygen in your blood.

Taking more about proteins based on biomolecule concept, Proteins are large biomolecules and macromolecules that comprise one or more long chains of amino acid residues. Proteins perform a vast array of functions within organisms, including catalysing metabolic reactions, DNA replication, responding to stimuli, providing structure to cells and organisms, and transporting molecules from one location to another. Proteins differ from one another primarily in their sequence of amino acids, which is dictated by the nucleotide sequence of their genes, and which usually results in protein folding into a specific 3D structure that determines its activity.

Proteins are assembled from amino acids using information encoded in genes. Each protein has its own unique amino acid sequence that is specified by the nucleotide sequence of the gene encoding this protein. The genetic code is a set of three-nucleotide sets called codons and each three nucleotide combination designates an amino acid.

The process of synthesizing a protein from an mRNA template is known as translation. The mRNA is loaded onto the ribosome and is read three nucleotides at a time by matching each codon to its base pairing anticodon located on a transfer RNA molecule, which carries the amino acid corresponding to the codon it recognizes. The enzyme aminoacyl tRNA synthetase "charges" the tRNA molecules with the correct amino acids. The growing

polypeptide is often termed the nascent chain. Proteins are always biosynthesized from N-terminus to C-terminus.

Proteins are the chief actors within the cell, said to be carrying out the duties specified by the information encoded genes. With the exception of certain types of RNA, most other biological molecules are relatively inert elements upon which proteins act. Proteins make up of half the dry weight of an Escherichia coli cell, whereas other macromolecules such as DNA and RNA make up only 3% and 20% respectively. The set of proteins expressed in a particular cell or cell type is known as its proteome.



PROTEIN AS FOOD:

Protein is a key part of any diet. The average person needs about 7 grams of protein every day for every 20 pounds of body weight. Because protein is found in an abundance of foods, many people can easily meet this goal. However, not all protein “packages” are created equal. Because food contain a lot more than protein, its important to pay attention to what else is coming with it. Animal based foods (meat, poultry, fish, eggs and dairy foods) tend to be good sources of complete protein, while plant-based foods (fruits, vegetables, grains, nuts and seeds) often lack one or more essential amino acid.

Whey protein is a mixture of proteins isolated from whey, the liquid material created as a by-product of cheese production. The proteins consist of α -lactalbumin, β -lactoglobulin, serum albumin and immunoglobulins. Glycomacropeptide also makes up the third largest component but is not a protein. Whey protein is commonly marketed as a protein supplement, and various health claims have been attributed to it.

Whey is left over when milk is coagulated during the process of cheese production, & contains everything that is soluble from milk after the pH is dropped to 4.6 during the coagulation process. It is a 5% solution of lactose in water and contains the water soluble proteins of milk as well as some lipid content. Processing can be done simple drying, or the relative protein content can be increased by removing the lactose, lipids and other non-protein materials. The primary usage of whey protein supplements is for muscle growth and development. Eating whey protein supplements before exercise will not assist athletic performance, but it will enhance the body's protein recovery and synthesis after exercise because it increases the free amino acids in the body's free amino acid pool.

MUSCLE & STRENGTH: HOW WHEY PROTEIN IS MADE

How does your post-workout shake transform from milk to high quality whey?



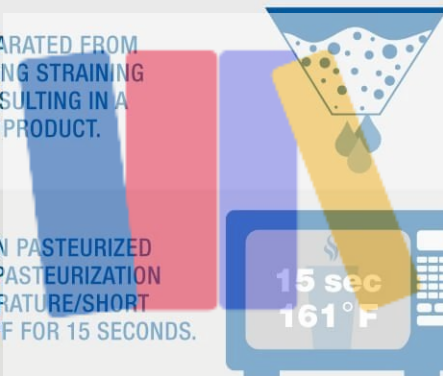
1

WHEY PROTEIN CAN BE MANUFACTURED DIRECTLY FROM MILK OR EVEN CHEESE.



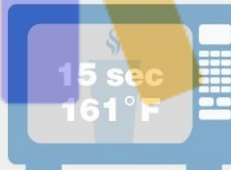
2

WHEY IS SEPARATED FROM THE MILK USING STRAINING METHODS RESULTING IN A LIQUID WHEY PRODUCT.



3

WHEY IS THEN PASTEURIZED USING HTST PASTEURIZATION (HIGH TEMPERATURE/SHORT TIME) AT 161°F FOR 15 SECONDS.



4

ION EXCHANGE OR MECHANICAL FILTERING IS PERFORMED TO ISOLATE THE PROTEIN FROM THE WHEY LIQUID.



5

THE KIND AND SIZE OF FILTER YOU USE AND THE NUMBER OF TIMES YOU FILTER IT DETERMINE WHAT TYPE OF WHEY YOU PRODUCE. THE FINER THE FILTER AND THE MORE TIMES YOU FILTER IT RESULTS IN A PURER PRODUCT.



6

THE END RESULT IS A WHEY PRODUCT (TYPICALLY ISOLATE OR CONCENTRATE) DEPENDING ON THE FILTERING PROCESS.



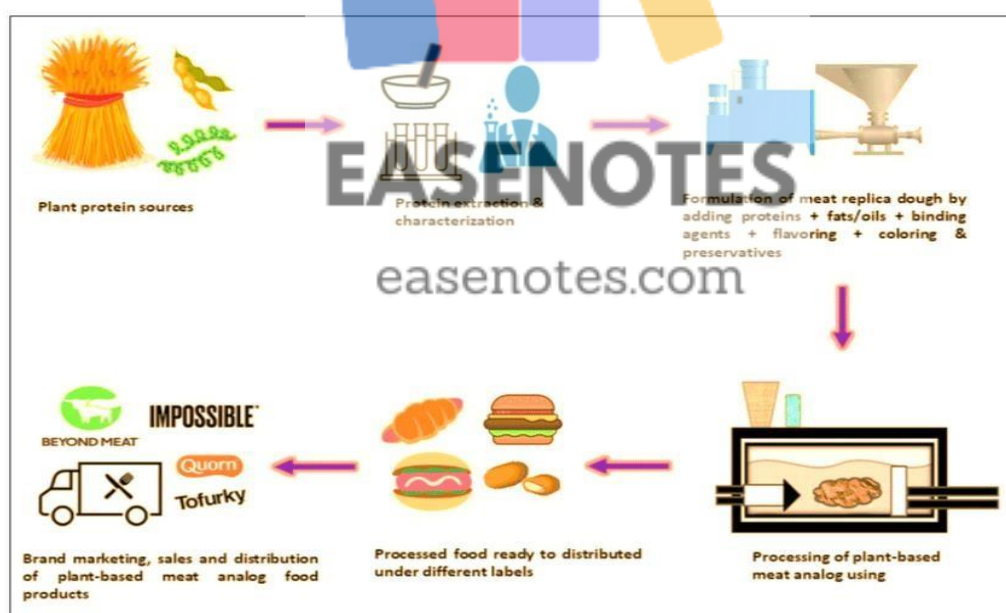
MUSCLEANDSTRENGTH.com



Meat analogues find raising interest of many consumers who are looking for indulgent, healthy, low environmental impact, ethical, cost-effective, and/or new food products. High moisture extrusion cooking enables the production of fresh, premium meat analogues that are texturally like muscle meat from plant or animal proteins. The appearance and eating sensation are similar to cooked meat while high protein content offers a similar nutritional value. This article focuses on plant-based meat analogues and covers process and product-related aspects including ingredients and structure formation, flavour, taste and nutritional value, postextrusion processing, packaging and shelf life, consumer benefits, and product-related environmental impacts.

Meat analogues, can be defined as products that mimic meat in its functionality, bearing similar appearance, texture, and sensory attributes to meat. Production of meat analogues has been on the increase, targeted at satisfying consumers' desire for indulgent, healthy, low environmental impact, and ethical meat substitutes.

The factors that lead to this shift is due to low fat and calorie foods intake, flexitarians, animal disease, natural resources depletion, and to reduce greenhouse gas emission. Currently, available marketed meat analog products are plant-based meat in which the quality (ie: texture or taste) are similar to the conventional meat. The ingredients used are mainly soy proteins with novel ingredients added, such as mycoprotein and soy leghemoglobin.



(plant protein marketing process)

PLANT BASED PROTEINS:

Plant protein is simply a meaningful food source of protein which is from plants. This group can include pulses, tofu, soya, tempeh, seitan, nuts, seeds, certain grains and even peas.

Pulses are a large group of plants, which include chickpeas, lentils, beans(such as black, kidney and adzuki beans) and split peas.

Plant proteins are highly nutritious – not only as good sources of protein, but also because they provide other nutrients such as fibre, vitamins and minerals. Our intake of fibre tends to be too low, however by incorporating certain plant proteins into your diet, such as pulses, peas and nuts, you can easily boost your fibre intake.

Consumer demand for plant protein- based products is high and expected to grow considerably in the next decade. Factors contributing to the rise in popularity of plant proteins include,

1. potential health benefits associated with increased intake of plant-based diets.
2. consumer concerns regarding adverse health effects of consuming diets high in animal protein (eg: increased saturated fat)
3. increased consumer recognition of the need to improve the environmental sustainability of food production.
4. Ethical issues regarding the treatment of animals.
5. General consumer view of protein as a “positive” nutrient (more is better). While there are health and physical function benefits of diets higher in plant-based protein, the nutritional quality of plant proteins may be inferior in some respects relative to animal proteins.

LIPIDS

Discussing about another important biomolecule, lipids, are a broad group of naturally occurring molecules which includes fat, waxes, sterols, fat-soluble vitamins(such as vitamin A,D,E and K), monoglycerides, diglycerides, phospholipids and others. The functions of lipids include storing energy, signaling, and acting as structural components of cell membranes. Lipids have applications in the cosmetic and food industries and in nanotechnology.

Lipids may be broadly defined as hydrophobic or amphiphilic small molecules; the amphiphilic nature of some lipids allows them to form structures such as vesicles, multilamellar/ unilamellar liposomes, or membranes in an aqueous environment. Biological lipids originate entirely or in part from two distinct types of biochemical subunits or “building-blocks” ketoacyl and isoprene groups. Using this approach, lipids may be divided into eight categories: fatty acyls, glycerolipids, glycerophospholipids, sphingolipids, saccharolipids, and polyketides (derived from condensation of ketoacyl subunits); and sterol lipids and prenol lipids (derived from condensation of isoprene subunits).

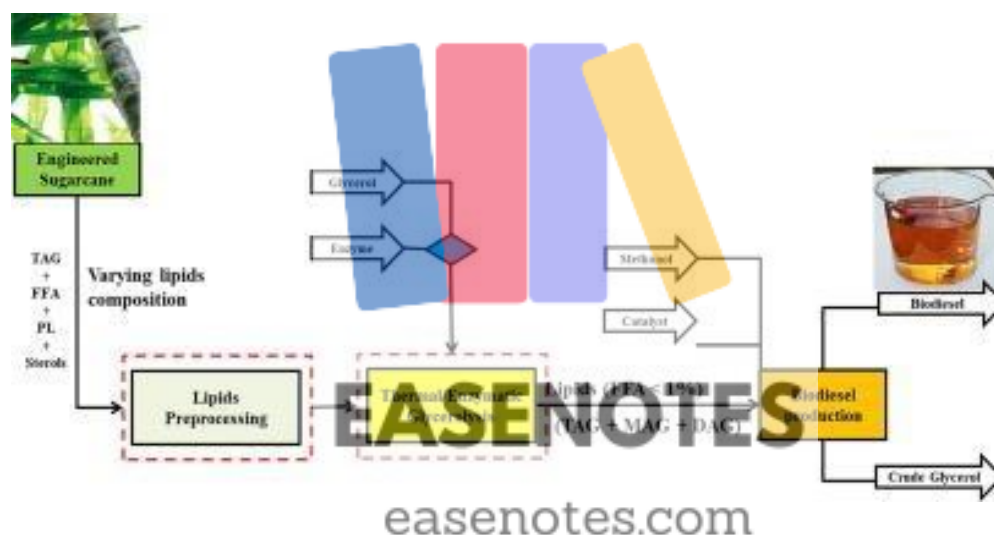
A biological membrane is a form of lamellar phase lipid bilayer. The formation of lipid bilayers is an energetically preferred process when the glycerophospholipids described above are in an aqueous environment. This is known as the hydrophobic effect.

Some of applications are: within the body, lipids function as an energy reserve, regulate hormones, transmit nerve impulses, cushion vital organs, and transport fat-soluble

nutrients. Fat in food serves as an energy source with high caloric density, adds texture and taste, and contributes to society.

Lipid obtained from food waste was used as a potential feedstock for biodiesel production using both a chemical catalyst and biocatalyst. Base (KOH) catalyzed transesterification of the lipid allowed a 100% conversion of biodiesel in 2 h; whereas Novozyme-435 yielded 90% biodiesel in 24h. So lipids are having a main application in biodiesel production.

Biodiesel demonstrates an animal fat-based or vegetable oil diesel fuel, including long chain- alkyl (methyl, ethyl, or propyl) esters. Biodiesel is generally made by esterifying lipids (eg: soybean oil, vegetable oil, and animal fat (tallow)) with an alcohol generating fatty acid esters. Biodiesel is suggested to be utilized in standard diesel engines and is thus well defined from the vegetable and waste oils used to fuel converted diesel engines. Biodiesel can be used singly or blended with gasoline in any proportions. Biodiesel blends can also be utilized as heating oil.



(Biodiesel Production)

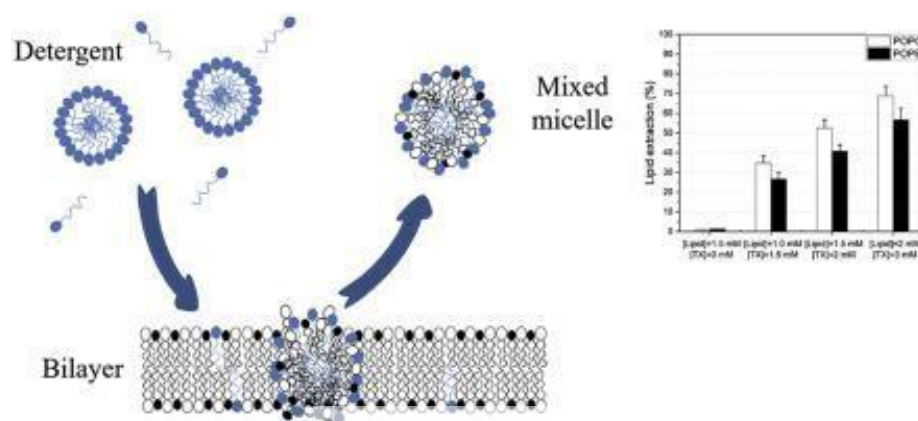
Lipids also has an interesting application in cleaning agents. Considering the case of,

DETERGENTS:

The hydrophobic end of the phospholipid bilayer stays away from the water. This avoids the dissolution of cell membrane in water. But the detergent can bind to the hydrophobic end of the cell membrane and from a solution with water, thus breaking the cell membrane barrier.

Detergent monomers solubilize membrane proteins by partitioning into the membrane bilayer. With increasing amounts of detergents, membranes undergo various stages of solubilization. The initial stage is lysis or rupture of the membrane.

While lipids also have the same general structure as detergents – a polar hydrophilic head group and a nonpolar hydrophobic tail- lipids differ from detergents in the shape of monomers, in the type of aggregates formed in solution, and in the concentration range required for aggregation.



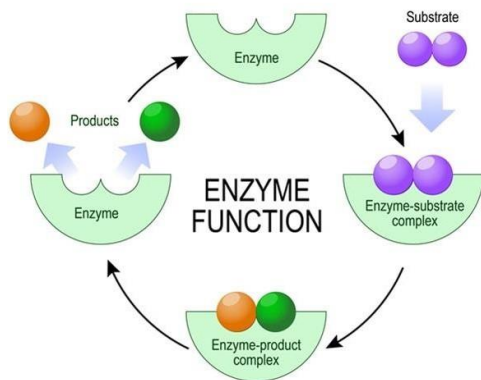
ENZYMES:

Are another important biomolecule, which are proteins that help speed up metabolism, or the chemical reactions in our bodies. They build some substances and break others down. All living things have enzymes. Our bodies naturally produce enzymes. The six kinds of enzymes are **hydrolases, oxidoreductases, lyases, transferases, ligases, and isomerases.**

Enzymes perform the critical task of lowering a reactions activation energy – that is the amount of energy that must be put in for the reaction to begin. Enzymes work by binding to reactant molecules and holding them in such a way that the chemical bond-breaking and bond-forming processes take place more readily.

Due to their high specificity, simplicity, and scalability, enzyme-based biosensors represent a fast, precise and continuous monitoring of analytes. Additionally, the high specificity of enzymes enhances the ability to detect lower analyte concentration limits. So enzymes are used in biosensors.

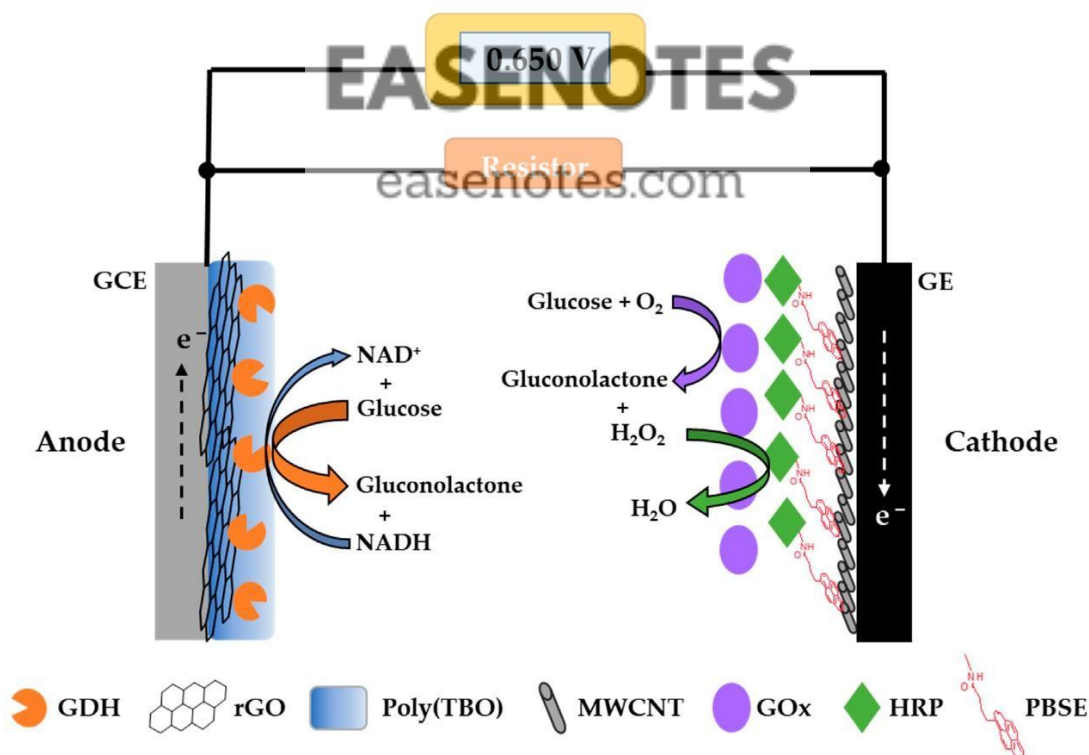
Biosensors are employed in applications such as disease monitoring, drug discovery, and detection of pollutants, disease causing micro-organisms and markers that are indicators of a disease in bodily fluids(blood, urine, saliva, sweat). Various types of biosensors being used are enzyme-based, immunosensors, DNA biosensors, thermal and piezoelectric biosensors. There are wide variety of enzymes used in biosensors. One such enzyme is glucose oxidase, mainly in amperometric glucose biosensor.



(function of enzymes)

GLUCOSE OXIDASE IN BIOSENSORS:

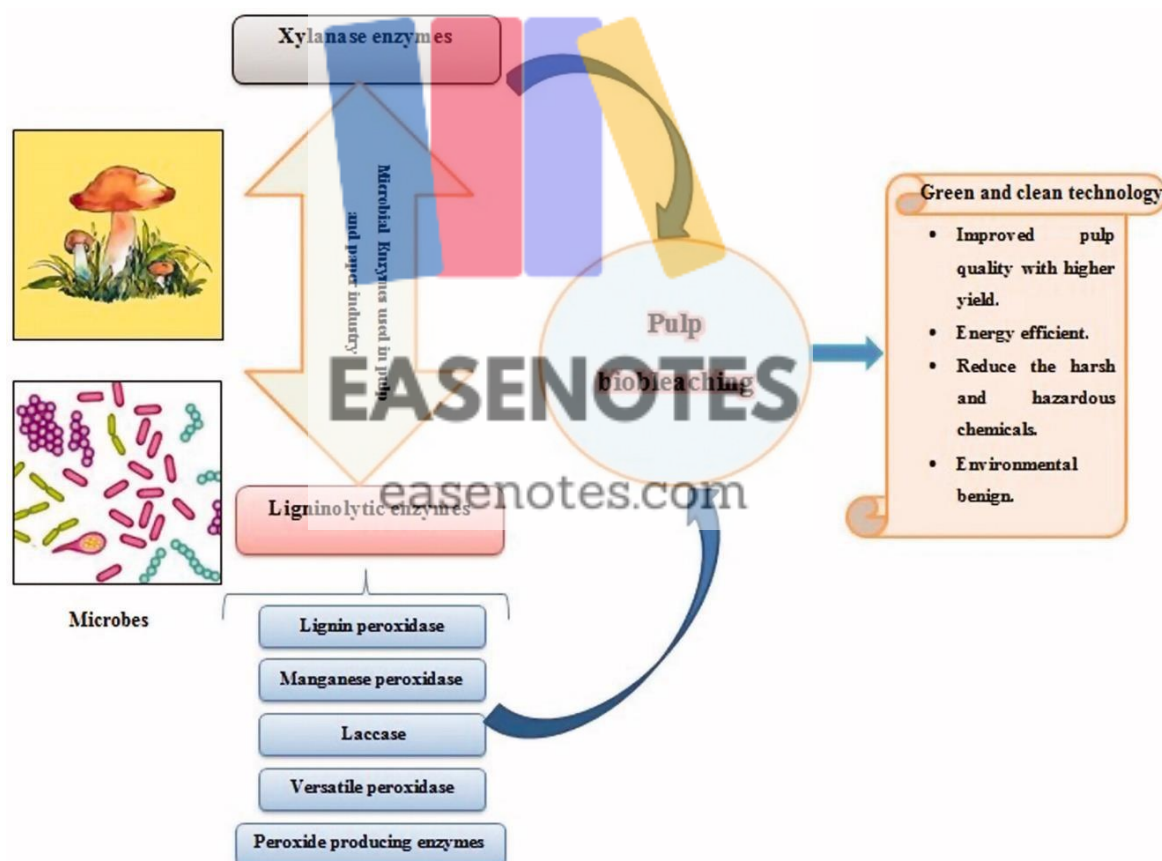
Glucose oxidase (Gox) is widely used enzyme in glucose biosensor due to its better stability and relatively inexpensive. Gox catalyses the redox reaction and transfer electrons from enzyme active sites to electrode for glucose level analysis in blood samples. Amperometric glucose biosensor was fabricated by immobilizing glucose oxidase (GOx), the most popular enzyme used for glucose detection, is able to reduce oxygen to hydrogen peroxide while at the same time transforming glucose to d-glucono-1, 5-lactone. Quantification of glucose can be achieved based on their the detection of the hydrogen peroxide produced or the oxygen consumed.



LIGNOLYTIC ENZYME IN BIO BLEACHING:

Ligninolytic enzymes play a key role in degradation and detoxification of lignocellulosic waste in environment. The major ligninolytic enzymes are laccase, lignin peroxidase, manganese peroxidase and versatile peroxidase. Ligninolytic fungi and enzymes (ie, laccase, manganese peroxidase, and lignin peroxidase) have been applied recently in the production of second-generation biofuels.

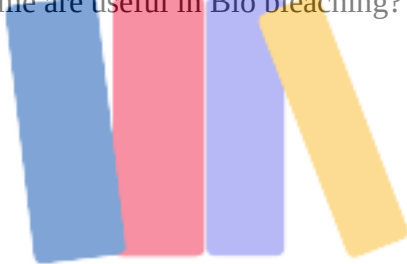
White-rot fungi are the main producers of lignin-oxidising enzymes. These fungi secrete a number of oxidative enzymes and some hitherto unknown substances (mediators) into their environment together effecting a slow but continuous degradation. The most important lignin- oxidising enzymes are lignin peroxidases, manganese peroxidases and laccases. Lignin peroxidase and manganese peroxidase appear to constitute a major component of the ligninolytic system.



(Ligninolytic enzyme in bio bleaching)

QUESTION BANK MODULE 2

1. What are Biomolecules? Write different types of Biomolecules.
2. What are Carbohydrates? Briefly explain main three types of Carbohydrates.
3. What is Cellulose-Based water filters? Write different types filtration methods.
4. Distinguish between PLA and PHA bioplastics.
5. Write a note on Nucleic Acid.
6. How DNA vaccines are useful for Rabies?
7. How RNA vaccines are useful for Covid-19?
8. Briefly explain DNA fingerprinting.
9. What are Proteins? How proteins are useful for human bodies?
10. Write a short note on Whey Protein.
11. Explain Meat analogues and Plant based proteins.
12. What are Lipids? Explain the production of biodiesel from lipids.
13. Write a note on Detergents.
14. What are Enzymes? Explain Biosensors.
15. Briefly explain Glucose Oxidase in Biosensors.
16. How lignolytic enzyme are useful in Bio bleaching?



EASENOTES

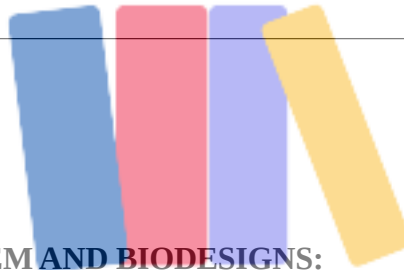
easenotes.com

BIOLOGY FOR ENGINEERS

MODULE 3

HUMAN ORGAN SYSTEM AND BIODESIGNS (QUALITATIVE)

Brain as a CPU system (architecture, CNS and Peripheral Nervous System, signal transmission, EEG, Robotic arms for prosthetics. Engineering solutions for Parkinson's disease). Eye as a Camera system (architecture of rod and cone cells, optical corrections, cataract, lens materials, bionic eye). Heart as a pump system (architecture, electrical signalling - ECG monitoring and heart related issues, reasons for blockages of blood vessels, design of stents, pace makers, defibrillators). Lungs as purification system (architecture, gas exchange mechanisms, spirometry, abnormal lung physiology - COPD, Ventilators, Heart-lung machine). Kidney as a filtration system (architecture, mechanism of filtration, CKD, dialysis systems).

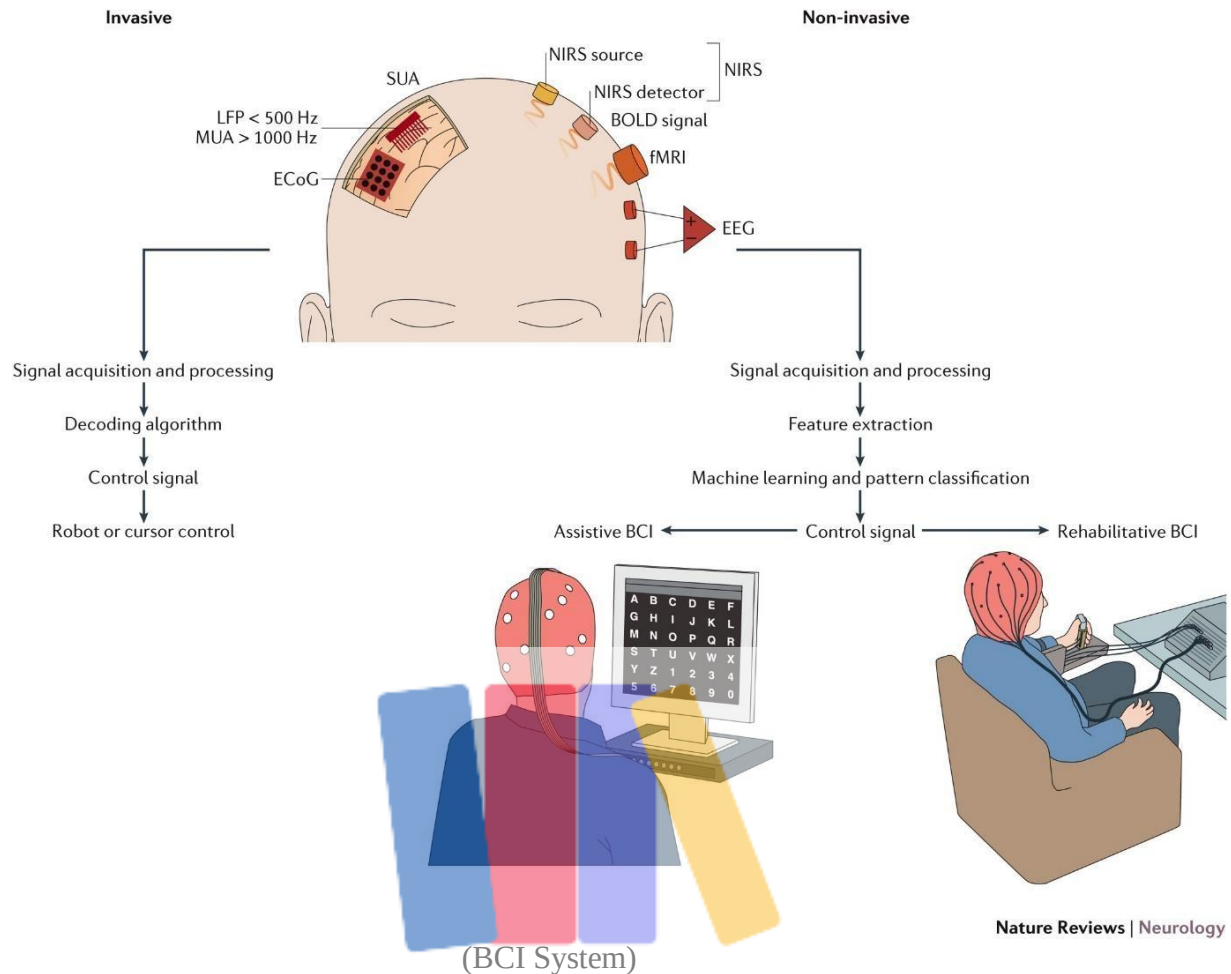


HUMAN ORGAN SYSTEM AND BIODESIGNS:

The human body is a biological machine made of body systems, groups of organs that work together to produce and sustain life. Organ systems are skeletal system, muscular system, cardiovascular system, Respiratory system, Nervous system, Digestive system, Urinary system, Endocrine system, Lymphatic system, reproductive system, Integumentary system.

Biodesign is the use of living organisms in design. Its processes can be used in the creation of fashion, textiles, furniture, and architecture. Non profits, design companies and universities around the world, including UC Davis, increasingly implement biodesign practices into research and product development.

Abio-design embodies an emerging design movement which incorporates the use of living materials, or moist media, such as fungi, algae, yeast, bacteria, and cultured tissue. This can be as part of standard crafting methods or the more complex fields of biomimicry and synthetic biology. The idea is to create a product whose properties are enhanced as a result of the use of these living materials. Some examples are explained below:



EASENOTES

BRAIN AS A CPU SYSTEM:

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Both CPU and brain use electrical signals to send messages. The brain uses chemicals to transmit information. The computer uses electricity. Even though electrical signals travel at high speeds in the nervous system, they travel even faster through the wires in a computer. Both transmit information.

A BCI system is a computer-based system that takes brain signals, analyses them and translates them into commands that are relayed to a device to trigger a desired action. A BCI system does not use peripheral nervous and head muscles. The CNS (Central Nervous System), for example is used to measure signals produced by the central nervous system. Thus, for example a sensor that is activated by the voice or the movement of a muscle is not a BCI system. Also an EEG is not BCI itself. Because it only records brain signals but it does not produce an output that acts on the user's environment. It is also wrong to think that BCI is a mind-reader. They do not export information from unsuspecting users or users unwillingly using the system. They allow users to act in their environment when they want it by reading their brain signals rather than muscles. The user and the BCI system. The BCI then translates these commands and transmits them into an output device. Brain computer interfaces have contributed to various areas of research. Applications that are about medicine, neuro-technology and smart environment,

neuro-marketing and advertising., education and self-regulation, games and entertainment, as well as security and identification.

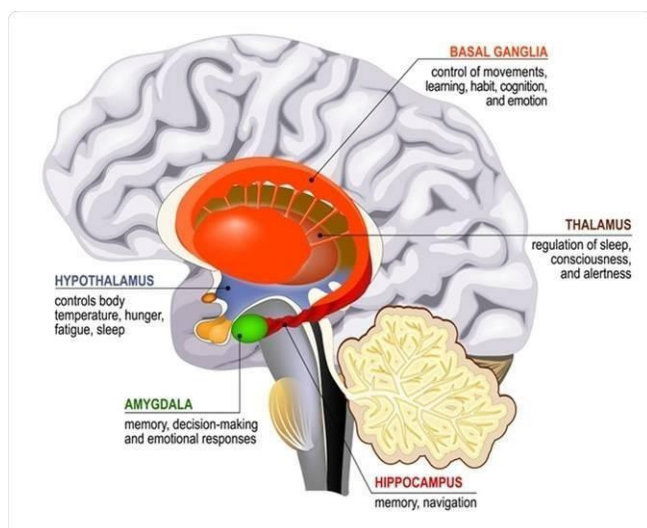
The human brain can be thought of as a highly sophisticated and complex information processing system, similar to a computer's Central Processing Unit (CPU). Both the brain and CPU receive and process inputs, store information, and perform calculations to produce outputs. However, there are significant differences between the two, such as the way they store and process information and the fact that the human brain has the ability to learn and adapt, while a computer's CPU does not. Additionally, the human brain is capable of performing tasks such as perception, thought, and emotion, which are beyond the scope of a computer's CPU.

Comparison between Brain and Computer.

Basis for Comparison	Brain	Computer
Construction	Neurons and synapses	ICs, transistors, diodes, capacitors, transistors, etc.
Memory growth	Increases each time by connecting synaptic links	Increases by adding more memory chips
Backup systems	Built-in backup system	Backup system is constructed manually
Memory power	100 teraflops (100 trillion calculations/seconds)	100 million megabytes
Memory density	10^7 circuits/cm ³	10^{14} bits/cm ³
Energy consumption	12 watts of power	Gigawatts of power
Information storage	Stored in electrochemical and electric impulses.	Stored in numeric and symbolic form (i.e. in binary bits).
Size and weight	The brain's volume is 1500 cm ³ and weight is around 3.3pounds.	Variable weight and size form few grams to tons.
Transmission of information	Uses chemicals to fire the action potential in the neurons.	Communication is achieved through electrical coded signals.
Information processing power	Low	High
Input/output equipment	Sensory organs	Keyboards, mouse, web cameras, etc.
Structural organization	Self-organized	Pre-programmed structure
Parallelism	Massive	Limited
Reliability and damageability properties	Brain is self-organizing, self-maintaining and reliable.	Computers perform a monotonous job and can't correct itself.

Architecture:

Similarly, a computer's CPU also has memory units for storing information, and the human brain has several regions dedicated to memory storage, including the hippocampus and amygdala. Figure: Limbic system. Cross section of the human brain.

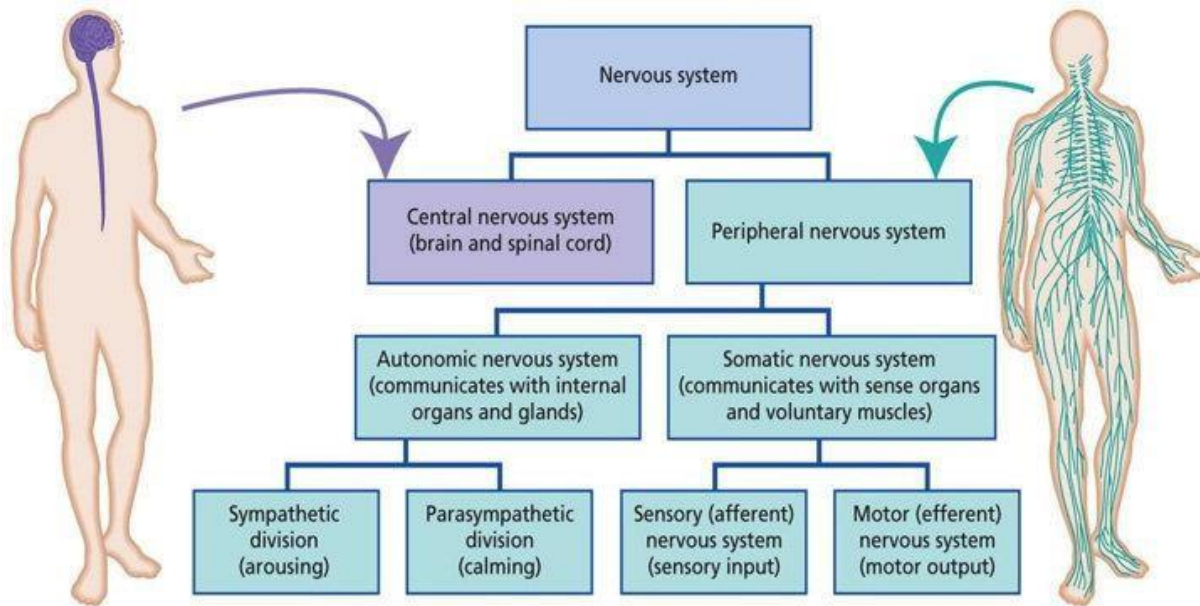


While the comparison between the human brain and a computer's CPU can provide useful insights, it is important to note that the human brain is a vastly more complex and capable system, with many functions that are still not fully understood.

The nervous system has two main parts : **The central nervous system (CNS)** made up of the brain and spinal cord. The **Peripheral Nervous System(PNS)** is made up of nervous that branch off from the spinal cord and extended to all parts of the body.

1)CNS: CNS includes the brain and spinal cord. The brain is the body's "control centre." The CNS has various centers located within it that carry out the sensory, motor and integration of data. These centers can be subdivided to Lower Centers (including the spinal cord and brain stem) and higher centers communicating with the brain via effectors.

2)PNS: PNS is a vast network of spinal and cranial nervous that are linked to the brain and the spinal cord. It contains sensory receptors which help in processing changes in the internal and external environment. This information is sent to the CNS via afferent sensory nervous. The PNS is then subdivided into the automatic nervous system and the somatic nervous system. The automatic has involuntary control of internal organs, blood vessels, smooth and cardiac muscles. The somatic has voluntary control of skin, bones, joints and skeletal muscles. The two systems function together, by way of nervous from the PNS entering and becoming part of the CNS, and vice versa.

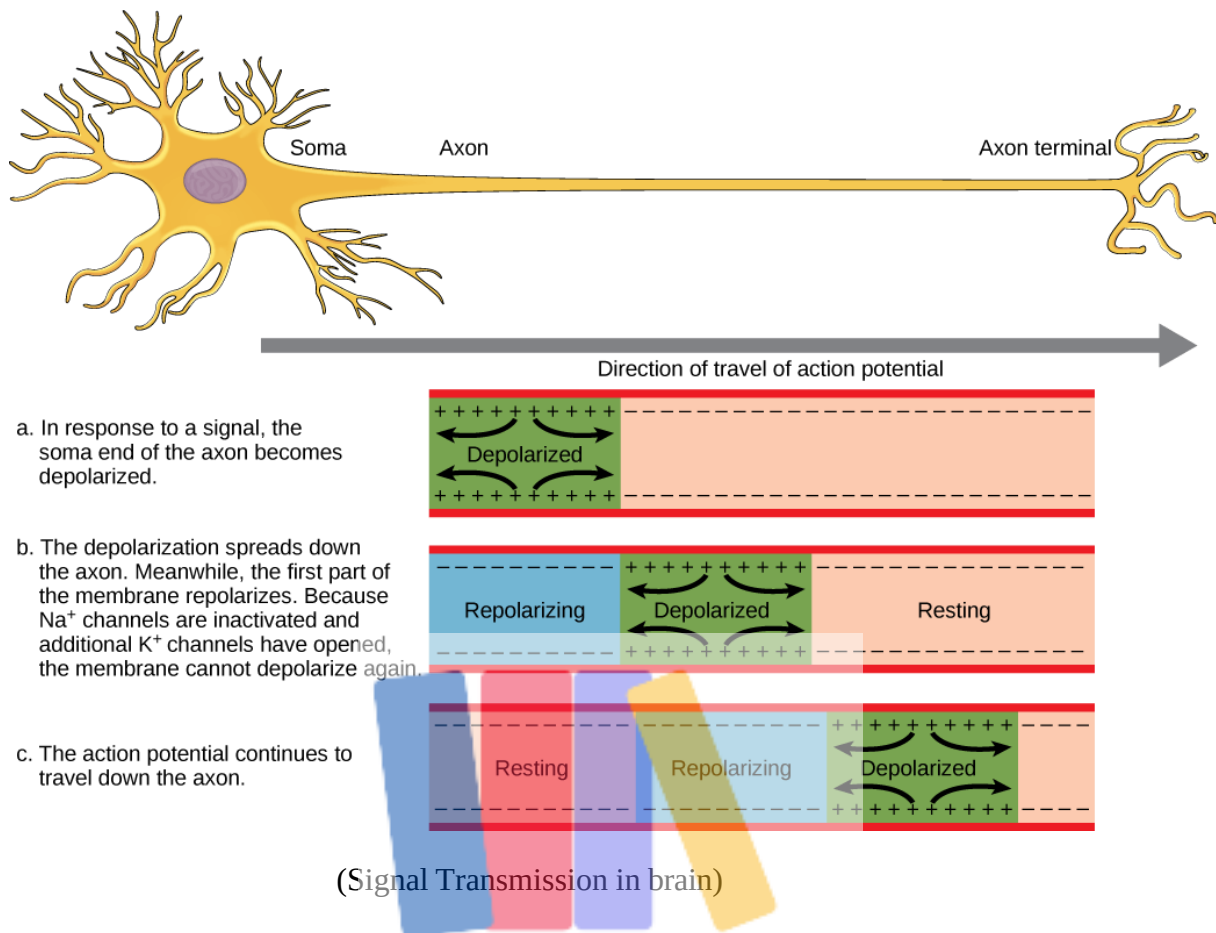


(CNS and PNS Nervous system)

SIGNAL TRANSMISSION:

A neuron sending a signal (a presynaptic neuron) releases a chemical called a neurotransmitter, which binds to a receptor on the surface of the receiving (postsynaptic) neuron. Neurotransmitters are released from presynaptic terminals, which may branch to communicate with several postsynaptic neurons.

Axon terminals are where neurotransmission begins. Hence, it is at axon terminals where the neuron sends its Output to other neurons. At electrical synapses, the Output will be the electrical signal itself. At chemical synapses, the Output will be neurotransmitter. The correct outline for the sequence of transmission of an electrical impulse through a neuron is dendrites, cell body, axon, axon terminal.



ELECTRO ENCEPHALO GRAM [EEG]:

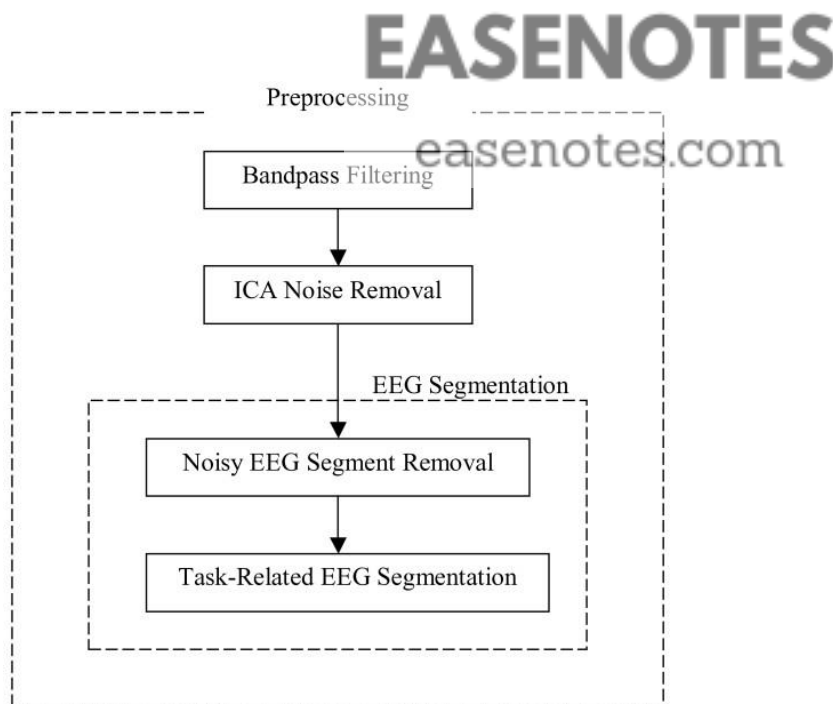
An electroencephalogram (EEG) is a test that measures electrical activity in the brain using small metal discs (electrodes) attached to the scalp. Brain cells communicate via electrical impulses and are active all the time even during asleep. This activity shows up as wavy lines on an EEG recording.

An EEG is one of the main diagnostic tests for epilepsy. An EEG can also play a role in diagnosing other brain disorders. An EEG can find changes in brain activity that might be useful in diagnosing brain disorders, especially epilepsy or another seizure disorder. An EEG might also be helpful for diagnosing or treating.

- Brain tumors
- Brain damage from head injury
- Brain dysfunction that can have a variety of causes (encephalopathy)
- Sleep disorders
- Inflammation of the brain (herpes encephalitis)
- Stroke
- Sleep disorders
- Creutzfeldt-Jakob disease

An EEG might also be used to confirm brain death in someone in a persistent coma. A continuous EEG is used to help find the right level of anaesthesia for someone in a medically induced coma. Voltage fluctuations measured by the EEG bio amplifier and electrodes allow the evaluation of normal brain activity including the posterior dominant rhythm (PDR), first described by Hans Berger. EEG can detect abnormal electrical discharges such as sharp waves, spikes or spike and wave complexes that are seen in people with epilepsy. Thus it is often used to inform the medical diagnosis. EEG can detect the onset and spatio-temporal evolution of seizures and the presence of status epilepticus. It is also used to help diagnose sleep disorders, depth of anaesthesia, coma, encephalopathies, cerebral hypoxia after cardiac arrest, and brain death. EEG used to be a first line method of diagnosis for tumours, stroke and other focal brain disorders but this use has decreased with the advent of high-resolution anatomical imaging techniques such as magnetic resonance imaging (MRI) and computed tomography (CT). Despite limited spatial resolution, EEG continuous to be a valuable tool for research and diagnosis. It is one of the few mobile techniques available and offers millisecond-range temporal resolution which is not possible with CT, PET or MRI.

Derivatives EEG technique include evoked potentials (EP), which involves averaging the EEG activity time-locked to the presentation of a stimulus of some sort (visual, somatosensory, or auditory). Event-related potentials (ERPs) refer to averaged EEG responses that are time-locked to more complex processing of stimuli. This technique is used in cognitive science, cognitive psychology and psychophysiological research.



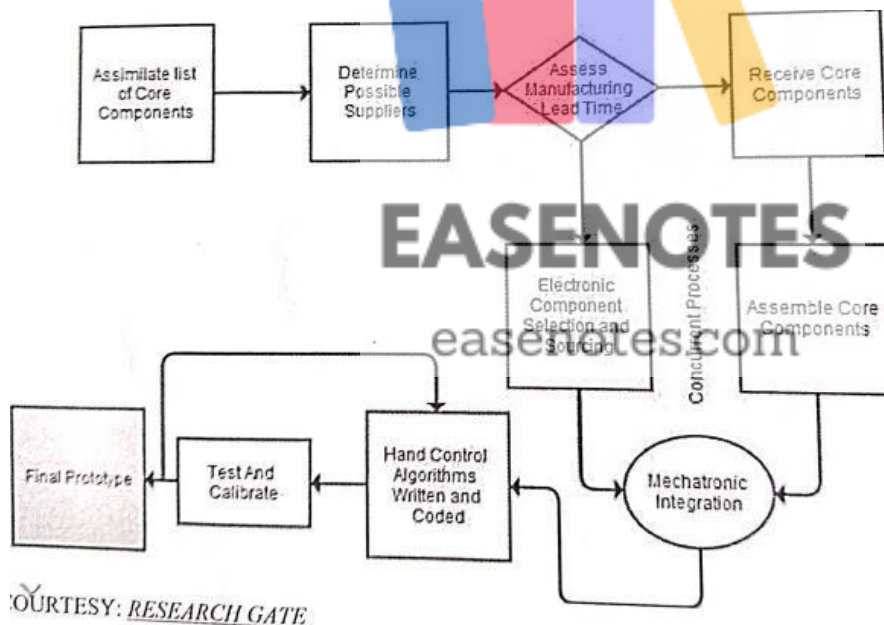
(EEG Process)

ROBOTIC ARMS FOR PROSTHETICS:

Robotic prosthetic limb is a well-established research area that integrates advanced mechatronics, intelligent sensing and control for achieving higher order lost sensorimotor functions while maintaining the physical appearance of amputated limb. Robotic prosthetic limbs are expected to replace the missing limbs of an amputee restoring the lost functions and providing aesthetic appearance. The main aspects are enhanced social interaction, comfortable amputee's life and productive amputee to the society. With the advancement of sensor technology, in the last few decades significant contributions have been made in this area.

Most current robotic prostheses work by recording from the surface of the skin-electrical signals from muscles left intact after an amputation. Some amputees can guide their artificial hand by contracting muscles remaining in the forearm that would have controlled their fingers.

If you are missing an arm or leg, an artificial limb can sometimes replace it. The device which is called a prosthesis. Can help you to perform daily activities such as walking, eating, or dressing. Robotic arms can be used to automate the process of placing goods or products onto pallets. By automating the process, palletizing becomes more accurate, cost-effective, and predictable. The use of robotic arms also frees human workers from performing tasks that present a risk of bodily injury.

**ENGINEERING SOLUTIONS FOR PARKINSON'S DESEASE:**

Parkinson's disease is a progressive disorder that affects the nervous system and the parts of the body controlled by the nerves. Symptoms start slowly. The first symptoms may be barely noticeable tremor in just one hand. Tremors are common, but the disorder may also causes stiffness or slowing of movement.

In Parkinson's disease, certain nerve cells (neurons) in the brain gradually breakdown or die.

Many of the symptoms are due to a loss of neurons that produce a chemical messenger in your brain called dopamine. When dopamine levels decrease, it causes a typical brain activity, leading to impaired movement and other symptoms of Parkinson's disease.

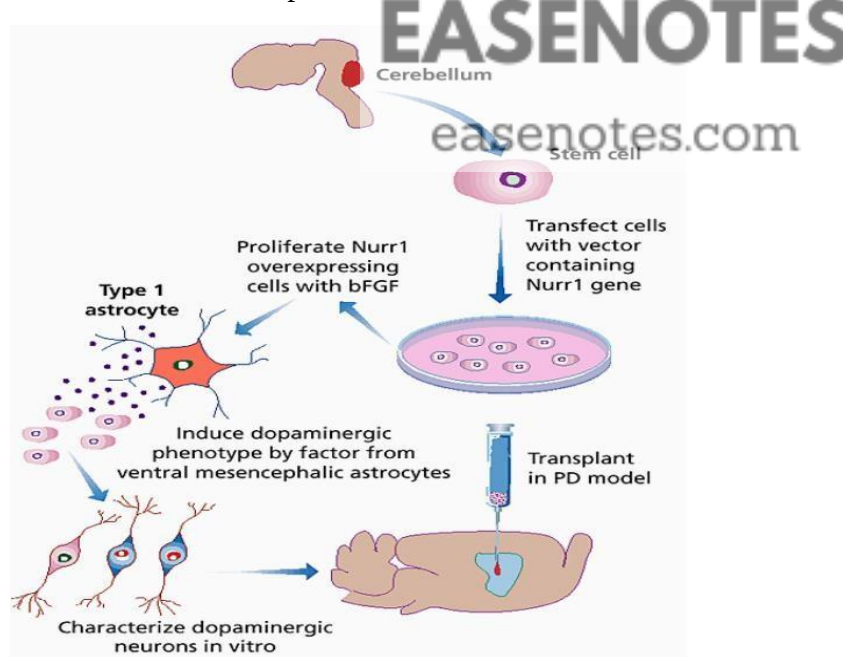
Parkinson's disease can't be cured, but medications can help control the symptoms often dramatically. In some more advanced cases, surgery may be advised. Your health care provider may also recommend lifestyle changes, especially ongoing aerobic exercise.

ENGINEERING SOLUTION TO THIS DISEASE ARE:

Deep brain stimulation(DBS) - This involves surgically implanting a neurotransmitter that sends electrical impulses to specific areas of our brain. This procedure has helped many people with Parkinson's reduce symptoms such as tremor, rigidity, and bradykinesia. There are six main types of medications available to treat symptoms of Parkinson disease; levodopa, dopamine agonists, and inhibitors of enzymes that inactivate dopamine (monoamine oxidase type B [MAO B] inhibitors and catechol-O-methyl transferase [COMT] inhibitors, anticholinergic drugs, and amantadine.

Engineering Neurons are another treatment method for this. Transplantation of embryonic neurons can restore functional dopaminergic neurons in the brains of patients with Parkinson's disease. But while, promising, cell transplantation therapy is still out of reach to most patients, in part because of the inaccessibility of human embryonic tissue.

First obtained neuronal stem cells from mouse cells transfected with a transcription factor that encourages cells to adopt a neuronal fate. They then co-cultured the cells with astrocytes, which release a factor that induces development into dopaminergic neurons. The engineered cells released dopamine and some maintained the characteristics of dopaminergic neurons for up to two weeks after implantation into mouse brain.

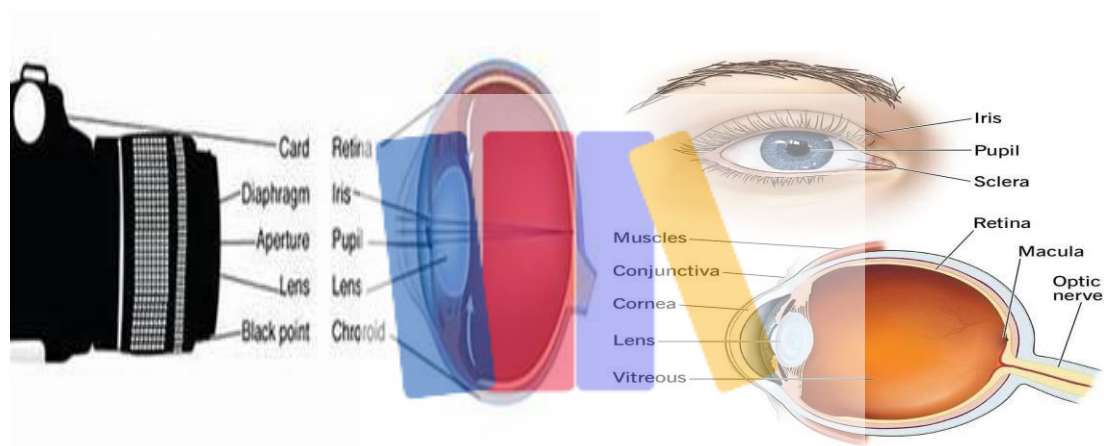


(Parkinson's disease solution)

EYE AS A CAMERA SYSTEM:

The human eye is a wonderful instrument, relying on refraction and lenses to form images. There are many similarities between the human eye and a camera. A diaphragm to control the amount of light that gets through to the lens. This is the shutter in a camera, and the pupil at the centre of the iris, in the human eye. A lens to focus the light and create an image. The image is real and inverted. In camera, film is used to record the image; in the eye, the image is focused on the retina, and a system of rods and cones is the end of an image-processing system that converts the image to electrical impulses and sends the information along the optic nerve to the brain.

The human eye can be analogized to a camera system, as both the eye and a camera capture light and convert it into an image



EASENOTES

The main components of the eye that correspond to a camera system include:

- **The Cornea:** This transparent outer layer of the eye functions like a camera lens, bending light to focus it onto the retina.
- **The Iris:** The iris functions like the diaphragm in a camera, controlling the amount of light that enters the eye.
- **The Pupil:** The pupil functions like the aperture in a camera, adjusting the size to control the amount of light entering the eye.
- **The Retina:** The retina functions like the camera film or sensor, capturing the light and converting it into electrical signals that are sent to the brain.
- **The Optic Nerve:** The optic nerve functions like the cable connecting the camera to a computer, transmitting the electrical signals from the retina to the brain.
- In both the eye and a camera, the captured light is transformed into an image by the lens and the light-sensitive component.
- The eye processes the image further, allowing for visual perception, while a camera stores the image for later use.

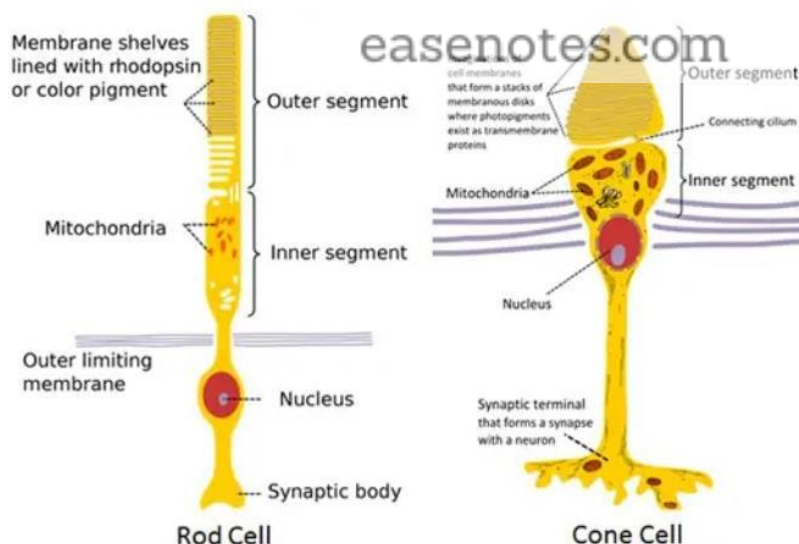
- It's important to note that the eye is much more complex than a camera and has several additional functions, such as adjusting for different levels of light and adjusting focus, that are not found in a camera.
- The eye also has the ability to perceive depth and color, as well as adjust to movements and provide a continuous, real-time image to the brain.

There are two photoreceptors: **RODS** and **CONES**

The photoreceptors are localized around an area near the centre of the retina called the macula, which is the functional centre of the retina. The fovea is located in the centre of the macula. The macula is responsible for high-resolution, colour vision, provided by different types of photoreceptors.

Photoreceptors in the retina are classified into two groups, named after their physical morphologies. Rod cells are highly sensitive to light and function in night vision, whereas cone cells are capable of detecting a wide spectrum of light photons and are responsible for colour vision. Rods and cones are structurally compartmentalized. They consist of five principal regions: Outer segment, connecting cilium, inner segment, nuclear region, Synaptic region, Rods are responsible for vision at low light levels (scotopic vision). They do not mediate colour vision and have a low spatial acuity.

Cones are active at higher light levels (photopic vision), are capable of colour vision and are responsible for high spatial acuity. The central fovea is populated exclusively by cones. There are 3 types of cones which we will refer to as the short- wavelength sensitive cones, the middle-wavelength sensitive cones and the long-wavelength sensitive cones or S-cone, M-cones, and L-cones for short.



OPTICAL CORRECTIONS:

A slight modification of geometrically correct lines (as of a building) for the purpose of making them appear correct to the eye.

Optical corrections refer to devices or techniques used to improve or correct vision problems caused by a refractive error in the eye.

Refractive errors occur when light entering the eye is not properly focused on the retina, leading to blurred vision. There are several types of refractive errors, including:

- Myopia (nearsightedness): Light is focused in front of the retina, making distant objects appear blurry.
- Hyperopia (farsightedness): Light is focused behind the retina, making near objects appear blurry.
- Astigmatism: Light is not focused evenly on the retina, leading to blurred or distorted vision.

The ability to see images or objects with clear, sharp vision results from light entering the eye. Light rays bend or refract when they hit the retina, sending nerve signals to the optic nerve, which then sends these signals to the brain. The brain processes them into images, allowing you to understand what you see. When these light rays bend incorrectly, it results in a refractive error and typically causes blurry or cloudy vision.

Since the primary cause of vision problem is caused by light bending incorrectly as it enters the eye, virtually any method of treatment that changes this can be categorised as a form of vision correction.

Eyeglasses and optical lenses- the most common types of corrective measures- are almost always recommended as the first course of treatment for vision problems. While they are considered a very basic method of vision correction, they are unable to control the refractive error from progressing. Patients whose vision worsens over time need new glasses or contacts. In these cases longer-term solutions are needed.

CATARACT:

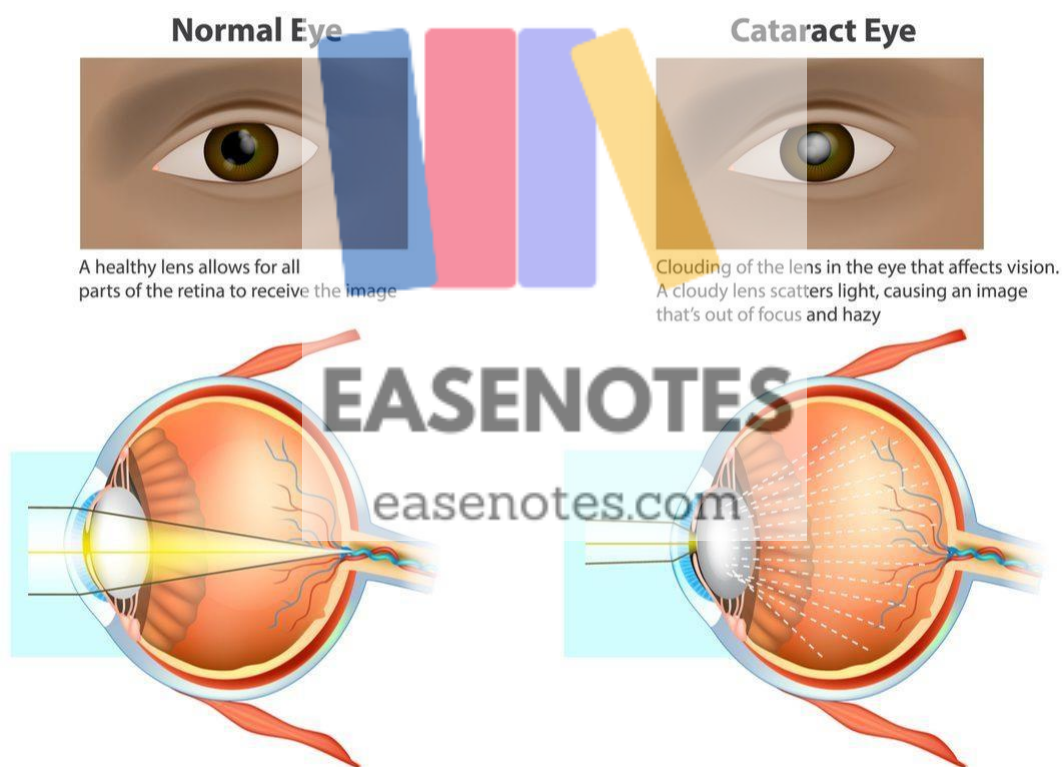
A cataract is a clouding of the normally clear lens of the eye. At first, the cloudiness in your vision caused by a cataract may affect only a small part of the eye's lens and you may be unaware of any vision loss. As the cataract grows larger, it clouds more of our lens and distorts the light passing through the lens. This may lead to more noticeable symptoms. A cataract is a cloudy lens. The lens is positioned behind the coloured part of eye (iris). The lens focuses light that passes into our eye, producing clear, sharp images on the retina, the light sensitive membrane in the eye that functions like the film in a camera.

As the cataract continues to develop, the clouding becomes denser. A cataract scatters and blocks the light as it passes through the lens, preventing a sharply defined image from reaching your retina. As a result, your vision becomes blurred. Cataracts generally develop in both eyes, but not always at the same rate. The cataract in one eye may be more advanced than the other, causing a difference in vision between eyes. Cataracts may be partial or complete, stationary or

progressive, hard or soft. Histologically, the main types of age-related cataracts are nuclear sclerosis, cortical, and posterior subcapsular.

Nuclear sclerosis is the most common type of cataract and involves the central or nuclear part of the lens. This eventually becomes hard or sclerotic, due to condensation on the lens nucleus and the deposition of brown pigment within the lens. In its advanced stages, it is called brunescant cataract. In early stages, an increase in sclerosis may cause an increase in refractive index of the lens. This causes a myopic shift (lenticular shift) that decreases hyperopia and enables presbyopic patients to see at near without reading glasses. This is only temporary and is called second sight.

An immature cataract has some transparent protein, but with a mature cataract, all the lens protein is opaque. In a hyper mature or Morgagnian cataract, the lens proteins have become liquid. Congenital cataract, which may be detected in adulthood, has a different classification and includes lamellar, polar, and sutural cataracts.



(Normal Eye and Cataract Eye)

LENS MATERIALS:

Corrective spherocylindrical lenses are commonly used to treat refractive errors such as myopia, hyperopia, presbyopia, and astigmatism. Both lenses and prisms are also frequently

used to improve eye alignment and treat diplopia in strabismus. Eye glasses also serve an important role in protecting the eyes from physical trauma and harmful radiation. Lenses can be produced using a variety of materials and designed with several optical profiles to optimize use in specific applications. Critical lens properties include refractive index, Abbe number (chromatic dispersion), specific gravity, and ultraviolet absorption.

The most common lens material is of course optical glass, but crystals and plastics are frequently used, while mirrors can be made of essentially anything that is capable of being polished. There are 5 main types of lens materials for eyeglasses and sunglasses. Each type of lens material can help correct refractive errors such as nearsightedness, farsightedness, astigmatism, or presbyopia.

Types of lens materials:

1. **CR-39:** The most used plastic lens material for years was CR-39. It was first developed as a replacement for glass lenses during World War II. It still has 55% of world market at age 60. The patent was awarded to Muskat and Strain of Pittsburgh Plate Glass Company (now named PPG) in 1946. CR-39 is available in all lens styles and from multiple manufactures. The basic monomer comes from PPG, and then each company adds their own materials to create their lenses. Advantages include light weight, good optical properties and tinting well. Disadvantages of CR-39 are that it is the thickest material and scratches easily.
2. **Crown Glass:** Is the most commonly used clear glass for ophthalmic lenses. In general, glass is the most durable material used for lenses. Crown glass is used mainly for single vision lenses and the distance carrier for most glass bifocals and trifocals. It has an index of refraction of 1.523 and an abbe value of 59. It is approximately 4% thinner than CR-39 resin lenses and is 40% heavier than polycarbonate lenses and is slightly lighter than high index glass. It blocks out about 10% of UV light.
3. **Flint Glass:** It uses lead oxides in its chemical make up to increase its index of refraction to approximately 1.58 to 1.69. Its abbe value ranges from 30 to 40. This material is relatively soft, displays a brilliant luster and has chromatic aberration. Although it was used in the past as a single vision alternative for higher Rx lenses, its use today is often limited to segments for some fused bifocals. The advantages of glass lenses include optical clarity, resistance to scratches and it is the least susceptible to chemicals. The disadvantages include that it is the heaviest material and it is less impact resistant than other materials.
4. **Polycarbonate Lenses:** Polycarbonate lenses were first developed by a company named Gentex. Polycarbonate is a thermoplastic which means it is moldable under sufficient heat. In the 1950's it was marketed under the name Lexan and due to its extraordinary resistance to impact was originally manufactured for safety devices.

BIONIC EYES:

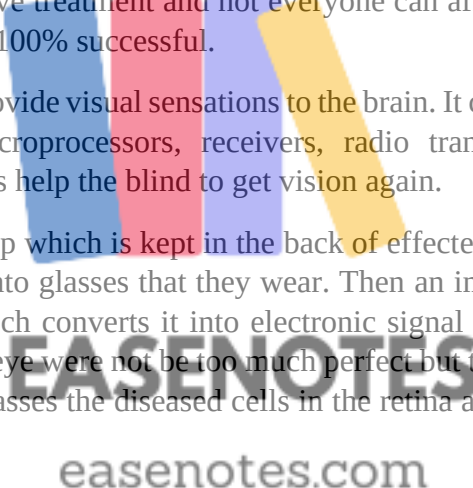
Bionic eye, electrical prosthesis surgically implanted into a human eye in order to allow for the transduction of light (the change of light from the environment into impulses the brain can process) in people who have sustained severe damage to the retina.

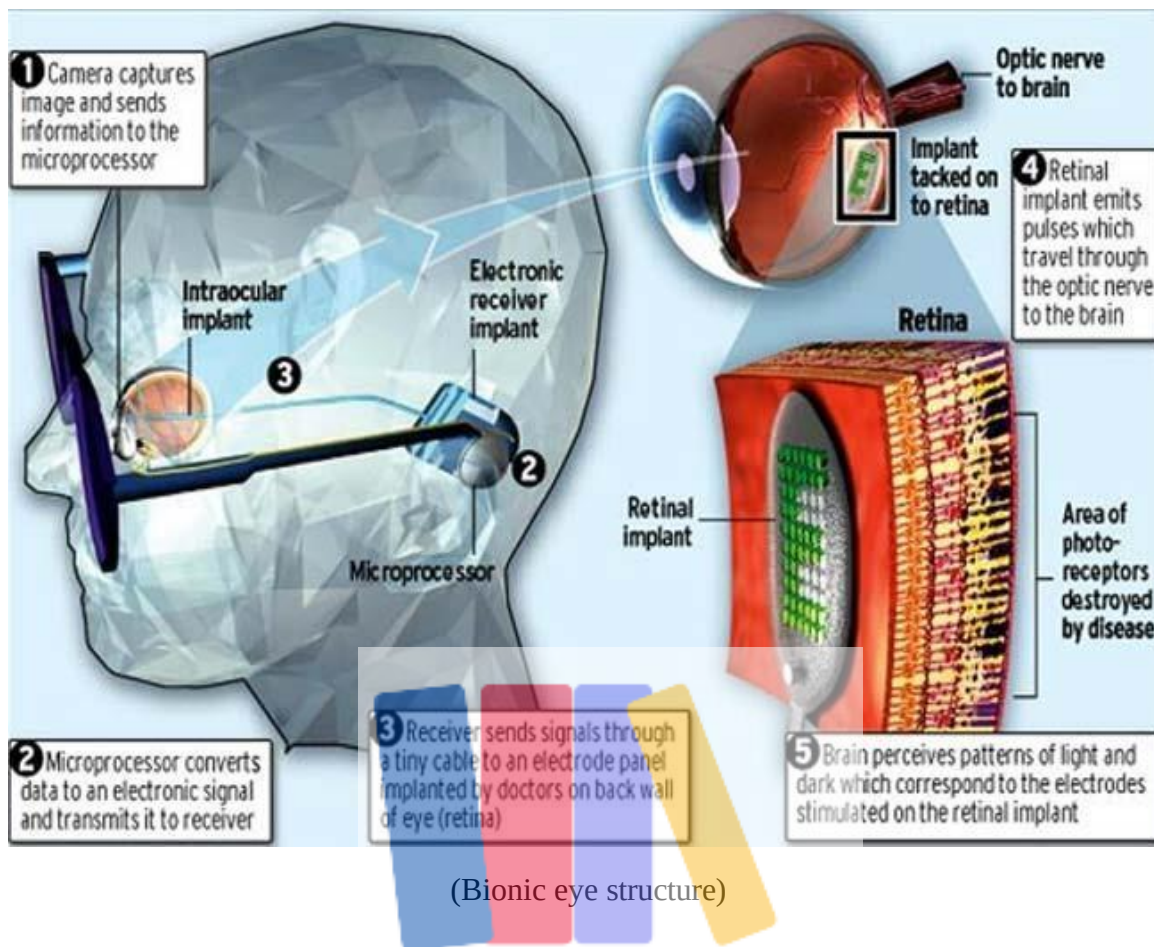
The bionic eye comprises an external camera and transmitter and an internal microchip. The camera is mounted on a pair of eyeglasses, where it serves to organize the visual stimuli of the environment before emitting high-frequency radio waves. The stimulator microchip consists of an electrode array that is surgically implanted into the retina. That functions as an electrical relay in place of degenerated retinal cells. The radio waves that are emitted by the external camera and transmitter are received by the stimulator, which then fires electrical impulses. The impulses are relayed by the few remaining retinal cells and are transduced as normal to the optic nerve pathway, resulting in vision.

The bionic vision system consists of camera, attached to a pair of glasses, which transmits high-frequency radio signals to a microchip implanted in the retina. Electrodes on the implanted chip convert these signals into electrical impulses to stimulate cells in the retina that connect to the optic nerve. It is an expensive treatment and not everyone can afford it. Since research is still going on results are yet not 100% successful.

It is an artificial eye which provides visual sensations to the brain. It consists of electronic systems having image sensors, microprocessors, receivers, radio transmitters and retinal chips. Technology provided by this helps the blind to get vision again.

It consists of a computer chip which is kept in the back of affected person's eye and linked with a mini video camera built into glasses that they wear. Then an image captured by the camera is focused to the chip which converts it into an electronic signal that the brain can interpret. The images produced by bionic eyes were not too much perfect but they could be clear enough to recognize. The implant bypasses the diseased cells in the retina and goes through the remaining possible cells.





The device consists of 3,500 micro photodiodes which are set at the back part of the retina. The electrical signal which is sent to brain is obtained from these miniature solar cells array as they convert the normal light to electrical signal.

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(Bionic Eyes)

HEART AS A PUMP SYSTEM:

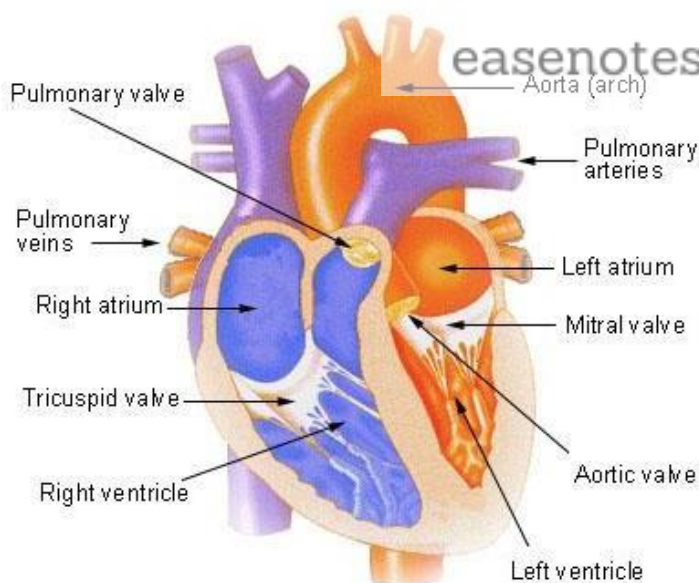
Heart is a sort of like a pump, or two pumps in one. The right side of our heart receives blood

from the body and pumps it to the lungs. The left side of the heart does the exact opposite. It receives blood from the lungs and pumps it out to the body. While an LVAD consists of thick tubes and a pump connected externally to the heart muscle and aorta, percutaneous heart pumps place a much smaller tube inside the heart's chambers. These tiny heart pumps are placed in the heart via a thin tube called a catheter that is threaded through a puncture site in the skin. The human heart is very strong and is capable of pumping blood up to 30 feet distance. An average heart beats maximum of 70-80 beats per minute and is considered healthy. The efficiency of the heart can be maintained and improved by performing physical activity. The heart is called double pump because each side pumps blood to a different circulation. Deoxygenated blood from the body drains to the right side of the heart. This is the first pump that sends blood to the lungs, called pulmonary circulation, where it becomes oxygenated and releases carbon dioxide.

Here is what happens as blood flows through the heart and lungs:

The blood first enters the right atrium. The blood then flows through the tricuspid valve into the right ventricle. When the heart beats, the ventricle pushes blood through the pulmonic valve into the pulmonary artery. The pulmonary artery carries blood to the lungs where it “picks up” oxygen. It then leaves the lungs to return to the heart through the pulmonary vein. The blood enters the left atrium. It drops through the mitral valve into the left ventricle. The left ventricle then pumps blood through the aortic valve and into the aorta. The aorta is the artery that feeds the rest of the body through a system of blood vessels. Blood returns to the heart from the body via two large blood vessels called the superior vena cava and the inferior vena cava. This blood carries little oxygen, as it is returning from the body where oxygen was used. The vena cava pump blood into the right atrium and the cycle begins all over again.

Internal View of the Heart



The human heart is a four-chambered muscular organ shaped and sized roughly like a man's closed fist with two-thirds of the mass to the left of midline. The heart is enclosed in a pericardial sac that is lined with the parietal layers of a serous membrane. The visceral layer of the serous membrane forms the epicardium.

The myocardium of the heart wall is a working muscle that needs a continuous supply of oxygen and nutrients to function efficiently.

For this reason, cardiac muscle has an extensive network of blood vessels to bring oxygen to the contracting cells and to remove waste products.

ELECTRICAL SIGNALING:

The sinus node generates an electrical stimulus regularly, 60 to 100 times per minute under normal conditions. The atria are then activated. The electrical stimulus travels down through the conduction pathways and causes the heart's ventricles to contract and pump out blood.

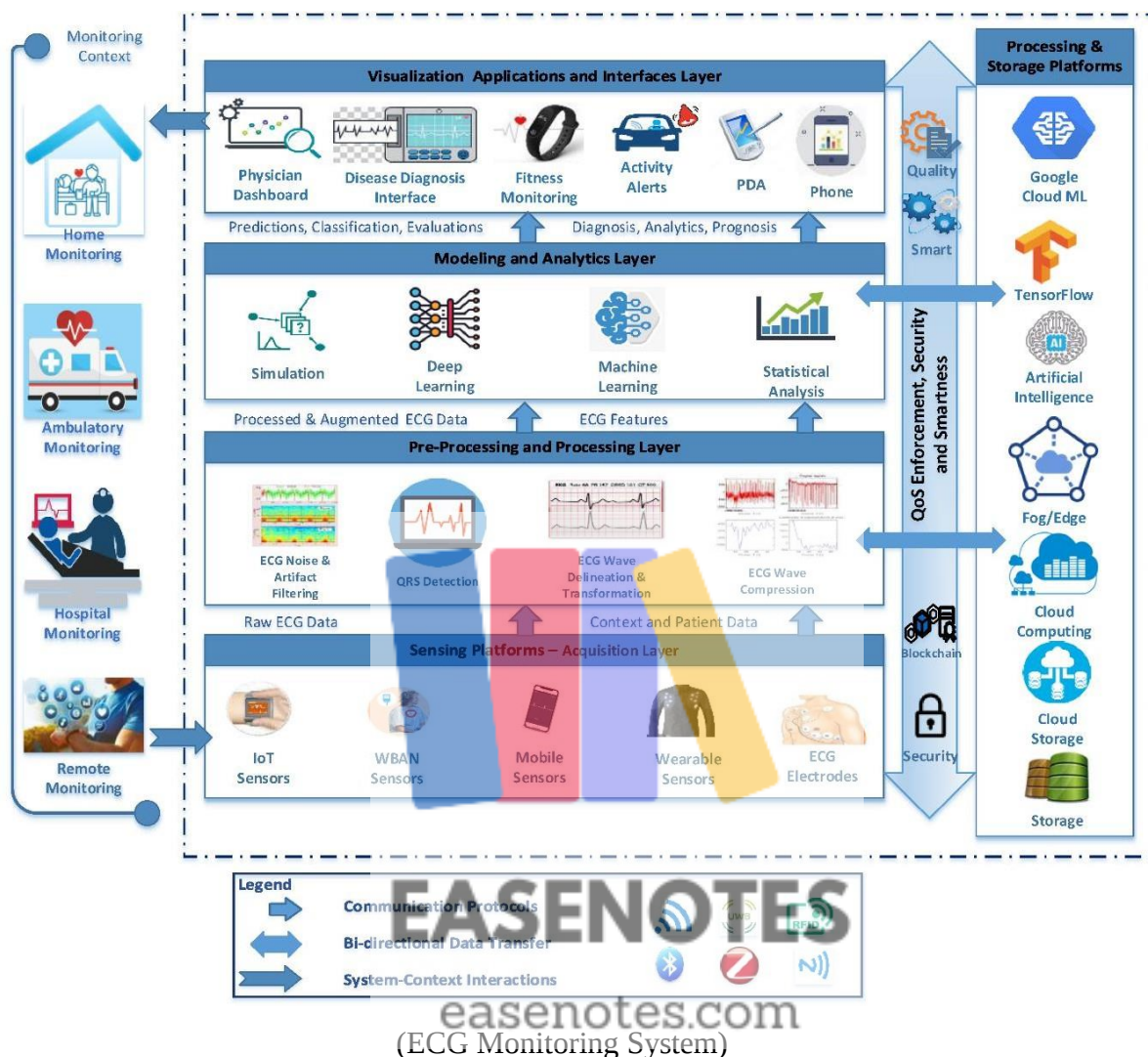
ECG MONITORING:

ECG monitoring systems have been developed and widely used in the healthcare sector for the past few decades and have significantly evolved over time due to the emergence of smart enabling technologies.

Nowadays, ECG monitoring systems are used in hospitals, homes, outpatient ambulatory settings, and in remote contexts. They also employ a wide range of technologies such as IoT, edge computing and mobile computing. In addition, they implement various computational settings in terms of processing frequencies, as well as monitoring schemes. They have also evolved to serve purposes and targets other than disease diagnosis and control, including daily activities, sports, and even mode-related purposes.

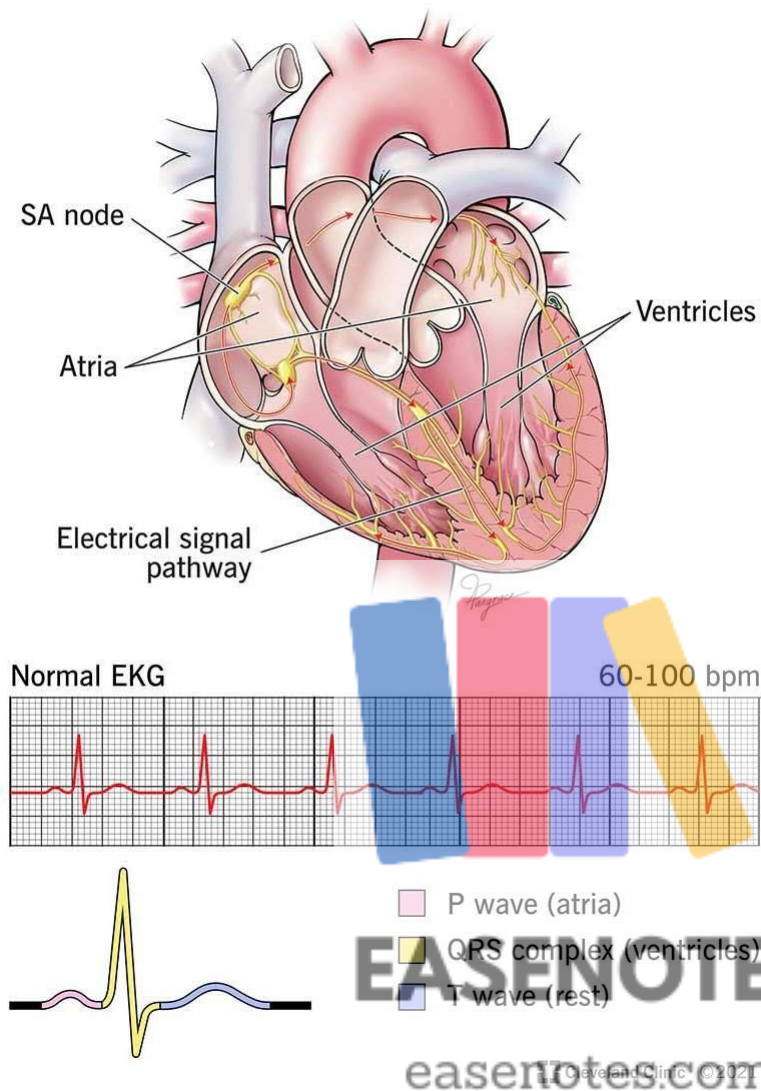
This massive diversity in ECG monitoring systems contexts, technologies, computational schemes, and purposes makes it hard for researchers and professionals to design, classify, and analyse ECG monitoring systems. Some efforts attempted to provide a common understanding of ECG monitoring systems processes, guiding the design of efficient monitoring systems. However, these studies lack comprehensiveness and completeness. They work for specific contexts, serve specific targets or are suitable for specific technologies.

They do not consider the latest technologies trends and they target very narrow research niches, such as wearable sensors, mobile sensors, disease diagnosis, heartbeat detection, emotion recognition, or ECG compression methods. Hence, there is a need to provide a comprehensive, expert-verified taxonomy of ECG monitoring systems, a common architecture, and a complete set of processes to guide the classification, analysis, and design of these systems.



The existing ECG classification algorithms usually include signal preprocessing, such as wavelet transform and manual feature extraction, but the amount of computation will increase the delay of the real time classification system. In recent years, deep learning algorithm with their advantages of automatic learning features is increasingly used in the field of health care, such as medical image recognition and segmentation, time series data monitoring and analysis. At present, the outstanding algorithm can establish an end-to-end DNN network to learn the characteristics of ECG records by using the extensive digital characteristics of ECG data, which saves lots of signal pre processing steps. Because the performance of DNN increases with the amount of training data, this method can make good use of the extensive digitalization of ECG data. According to WHO , around 18 million people die yearly due to heart disease globally. Due to the increasing prevalence of cardiac diseases, people are prone to prevent devastating event from happening. They are used to diagnose a patients cardiac condition.

Electrocardiogram (EKG)



(Electrocardiogram System)

Electrocardiography (ECG) is a quick and easily accessible method for diagnosis and screening of **cardiovascular diseases** including **heart failure(HF)**. Artificial Intelligence (AI) can be used for semi-automated ECG analysis. The aim of this evaluation was to provide an overview of AI use in HF detection from ECG signals and to perform a meta-analysis of available studies.

Evaluation of symptoms suggestive of HF currently demands physicians to evaluate various parameters including imaging and laboratory data and the electrocardiogram (ECG).

Besides a standard examination that includes an ECG, imaging information, such as echocardiography or magnetic resonance imaging, is seen as gold standard in diagnosis of HF. The ECG is a well-established, quick and easily accessible method for diagnosis and careening of various cardiovascular diseases. It provides specific features that indicate presence of HF or prognosis in HF patients especially to rule out HF in case of a normal ECG. However, use of an ECG as primary diagnostic instrument often only yields insufficient diagnostic specificity.

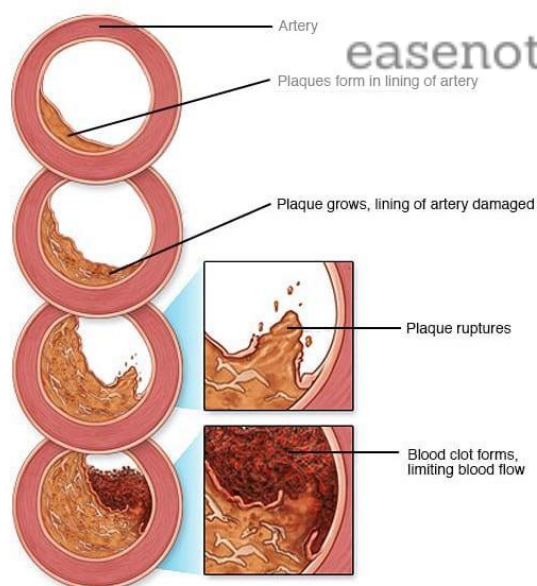
Further, general practitioner-based ECG reporting has varying results, introducing further diagnostic uncertainty.

Devices providing medically relevant information generated directly by individuals outside the healthcare system such as smart phones with health applications or wearables including smartwatches are an emerging trend. This development promises that a growing number of ECG data generated at home will be available for a diagnostic screening. Management of this quantity of data however, might be challenge for the individual healthcare professional, as well as for the healthcare system itself. The potentially beneficial use of artificial Intelligence (AI) in cardiology in general has been discussed already. AI might also be able to help in interpretation of ECG signals and could therefore used to analyse ECG data in specific cases and on a large scale for early identification of cardiovascular diseases such as HF.

REASONS FOR BLOCKAGES OF BLOOD VESSELS:

Coronary artery disease is a common heart condition. The major blood vessels that supply the heart (coronary arteries) struggle to send enough blood, oxygen and nutrients to the heart muscle. Cholesterol deposits (plaques) in the heart arteries and inflammation are usually the cause of coronary artery disease.

Signs and symptoms of coronary artery disease occur when the heart doesn't get enough oxygen rich blood. If you have coronary artery disease, reduced blood flow to the heart can cause chest pain (angina) and shortness of breath. A complete blockage of blood flow can cause a heart attack. Coronary artery disease starts when fats, cholesterol and other substances collect on the inner walls of the heart arteries. This condition is called atherosclerosis. The build up is called plaque. Plaque can cause the arteries to narrow, blocking blood flow. The plaque can also burst, leading to a blood clot.



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(Blockages of blood vessels)

Besides high cholesterol, damage to the coronary arteries may be caused by; Diabetes or insulin resistance, High blood pressure, Not getting enough exercise (sedentary lifestyle), smoking or tobacco use.

DESIGN OF STENTS:

A stent is a tiny tube that can play a big role in treating heart disease. It helps keep our arteries the blood vessels that carry blood from our heart to other parts of the body, including the heart muscle itself open.

Most stents are made out of wire mesh and are permanent some are made out of fabric. These are called stent grafts and are often used for larger arteries. Others are made of a material that dissolves and that our body absorbs over time. They're coated in medicine that slowly releases into our artery to prevent it from being blocked again.

Why would You Need a Stent??

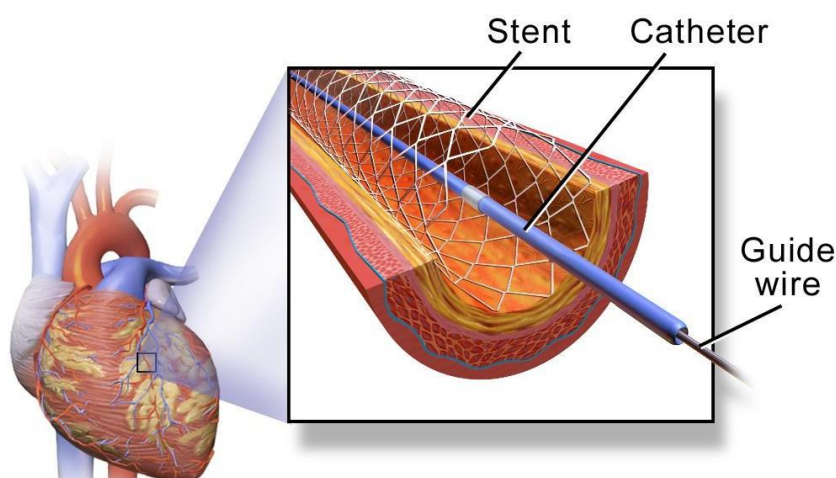
If a fatty substance called plaque builds up inside an artery, it can reduce blood flow to heart. This is called coronary heart disease and it can cause chest pain. The plaque can also cause a blood clot that blocks blood flowing to our heart, which may lead to a heart attack. By keeping an artery open, stents lower your risk of chest pain. They can also treat a heart attack that's in progress.

Doctor usually inserts a stent using a minimally invasive procedure. They will make a small incision and use a catheter to guide specialized tools through blood vessels to reach the area that needs a stent. One of those tools may have a camera on the end to help your doctor guide the stent. During the procedure, doctor may also use an imaging technique called an angiogram to help guide the stent through the vessel. Using the necessary tools, doctor will locate the broken or blocked vessel and install the stent. They will remove the instruments from your body and close the incision.

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Stent in Coronary Artery



DESIGN:

Most of these stents are constructed from a nickel titanium alloy. Balloon expandable stents are susceptible to permanent deformation when they are compressed extrinsically, which is not an issue in the coronary tree. Self-expanding stents do not have this limitation. Furthermore, self-expanding stents have less axial stiffness and are thus more flexible and will conform to the shape of the stent. Balloon expandable stents, by virtue of their design, resist expansion by balloon, but they have less acute recoil when they are placed in a poorly compliant lesion. However, after the initial deployment, the stent is at its maximal diameter and cannot get larger, whereas a self-expanding stent that is appropriately oversized for the vessel will exhibit a chronic outward force on the lesion and may lead to a larger lumen over time. For the reasons above, there are some coronary lesions where balloon expandable stents are not ideal, such as aneurysmal, ectatic vessels, thrombus laden vessels and vessels that are tapering with a large size mismatch between distal reference and proximal reference vessels.

PACE MAKERS:

A pacemaker is a small device that's placed (implanted) in the chest to help control the heartbeat. It's used to prevent the heart from beating too slowly. Implanting a pacemaker in the chest requires a surgical procedure. A pacemaker is also called a cardiac pacing device.

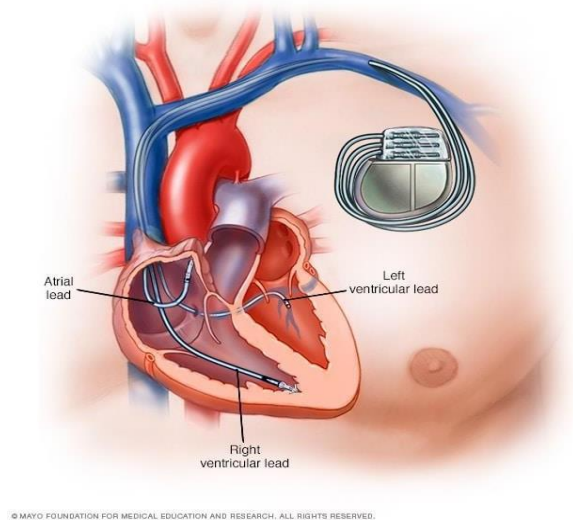
Types:

Single chamber pacemaker. This type usually carries electrical impulses to the right ventricle of heart.

Dual chamber pacemaker. This type carries impulses to the right ventricle and the right atrium of heart to help control the timing of contractions between the two chambers.

Biventricular pacemaker. Biventricular pacing, also called cardiac resynchronization therapy is for people who have heart failure and heartbeat problems. This type of pacemaker stimulates both of the lower heart chambers (the right and left ventricles) to make the heart beat more efficiently. A **pacemaker** is implanted to help control our heartbeat. Doctor may recommend a temporary pacemaker when you have a slow heartbeat (bradycardia) after a heart attack, surgery or medication overdose but our heartbeat is otherwise expected to recover. A pacemaker may be implanted permanently to correct a chronic slow or irregular heartbeat or to help treat heart failure. Pacemakers work only when needed. If heartbeat is too slow, the pacemaker sends electrical signals to our heart to correct the beat.

Some newer pacemakers also have sensors that detect body motion or breathing rate and signal the devices to increase heart rate during exercise, as needed.

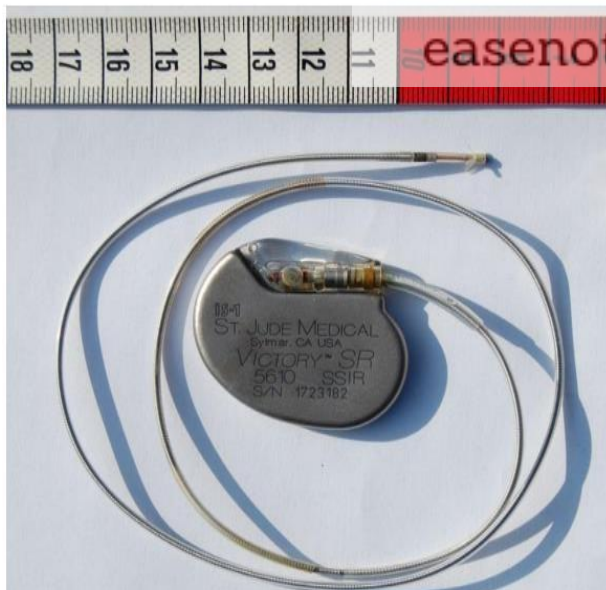


(pacemakers)

A pacemaker has two parts:

1. **Pulse Generator:** this small metal container houses a battery and the electrical circuitry that controls the rate of electrical pulses sent to the heart.
2. **Leads (electrodes):** one to three flexible, insulated wires are each placed in one or more chambers of the heart and deliver the electrical pulses to adjust the heart rate. However, some newer pacemakers don't require leads. These devices, called leadless pacemakers are implanted directly into the heart muscle.

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(pulse generator)

DEFIBRILLATORS:

Defibrillators are devices that send an electric pulse or shock to the heart to restore a normal heartbeat. They are used to prevent or correct an arrhythmia, an uneven heartbeat that is too slow or too fast. If the heart suddenly stops defibrillators can also help it beat again. Different types of defibrillators work in different ways. Automated External Defibrillators (AEDs), which are now found in many public spaces are used to save the lives of people experiencing cardiac arrest. Even untrained bystanders can use these devices in an emergency.

Other defibrillators can prevent sudden death among people who have a high risk of a life-threatening arrhythmia. They include implantable cardioverter defibrillators (ICDs), which are surgically placed inside our body, and wearable cardioverter defibrillators (WCDs). Which rest on the body. It can take time and effort to get used to living with a defibrillator, and it is important to be aware of possible complications.

There are three types of defibrillators: AEDs, ICDs, and WCDs.

An **AED** is a lightweight, battery-operated, portable device that checks the heart's rhythm and sends a shock to the heart to restore normal rhythm. The device is used to help people having cardiac arrest.

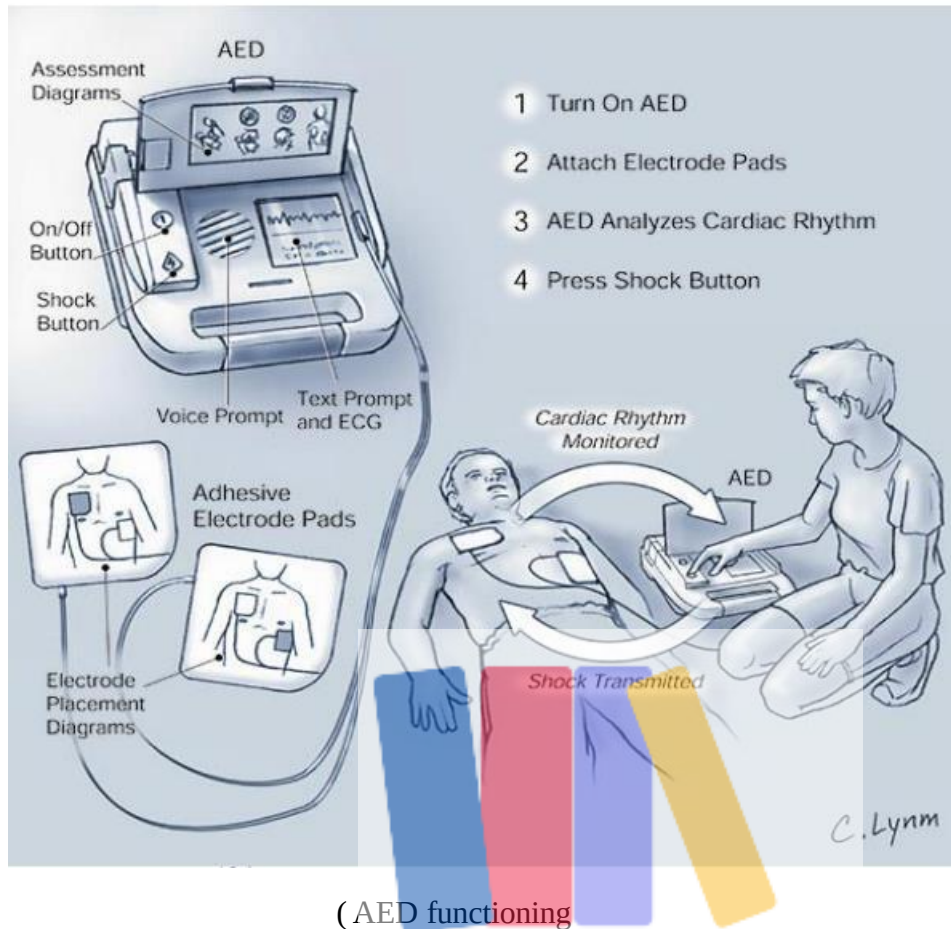
Sticky pads with sensors, called **electrodes**, are attached to the chest of someone who is having cardiac arrest. The electrodes send information about the person's heart rhythm to a computer in the AED. The computer analyses the heart rhythm to find out whether an electric shock is needed. If it is needed, the electrodes deliver the shock.

ICDs are placed through surgery in the chest or stomach area, where the device can check for arrhythmias. Arrhythmias can interrupt the flow of blood from our heart to rest of the body or cause our heart to stop. The ICD sends a shock to restore a normal heart rhythm.

An ICD can give off a low-energy shock that speeds up or slows down an abnormal heart rate, or a high-energy shock to correct a fast or irregular heartbeat. If low energy shocks do not restore normal heart rhythm, the device may switch to high-energy shocks for defibrillation. ICDs are similar to pacemakers, but pacemakers deliver only low-energy electrical shocks. ICDs have a generator connected to wires that detect our heart's beats and deliver a shock when needed.

The ICD can also record the heart's electrical activity and heart rhythms. The recordings can help our healthcare provider fine-tune the programming of the device so it works better to correct irregular heartbeats. The device is programmed to respond to the type of arrhythmia you are most likely to have.

WCDs have sensors that attach to the skin. They are connected by wires to a unit that checks our heart's rhythm and delivers shocks when needed. Like an ICD the WCD can deliver low and high-energy shocks. The device has a belt attached to a vest that is worn under our clothes. Provider fits the device to your size. It is programmed to detect a specific heart rhythm.

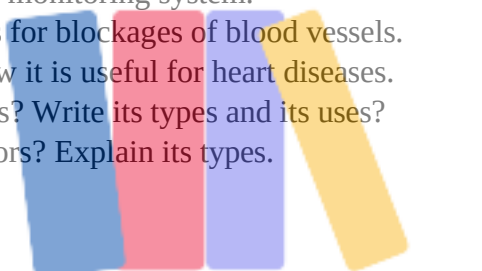


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QUESTION BANK MODULE 3 (PART 1)

1. Write a note on Human organ system and Bio design.
2. Explain how brain acts as a CPU system.
3. What are the two nervous system in human body? Explain each.
4. How signal transmission occurs in nervous system.
5. Briefly explain EEG process and its application.
6. Write a note on Robotic arms for prosthetics.
7. Briefly explain engineering solutions for Parkinson's disease.
8. How we can relate eye as a camera system.
9. Write optical corrections for eye.
10. What is Cataract eye? Write a Reason for causing the cataract and solution for the same.
11. What are the lens material using for refractive errors in the eye.
12. Write a brief note on Bionic eyes.
13. Explain Heart as a Pump system.
14. Briefly explain ECG monitoring system.
15. What are the reasons for blockages of blood vessels.
16. What are stents? How it is useful for heart diseases.
17. What are pacemakers? Write its types and its uses?
18. What are Defibrillators? Explain its types.

The logo for EASENOTES features four vertical bars of different colors (blue, red, purple, and yellow) arranged in a slightly staggered, overlapping manner. Below the bars, the word "EASENOTES" is written in a bold, grey, sans-serif font.

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LUNGS AS A PURIFICATION SYSTEM:

Introduction:

Every cell in our body needs oxygen to live. The air we breathe contains oxygen and other gases. The respiratory system's main job is to move fresh air into our body while removing waste gases. Once in the lungs, oxygen is moved into the bloodstream and carried through our body. At each cell in our body, oxygen is exchanged for a waste gas called carbon dioxide. our bloodstream then carries this waste gas back to the lungs where it is removed from the bloodstream and then exhaled. Lungs and the respiratory system automatically perform this vital process, called gas exchange. In addition to gas exchange, your respiratory system performs other roles important to breathing. These include:

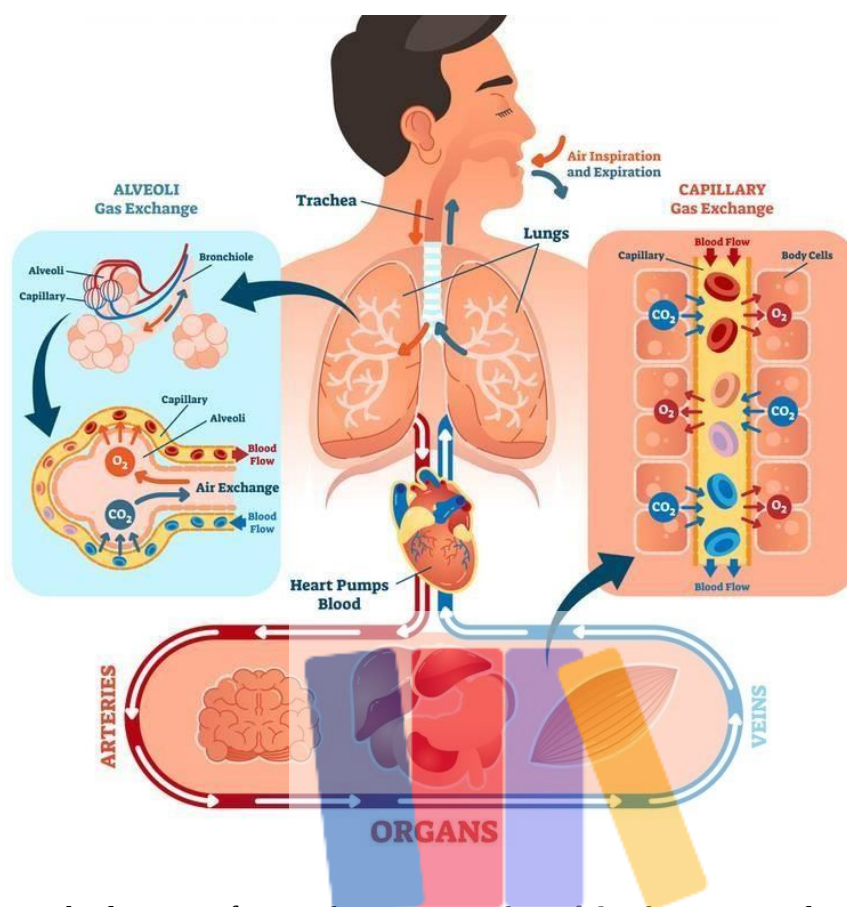
1. Bringing air to the proper body temperature and moisturizing it to the right humidity level.
2. Protecting our body from harmful substances. This is done by coughing, sneezing, filtering or swallowing them.
3. Supporting our sense of smell.

Lungs help in the purification of blood. Arteries carry pure oxygenated blood from the heart to other parts of the body. Veins carry impure venous blood back from other parts of the body to the right side of the heart. This impure blood goes to the lungs for purification.

When breathe is inhaled, oxygen from the air comes in contact with the impure blood and the blood takes up oxygen. The waste matter in the blood releases carbonic acid and the blood is purified. The purified blood is carried to the heart by veins.

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The lung purifies air by removing harmful substances and adding oxygen to the bloodstream. The process of purifying air in the lungs can be described as follows:

- **Filtration:** The nose and mouth serve as a first line of defense against harmful substances in the air, such as dust, dirt, and bacteria. The tiny hairs in the nose, called cilia, and the mucus produced by the respiratory system trap these substances and prevent them from entering the lungs.
- **Moisturization:** The air is also humidified as it passes over the moist lining of the respiratory tract, which helps to keep the airways moist and prevent them from drying out.
- **Gas Exchange:** Once the air reaches the alveoli, the gas exchange process occurs, where oxygen diffuses across the thin alveolar and capillary walls into the bloodstream, and carbon dioxide diffuses in the opposite direction, from the bloodstream into the alveoli to be exhaled. This process ensures that the bloodstream is supplied with fresh, oxygen-rich air, while waste carbon dioxide is removed from the body.

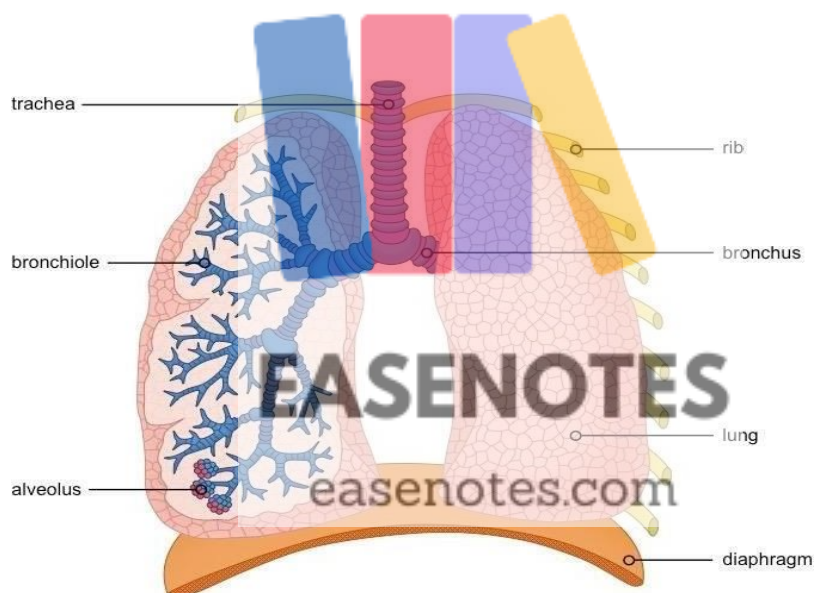
Overall, the lung serves as a vital purification system, filtering out harmful substances, adding oxygen to the bloodstream, and removing waste carbon dioxide. It plays a critical role in maintaining the body's homeostasis and supporting life.

Architecture:

The lung parenchyma is mainly comprised of numerous air-containing passages and intervening fine structures, corresponding to alveolar ductal lumens and alveoli, as well as alveolar septa and small pulmonary vessels occupying 10% of total parenchymal volume. The primary function of the lungs is gas exchange. However, the lungs perform several important non-respiratory functions that are vital for normal physiology.

The lung, with its unique ability to distend and recruit pulmonary vasculature, acts as a reservoir of blood, fine-tuning preload to the left heart to optimize cardiac output.

1. The lung acts as a filter against endogenous and exogenous emboli, preventing them from accessing system circulation.
2. Pulmonary epithelium forms the first line of defense against inhaled particles.
3. Pulmonary endothelial cells are responsible for the uptake, metabolism and biotransformation of several exogenous and endogenous substances.
4. Pulmonary metabolic capacity is easily saturated, but pulmonary endothelial binding of some drugs alters their pharmacokinetics.



The architecture of the lung is designed to maximize surface area for efficient gas exchange. The lung is divided into several parts, including the trachea, bronchi, bronchioles, and alveoli.

- **Trachea:** The trachea is the main airway that leads from the larynx (voice box) to the lungs. It is lined with cilia and mucus-secreting glands that help to filter out harmful substances and trap them in the mucus.
- **Bronchi:** The trachea branches into two main bronchi, one for each lung. The bronchi are larger airways that continue to branch into smaller airways called bronchioles.
- **Bronchioles:** The bronchioles are smaller airways that eventually lead to the alveoli.

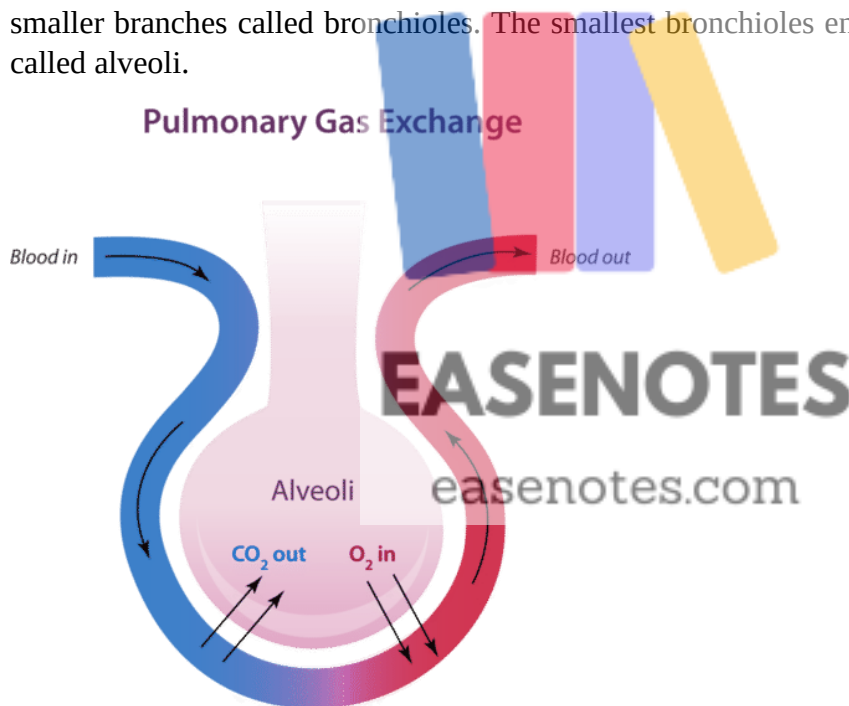
They are surrounded by tiny air sacs called alveoli, which are the sites of gas exchange.

- **Alveoli:** The alveoli are tiny air sacs that are lined with a network of capillaries. This close proximity of the alveoli and capillaries allows for efficient diffusion of oxygen and carbon dioxide between the air in the alveoli and the bloodstream.

Overall, the architecture of the lung is designed to provide a large surface area for gas exchange, while filtering out harmful substances and humidifying the air. The close proximity of the alveoli and capillaries, along with the moist lining of the respiratory tract, ensures that the air is properly purified and the bloodstream is supplied with fresh, oxygen-rich air.

GAS EXCHANGE MECHANISMS:

Air enters the body through the mouth or nose and quickly moves to the pharynx or throat. From there, it passes through the larynx, or voice box and enters the trachea. The trachea is a strong tube that contains rings of cartilage that prevent it from collapsing. Within the lungs, the trachea branches into a left and right bronchus. These further divide into smaller and smaller branches called bronchioles. The smallest bronchioles end in tiny air sacs. These are called alveoli.



(Pulmonary Gas Exchange)

The gas exchange mechanism in the lung involves the transfer of oxygen from the air in the alveoli to the bloodstream, and the transfer of carbon dioxide from the bloodstream to the air in the alveoli. This process is known as diffusion and occurs due to differences in partial pressures of oxygen and carbon dioxide.

- **Oxygen Diffusion:** The partial pressure of oxygen in the air in the alveoli is higher than the partial pressure of oxygen in the bloodstream. This difference creates a gradient that causes oxygen to diffuse from the alveoli into the bloodstream, where it binds to hemoglobin in red blood cells to form oxyhemoglobin.
- **Carbon Dioxide Diffusion:** The partial pressure of carbon dioxide in the

bloodstream is higher than the partial pressure of carbon dioxide in the air in the alveoli. This difference creates a gradient that causes carbon dioxide to diffuse from the bloodstream into the alveoli, where it is exhaled.

SPIROMETRY:

Spirometry(spy-ROM-uh-tree) is a common office test used to assess how well our lungs work by measuring how much air we inhale, how much we exhale, and how quickly we exhale. Spirometry is used to diagnose asthma, chronic obstructive pulmonary disease (COPD), and other conditions that affect breathing.

175 years have elapsed since John Hutchinson introduced the world to his version of an apparatus that had been in development for nearly two centuries, the spirometer.

Spirometers can be divided into two basic groups:

1. Volume-measurement devices (eg: wet and dry spirometers).
2. Flow-measurement devices(eg: pneumotachograph systems, mass flow meters).

Principle: The principle behind spirometry is to measure the volume of air that can be exhaled from the lungs in a given time period. By measuring the volume of air exhaled, spirometry can provide information about the functioning of the lungs and the ability of the lungs to move air in and out.

Working: Spirometry is performed using a spirometer, a device that consists of a mouthpiece, a flow sensor, and a volume sensor. The patient is asked to exhale as much air as possible into the spirometer, and the spirometer measures the volume and flow rate of the exhaled air. The volume of air exhaled is displayed on a graph called a flow-volume loop, which provides information about the lung function.



(Spirometer Instrument)

Interpretation Of Results

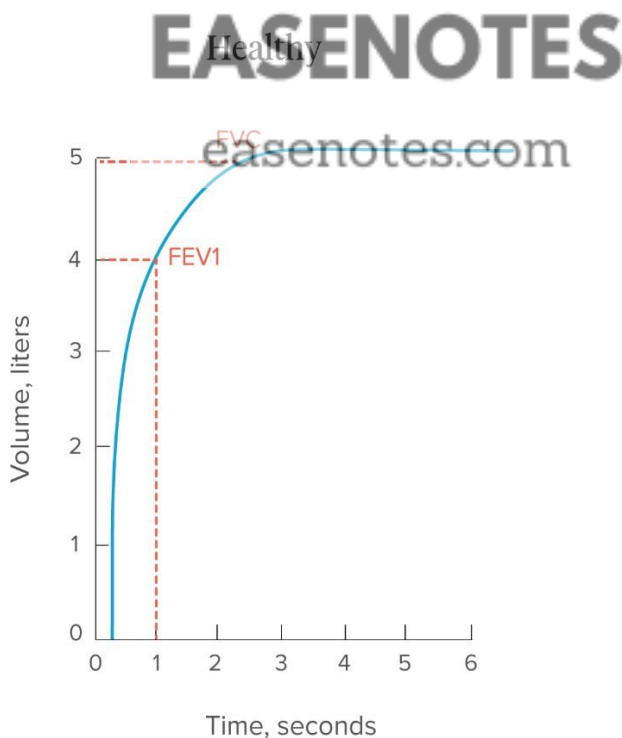
The results of spirometry can be used to determine if the lungs are functioning normally and to diagnose lung conditions. For example, a decrease in the volume of air exhaled or a decrease in the flow rate of the exhaled air can indicate a restriction in the airways, which can be a sign of a lung condition such as asthma or COPD.

Requirements of an acceptable spirometer are:

1. Spirometers must be able to accumulate volume for ≥ 15 s.
2. The measuring volume should be ≥ 8 L (body temperature and pressure, saturated).
3. The accuracy of reading should be at least $\pm 3\%$ (or ± 0.05 L) with flows from 0-14 L per s.
4. The total resistance to airflow at 14L per s should be < 1.5 cmH₂O per L per s (0.15kPa per L per s)

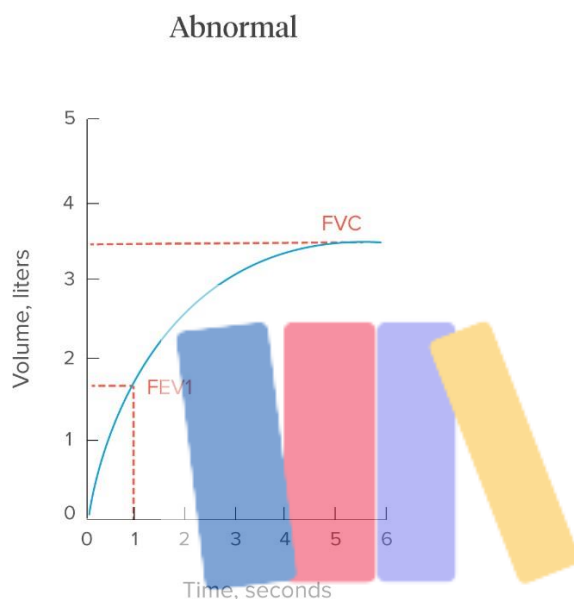
HOW TO CALCULATE THE NORMAL RATE OF RESPIRATION IN A SPIROMETER

The FEV₁/FVC Ratio (FEV₁%) parameter is calculated by dividing the measured FEV₁ value by the measured FVC value. The measured column shows the absolute (numerical) ratio, and the predicted column shows the ratio expressed as a percentage. In healthy adults of the same gender, height, and age the normal predicted percentage should be between 70% and 85%.



ABNORMAL LUNG PHYSIOLOGY:

Percentages lower than 70% are considered abnormal. This is an important measurement because obstructive disease such as COPD, chronic bronchitis and emphysema cause increased airway resistance to expiratory airflow, and may result in percentages of 45% to 60%. Restrictive diseases such as pulmonary fibrosis tend to reduce both FEV1 and FVC values, so the percentage can remain the normal range, or even increase.



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COPD:

Chronic obstructive pulmonary disease (COPD) is a chronic inflammatory lung disease that causes obstructed airflow from the lungs. Symptoms include breathing difficulty, cough, mucus(sputum) production and wheezing. It's typically caused by long-term exposure to irritating gases or particulate matter, most often from cigarette smoke. People with COPD are at increased risk of developing heart disease, lung cancer, and a variety of other conditions.

Emphysema and chronic bronchitis are the two most common conditions that contribute to COPD. These two conditions usually occur together and can vary in severity among individuals with COPD.

Symptoms:

COPD symptoms often don't appear until significant lung damage has occurred, and they usually worsen over time, particularly if smoking exposure continues.

Signs and symptoms of COPD may include:

- Shortness of breath, especially during physical activities
- Wheezing
- Chest tightness

- A chronic cough that may produce mucus(sputum) that may be clear, white, yellow or greenish
- Frequent respiratory infections
- Lack of energy
- Unintended weight loss (in later stages)
- Swelling in ankles, feet or legs

Tests may include:

- Lung (pulmonary) infection tests.
- Chest X-ray.
- CT Scan
- Arterial blood gas analysis.
- Laboratory tests.

Medications:

Several kinds of medications are used to treat the symptoms and complications of COPD. You may take some medications regularly and others as needed.

- Bronchodilators
- Inhaled steroids
- Combination inhalers
- Oral steroids
- Phosphodiesterase-4 inhibitors
- Theophylline
- Antibiotics



Mechanical ventilation is a lifesaving therapy in patients who have acute respiratory failure due to chronic obstructive pulmonary disease (COPD). Mechanical ventilation either invasive or non-invasive has an important role in the management of acute exacerbation of COPD (AECOPD).

VENTILATOR:

Mechanical ventilators are machines that act as bellows to move air in and out of your lungs. Your respiratory therapist and doctor set the ventilator to control how often it pushes air into your lungs and how much air you get. You may be fitted with a mask to get air from the ventilator into your lungs or you may need a breathing tube if your breathing problem is more serious. When we are ready to be taken off the ventilator, healthcare team will “wean” you or decrease the ventilator support until you can start breathing on your own.



Ventilators work by delivering pressurized air or oxygen into the lungs through a breathing tube or mask. The pressure can be adjusted to match the patient's needs and to help maintain adequate oxygen levels in the blood.

While ventilators can be lifesaving for individuals with acute respiratory failure, they also come with potential risks and complications. For example, prolonged use of a ventilator can increase the risk of ventilator-associated pneumonia, and patients may experience discomfort or pain from the breathing tube.

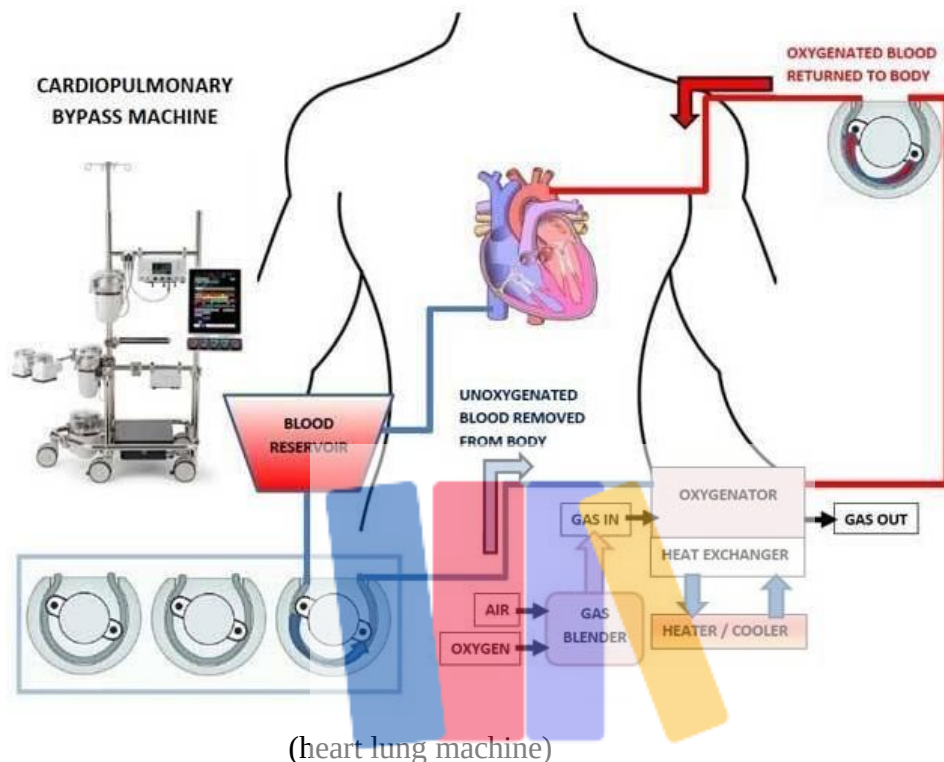
The use of ventilators is carefully monitored and managed by healthcare professionals to ensure that the patient receives the appropriate level of support while minimizing potential risks and complications.

HEART LUNG MACHINE:

A heart-lung machine, also known as a cardiopulmonary bypass machine, is a device used in cardiovascular surgery to temporarily take over the functions of the heart and lungs. The heart-lung machine is used during open-heart surgery, such as coronary artery bypass graft (CABG) surgery and valve replacement surgery, to support the patient's circulatory and respiratory functions while the heart is stopped.

The heart-lung machine works by circulating blood outside of the body through a series of tubes and pumps. Blood is taken from the body, oxygenated, and then returned to the body. This allows the heart to be stopped during the surgery without causing any harm to the patient.

The use of a heart-lung machine during surgery carries some risks, including the potential for blood clots, bleeding, and infections. Additionally, there may be some long-term effects on the body, such as cognitive decline, that are not yet fully understood. However, the use of a heart-lung machine has revolutionized the field of cardiovascular surgery, allowing for more complex procedures to be performed and greatly improving patient outcomes.



KIDNEY AS FILTRATION SYSTEM:

Introduction:

Kidneys remove wastes and extra fluid from the body. Kidneys also remove acid that is produced by the cells of the body and maintain a healthy balance of water, salts, and minerals- such as sodium, calcium, phosphorus and potassium in the blood. Without this balance, nerves, muscles and tissues in the body may not work normally.

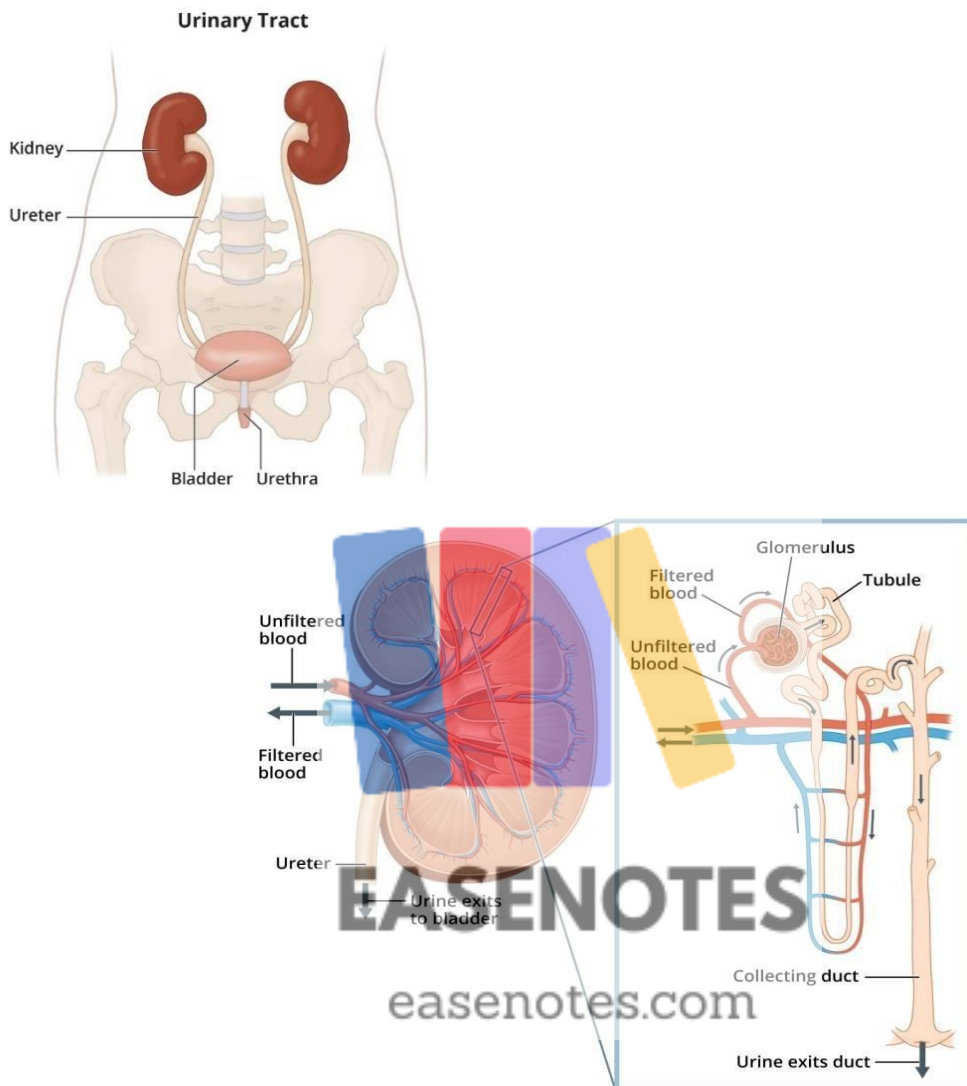
Kidneys also make hormones that help

- Control blood pressure
- Make red blood cells NIH external link
- Keeps bones strong and healthy.

Architecture:

The kidneys are two bean-shaped organs each about the size of a fist. They are located inst below the rib cage, one on each side of the spine. Healthy kidneys filter about a half cup of blood every minute, removing wastes and extra water to make urine. The urine flows from the kidneys to the bladder through two thin tubes of muscle called ureters, one on each side of

the bladder. Your bladder stores urine, kidneys, ureters, and bladder are part of our urinary tract.



(Representing kidney and Nephron)

The Nephron consists of several key structures:

- **Bowman's capsule:** This is a cup-shaped structure that surrounds the glomerulus and filters waste and excess fluid from the bloodstream into the renal tubule.
- **Glomerulus:** A network of tiny blood vessels within the Bowman's capsule that filters waste and excess fluid from the bloodstream.
- **Proximal convoluted tubule:** A segment of the renal tubule that reabsorbs important substances, such as glucose, amino acids, and electrolytes, back into the bloodstream.

- Loop of Henle: A U-shaped segment of the renal tubule that is critical for the reabsorption of ions and water.
- Distal convoluted tubule: A segment of the renal tubule that regulates the levels of electrolytes and other important substances in the bloodstream.
- Collecting duct: A series of ducts that collect the filtrate from the renal tubules and transport it to the renal pelvis, where it drains into the ureter and eventually into the bladder.

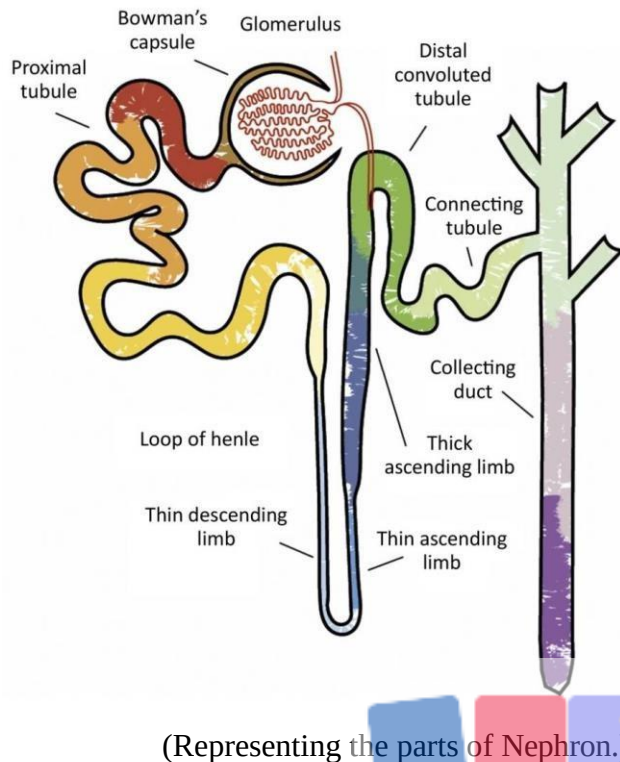
The nephrons are surrounded by a network of blood vessels, including the afferent arteriole and the efferent arteriole, which bring blood into and out of the glomerulus, respectively. The filtrate produced by the nephron passes through the renal tubules, where it is modified by reabsorption and secretion, before being eliminated from the body as urine.

MECHANISM OF FILTRATION:

The mechanism of filtration in the kidneys is a complex process that involves multiple steps to remove waste and excess fluids from the bloodstream. The following is a summary of the steps involved in the filtration process:

- Blood enters the kidney through the renal arteries and flows into tiny filtering units called glomeruli.
- At the glomerulus, the pressure in the blood vessels causes a portion of the plasma and dissolved substances to filter out and enter a structure called Bowman's capsule.
- In Bowman's capsule, the filtrate is then transferred into the renal tubules, which are the main filtering units of the kidneys.
- In the renal tubules, the filtrate passes through a series of specialized cells, such as proximal tubular cells and distal tubular cells, which reabsorb important substances such as glucose, amino acids, and electrolytes back into the bloodstream.
- At the same time, the renal tubules secrete waste products, such as urea and creatinine, back into the filtrate.
- Finally, the filtered fluid, now known as urine, is transported through the renal pelvis and ureters to the bladder, where it is eventually eliminated from the body.

This process of filtration, reabsorption, and secretion helps to maintain the proper balance of fluids and electrolytes in the body, as well as to remove waste and excess substances.



The blood vessel runs alongside the tubule. As the filtered fluid moves along the tubule, the blood vessel reabsorbs almost all of the water, along with minerals and nutrients our body needs.

How does blood flow through our kidneys?

Blood flows into the kidney through the renal artery. This large blood vessel branches into smaller and smaller blood vessels until the blood reaches the nephrons. In the nephron, Blood is filtered by tiny blood vessels of the glomeruli and then flows out of the kidney through the renal vein. Blood circulates through kidneys many times a day. In a single day, kidneys filter about 150 quarts of blood. Most of the water and other substances that filter through glomeruli are returned to the blood by the tubules. Only 1 to 2 quarts become urine. When the kidney doesn't function properly, chronic kidney disease occurs when a disease or condition impairs kidney function, causing kidney damage to worsen over several months or years.

Chronic Kidney Disease (CKD)

Chronic kidney disease includes conditions that damage your kidneys and decrease their ability to keep you healthy by filtering wastes from your blood. If kidney disease worsens, wastes can build to high levels in your blood and make you feel sick. You may develop complications like,

- High blood pressure
- Anemia (low blood count)

- Weak bones
- Poor nutritional health
- Nerve damage

Symptoms:

People with CKD may not feel ill or notice any symptoms. The only way to find out for sure if you have CKD is through specific blood and urine tests. These tests include the measurement of both the creatinine level in the blood and the protein in the urine.

Treatment:

Depending on the cause, some types of kidney disease can be treated. Often, though, chronic kidney disease has no cure. Treatment usually consists of measures to help control signs and symptoms, reduce complications and slow the progression of the disease. If your kidneys become severely damaged, you might need treatment for end-stage kidney disease.

DIALYSIS:

Dialysis is a procedure to remove waste products and excess fluid from the blood when the kidneys stop working properly. It often involves diverting blood to a machine to be cleaned. Dialysis is a medical treatment that helps to filter waste and excess fluids from the blood when the kidneys are unable to function properly. There are two main types of dialysis systems: hemodialysis and peritoneal dialysis.

There are 2 main types of dialysis:

- Haemodialysis involves diverting blood into an external machine, where it's filtered before being returned to the body. Hemodialysis is a procedure that uses a machine to clean the blood. During hemodialysis, blood is removed from the body, passed through a dialysis machine that filters out waste and excess fluids, and then returned to the body.
- Peritoneal dialysis involves pumping dialysis fluid into the space inside your abdomen (tummy) to draw out waste products from the blood passing through vessels lining the inside of the abdomen.

QUESTION BANK MODULE 3 (PART 2)

1. Explain briefly lungs as a purification system.
2. Explain gas exchange mechanism in lungs.
3. What is spirometry? Distinguish between normal and abnormal lung physiology.
4. What is COPD? Write its symptoms, tests and medications.
5. Write a short note on Ventilator.
6. Explain the working of Heart-Lung machine.
7. Briefly explain Kidney as filtration system with its mechanism of filtration.
8. What is CKD? Explain its symptoms and Dialysis process.



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BIOLOGY FOR ENGINEERS**MODULE 4****NATURE-BIOINSPIRED MATERIALS AND MECHANISMS (QUALITATIVE)**

Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircrafts), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro), Shark skin (Friction reducing swim suits), Kingfisher beak (Bullet train). Human Blood substitutes - hemoglobin-based oxygen carriers (HBOCs) and perfluorocarbons (PFCs).

ECHOLOCATION:

In nature's sonar system, echolocation occurs when an animal emits a sound wave that bounces off an object, returning an echo that provides information about the object's distance and size. Over a thousand species echolocate, including most bats, all-toothed whales, and small mammals. Human echolocation is the ability of humans to detect objects in their environment by sensing echoes from those objects and by actively creating sounds: for example, by tapping their canes, lightly stomping their feet, snapping their fingers, or making clicking noises with their mouths. People trained to orient by echolocation can interpret the sound waves reflected by nearby objects, accurately identifying their location and size.

Many blind individuals passively use natural environmental echoes to sense details about their environment: however, others actively produce mouth clicks and can gauge information about their environment using the echoes from those clicks. Both passive and active echolocation help blind individuals sense their environments.

Those who can see their environments often do not readily perceive echoes from nearby objects, due to an echo suppression phenomenon brought on by the precedence effect. However, with training, sighted individuals with normal hearing can learn to avoid obstacles using only sound, showing that echolocation is a general human ability.

A comparison of biological echolocation and technological echolocation is given below:

Biological Echolocation

- Found in various animals such as bats, dolphins, and some species of whales.
- Relies on the emission of sound waves, usually in the form of clicks or vocalizations.
- Animals emit sound waves and listen for the echoes produced when the sound waves bounce off objects in their environment.
- By analyzing the echoes, animals can determine the location, distance, and even the shape of objects around them.
- This ability is mainly used for navigation, hunting, and communication in the animal kingdom.
- Biological echolocation is a natural adaptation that has evolved over millions of years.

Technological Echolocation

- Replicates the concept of biological echolocation using technological devices.
- Utilizes sound waves, typically generated by artificial sources such as sonar or ultrasonic sensors.
- These devices emit sound waves and analyze the echoes that bounce back from objects.
- The information from the echoes is processed and interpreted by the technology to generate useful data, such as distance, location, and object recognition.
- Technological echolocation has applications in various fields, including navigation, robotics, obstacle detection, and medical imaging.
- It is a human-engineered solution inspired by the natural abilities of animals.

Principle of Echolocation

Both biological and technological echolocation rely on the same basic principles and have the same underlying purpose: to determine the location, distance, and shape of objects in the environment using sound waves and their echoes.

The principle of echolocation is based on the emission of sound waves and the interpretation of the echoes that bounce back from objects in the environment.



Figure representing echolocation in bats and dolphins.

Mechanics:

Vision and hearing are akin in that each interprets detections of reflected waves of energy. Vision processes light waves that travel from their source, bounce off surfaces throughout the environment and enter the eyes. Similarly, the auditory system processes sound waves as they travel from their source, bounce off surfaces, and enter the ears. Both neural systems can extract a great deal of information about the environment by interpreting the complex patterns of reflected energy that their sense organs receive. In the case of sound, these waves of reflected energy are referred to as echoes.

Comparing the Sound Emission and Reception in Biological Ecosystem and Technological Ecosystem

In biological systems, sound emission and sensory reception organs are specialized adaptations that allow animals to engage in echolocation. Technological systems, on the other hand, employ devices designed to replicate and enhance these abilities.

Here's a concise comparison of sound emission and sensory reception organs/devices in biological and technological systems:

	Biological System	Technological System
Sound Emission	Biological organisms, such as bats and cetaceans, have specialized sound emission organs to produce sounds for echolocation. Bats emit sounds using their larynx and modify the emitted sounds using structures like the nose leaf or mouth cavity. Dolphins and whales emit sounds through their blowholes, producing clicks or vocalizations.	Technological systems rely on artificial sound emission devices, such as speakers or transducers, to generate sound waves for echolocation. Ultrasonic sensors or sonar systems emit sound waves through these devices, typically using piezoelectric elements or transducers.
Sensory Reception	Biological organisms possess specialized sensory reception organs that allow them to detect and interpret the returning echoes. Bats have highly sensitive ears designed to detect and analyze ultrasonic frequencies. Dolphins and some whales also receive echoes through their lower jaw. The jawbone conducts sound vibrations to the middle ear, where they are converted into nerve impulses for interpretation by the brain.	Technological systems use sensors and receivers to capture and process the returning echoes. Ultrasonic sensors are commonly employed, which consist of a transducer that emits sound waves and receives the echoes. Sonar systems often incorporate hydrophones or other specialized underwater microphones to detect and interpret the echoes.

ULTRASONOGRAPHY:

Ultrasound:

Ultrasound refers to sound above the human audible limit of 20 kHz. Ultrasound of frequencies upto 10 MHz and beyond is used in medical diagnosis, therapy, and surgery. In investigative applications, an ultrasound source (transmitter) directs pulses into the body.

When the pulse encounters a boundary between organs or between two tissue regions of different densities, reflections of sound occur. By scanning the body with Ultrasound and detecting echoes generated by various organs, a sonogram of the internal structure(s) can be generated. The method is called diagnostic imaging by echolocation.



(Sonography of Kidney)

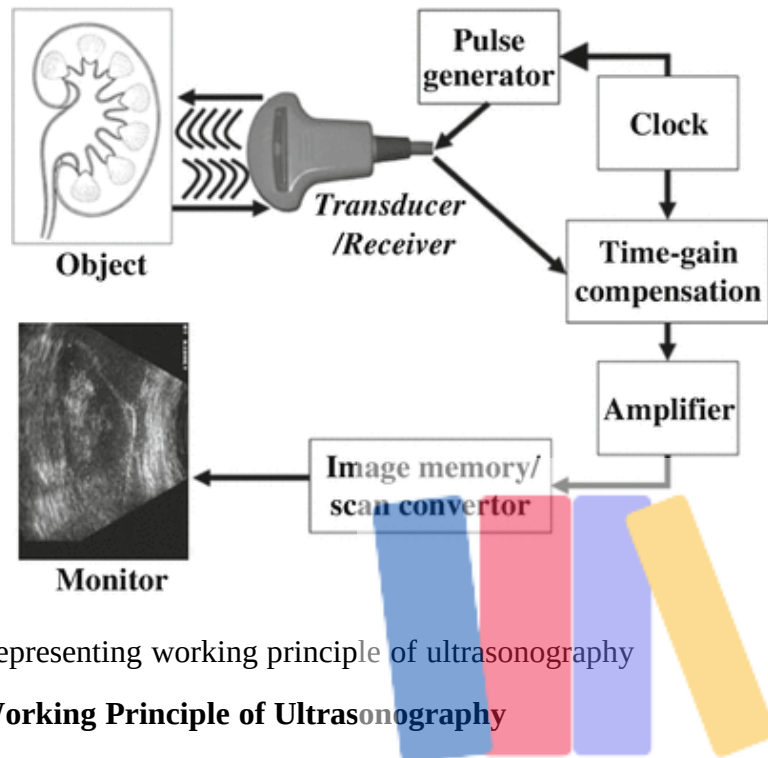
Diagnostic ultrasound, also called sonography or diagnostic medical sonography, is an imaging method that uses sound waves to produce images of structures within your body. The images can provide valuable information for diagnosing and directing treatment for a variety of diseases and conditions.

Ultrasonography is a medical imaging technique that uses high-frequency sound waves to produce images of the internal organs and tissues of the body. It is also known as ultrasound imaging or sonography.

The ultrasound machine emits high-frequency sound waves (usually in the range of 2 to 18 MHz) that travel through the body and bounce back off of the internal organs and tissues. The returning echoes are captured by the ultrasound machine and used to create images of the internal structures.

Ultrasonography is a non-invasive, safe, and painless imaging method that can be used to visualize a wide range of structures within the body, including the organs of the abdomen, pelvis, and chest,

as well as the uterus, fetus, and other soft tissues. It is commonly used in prenatal care to monitor the growth and development of the fetus and to diagnose any potential problems.



Representing working principle of ultrasonography

Working Principle of Ultrasonography

The working principle of ultrasonography is based on the reflection of high-frequency sound waves.

Transducer: An ultrasonography machine consists of a transducer that is used to emit and receive high-frequency sound waves. The transducer is placed in direct contact with the skin or inserted into the body through a gel.

Emission of sound waves: The transducer emits high-frequency sound waves (usually in the range of 2 to 18 MHz) into the body. These sound waves travel through the body and encounter different tissues and organs, which have different acoustic properties.

Reflection of sound waves: The sound waves encounter boundaries between different tissues and organs and bounce back, creating echoes. The strength of the echoes depends on the acoustic properties of the tissues and organs, such as density and stiffness.

Reception of echoes: The transducer in the ultrasonography machine receives the echoes and sends the information to a computer, which processes the data to create images.

Image formation: The computer uses the information from the echoes to create images of the internal organs and tissues of the body. The images are displayed on a screen, allowing the operator to see the structure and movement of the internal organs and tissues.

Advantages of Ultrasonography

Non-invasive: Ultrasonography does not involve any incisions or injections, making it a safe and convenient imaging method.

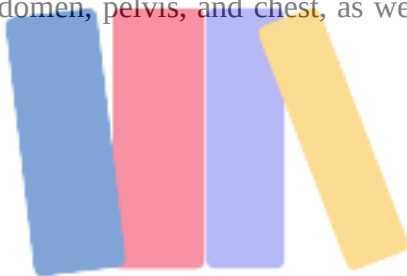
No ionizing radiation: Ultrasonography does not use ionizing radiation, making it a safer option for patients, especially pregnant women and children.

Real-time imaging: Ultrasonography provides real-time images that can be used to monitor the movement and function of internal organs and tissues in real-time.

Portable: Ultrasonography machines are portable and can be used in a variety of settings, making it a valuable tool for emergency and rural medicine.

Cost-effective: Ultrasonography is a cost-effective imaging method that does not require any special preparation or recovery time.

Versatile: Ultrasonography can be used to image a wide range of structures within the body, including the organs of the abdomen, pelvis, and chest, as well as the uterus, fetus, and other soft tissues.



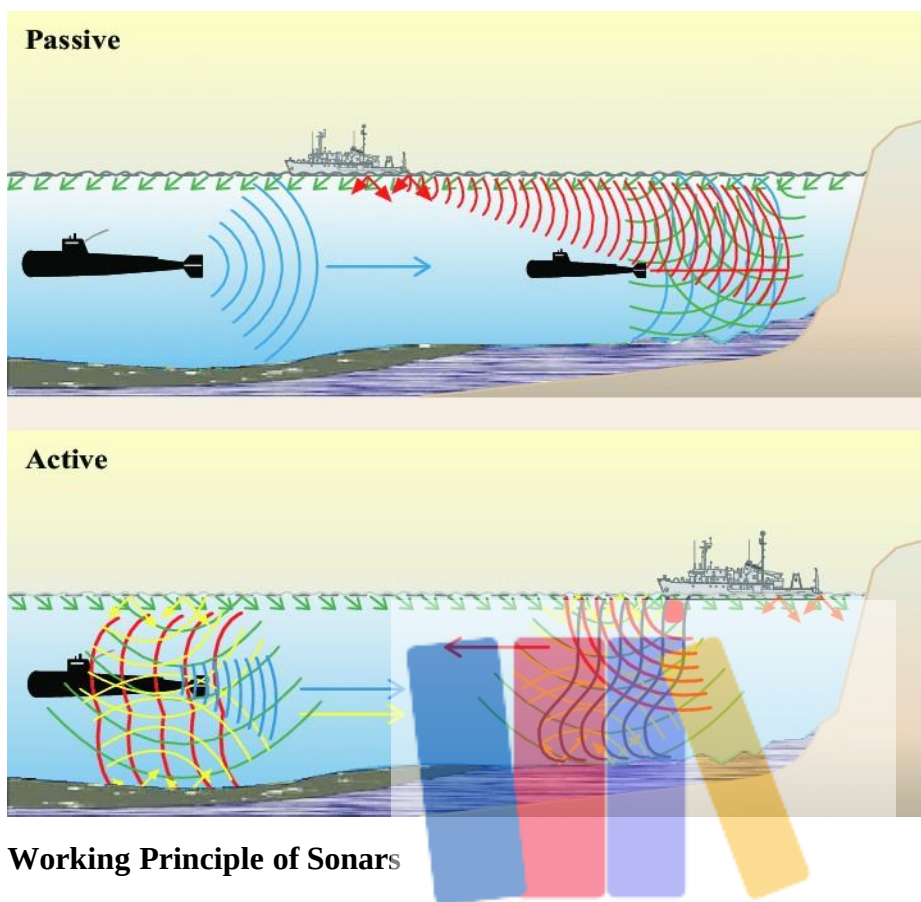
SONARS:

Sonar (Sound Navigation and Ranging or Sonic Navigation and Ranging) is a technique that uses sound propagation (usually underwater, as in submarine navigation) to navigate, measure distances (ranging), communicate with or detect objects on or under the surface of the water, such as other vessels.

“sonar” can refer to one of two types of technology:

Passive sonar means listening for the sound made by vessels

Active sonar means emitting pulses of sounds and listening for echoes.



Working Principle of Sonars

The working principle of sonar technology is based on the reflection of sound waves. Here's how it works:

Transmitter: A sonar system consists of a transmitter that produces and emits a series of sound pulses into the water. These sound pulses are typically in the form of high-frequency, low-power acoustic signals, known as "ping."

Propagation of sound waves: The sound pulses propagate through the water, traveling to the target object and bouncing back as echoes. The speed of sound in water is higher than in air, and it depends on the temperature, pressure, and salinity of the water.

Receiver: The sonar system also includes a receiver that listens for the returning echoes. The receiver is typically placed far away from the transmitter to minimize interference from the transmitted signals.

Calculation of range: The time it takes for the echoes to return to the receiver is used to calculate the range to the target object. The range is simply the product of the speed of sound in water and the time it takes for the echoes to return.

Determination of target properties: The frequency and pattern of the echoes are used to determine the properties of the target object, such as its size, shape, and composition. For example, a large,

solid object will produce a strong, low-frequency echo, while a small, porous object will produce a weaker, high-frequency echo.

Display of results: The results of the sonar measurement are typically displayed on a screen or other output device, allowing the operator to visualize the target object and its location.

Advantages of Sonar Technology

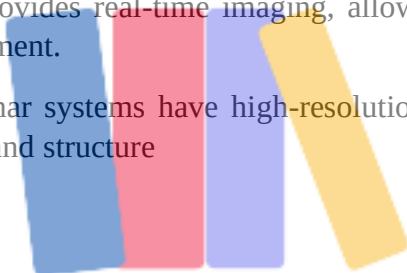
Versatility: Sonar technology is versatile and can be used in a variety of applications, such as underwater navigation, mapping, and imaging, as well as for military and scientific purposes.

Cost-effective: Compared to other underwater imaging technologies, sonar is relatively cost-effective and affordable.

Non-invasive: Unlike other imaging technologies, such as diving and remote-operated vehicles, sonar does not physically disturb the underwater environment, making it an ideal choice for environmental monitoring and scientific research.

Real-time imaging: Sonar provides real-time imaging, allowing operators to quickly and easily assess the underwater environment.

High resolution: Modern sonar systems have high-resolution capabilities, allowing for detailed images of underwater objects and structure

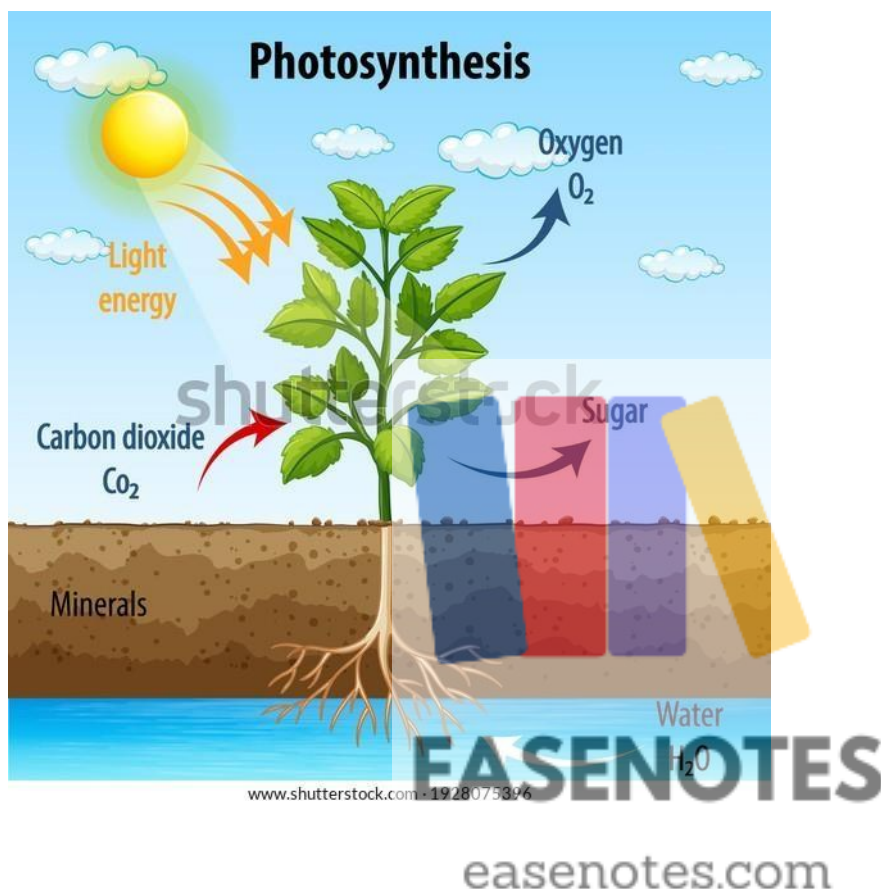


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PHOTOSYNTHESIS:

Most life on Earth depends on photosynthesis. The process is carried out by plants, algae, and some types of bacteria, which capture energy from sunlight to produce oxygen (O_2) and chemical energy stored in glucose (a sugar). Herbivores then Obtain this energy by eating plants, and carnivores obtain it by eating herbivores.

**The Process:**

During photosynthesis, plants take in carbon dioxide (CO_2) and water (H_2O) from the air and soil. Within the plant cell, the water is oxidized, meaning it loses electrons, while the carbon dioxide is reduced, meaning it gains electrons. This transforms the water into oxygen and the carbon dioxide into glucose. The plant then releases the oxygen back into the air, and stores energy within the glucose molecule. In plants, photosynthesis takes place in the chloroplasts of the cells located in the leaves.

The process starts with the absorption of light energy by pigments such as chlorophyll, which then excites electrons. These excited electrons are used to power the transfer of carbon dioxide into organic molecules, such as sugars and starches, through a series of chemical reactions. The end product of photosynthesis in plants is stored chemical energy in the form of organic compounds.

In some animals, such as algae, photosynthesis also takes place in chloroplasts. The process is essentially the same as in plants, with the absorption of light energy and the conversion of carbon dioxide into organic molecules.

Chlorophyll:

Inside the plant cell are small organelles called chloroplasts, which store the energy of sunlight. Within the thylakoid membranes of the chloroplast is a light-absorbing pigment called chlorophyll, which is responsible for giving the plant its green color. During photosynthesis, chlorophyll absorbs energy from blue- and red-light waves and reflects green-light waves, making the plant appear green.

PHOTOVOLTAIC CELLS:

WHAT IS PHOTOVOLTAIC?

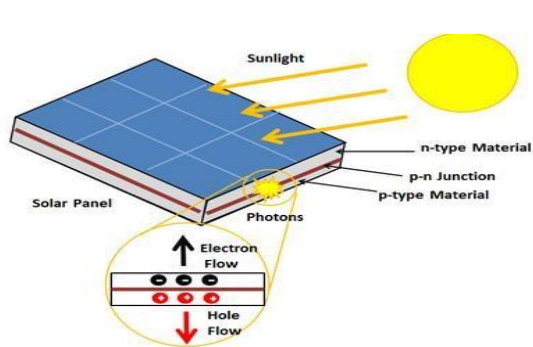
The sun's copious energy is captured by two engineering systems: photosynthetic plant cells and photovoltaic cells (PV). Photosynthesis converts solar energy into chemical energy, delivering different types of products such as building blocks, biofuels, and biomass; photovoltaics turn it into electricity which can be stored and used to perform work.

The connection between photosynthesis and photovoltaics lies in the conversion of light energy into usable forms of energy. In photosynthesis, light energy from the sun is converted into chemical energy stored in organic molecules, such as sugars and starches. In photovoltaics, light energy is converted into electrical energy.

Both photosynthesis and photovoltaics use the same basic principle of converting light energy into usable forms of energy, but the end products are different. In photosynthesis, the end product is stored chemical energy, while in photovoltaics, the end product is electrical energy. However, the similarities between photosynthesis and photovoltaics go beyond just the conversion of light energy. Both processes also involve the use of specialized components and materials, such as chlorophyll in photosynthesis and silicon in photovoltaics, to absorb and convert light energy into usable forms of energy.

The development of photovoltaics has been heavily influenced by the natural process of photosynthesis, and many researchers have sought to mimic and improve upon the efficiency and effectiveness of photosynthesis in order to develop more advanced and efficient photovoltaic systems. The study of photosynthesis has thus played a significant role in the development of sustainable energy systems and continues to be an important area of research in the field of renewable energy.





(Photovoltaic cell)

A solar cell or, photovoltaic cell is an electronic device that converts the energy of light directly into electricity by the photovoltaic effect, which is a physical and chemical phenomenon. It is a form of photoelectric cell, defined as a device whose electrical characteristics, such as current, voltage, or resistance, vary when exposed to light. Individual solar cell devices are often the electrical building blocks of photovoltaic modules, known colloquially as solar panels. The common single-junction silicon solar cell can produce a maximum open-circuit voltage of approximately 0.5 volts to 0.6 volts.

Application:

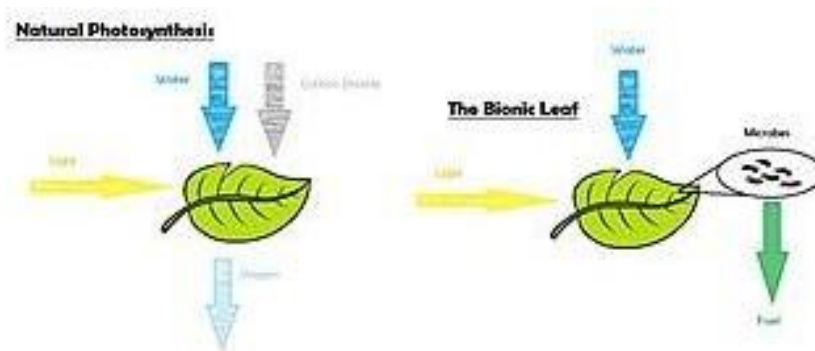
- Remote Locations
- Stand-Alone Power
- Power in space
- Building -Related Needs
- Military Uses.
- Transportation

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BIONIC LEAF:

The bionic Leaf is a biomimetic system that gathers solar energy via photovoltaic cells that can be stored or used in several different functions. Bionic leaves can be composed of both synthetic (metals, ceramics, polymers, etc.) and organic materials (bacteria), or solely made of synthetic materials. The Bionic Leaf has the potential to be implemented in communities, such as urbanized areas to provide clean air as well as providing needed clean energy.





Mechanics:

Natural Photosynthesis vs. The Bionic Leaf at its simplest form.

In natural photosynthesis, photosynthetic organisms produce energy-rich organic molecules from water and carbon dioxide by using solar radiation. Therefore, the process of photosynthesis removes carbon dioxide, a greenhouse gas, from the air. Artificial photosynthesis, as performed by the Bionic Leaf, is approximately 10 times more efficient than natural photosynthesis. Using a catalyst, the Bionic Leaf can remove excess carbon dioxide in the air and convert that to use alcohol fuels, like isopropanol and iso butanol.

The efficiency of the Bionic Leaf's artificial photosynthesis is the result of bypassing obstacles in natural photosynthesis through its artificiality. In natural systems, numerous energy conversion bottlenecks limit the overall efficiency of photosynthesis. As a result, most plants do not exceed 1% efficiency and even microalgae grown in bioreactors do not exceed 3%. Existing artificial photosynthetic solar-to-fuels cycles may exceed natural efficiencies but cannot complete the cycle via carbon fixation. When the catalysts of the Bionic Leaf are coupled with the bacterium *Ralstonia eutropha*, this results in a hybrid system capable of carbon dioxide fixation. This system can store more than half of its input energy as products of carbon dioxide fixation. Overall, the hybrid design allows for artificial photosynthesis with efficiencies rivaling that of natural photosynthesis.

Components of Bionic Leaf

A bionic leaf is a biohybrid system that mimics the natural process of photosynthesis to convert sunlight into chemical energy. It typically consists of several key components that work together to facilitate this conversion. Here are the main components of a bionic leaf:

Photosynthetic Organism: The bionic leaf utilizes a photosynthetic organism, such as a cyanobacterium or a genetically modified plant, as the primary component. This organism contains chlorophyll or other light-absorbing pigments that capture solar energy and initiate the photosynthetic process.

Light Harvesting System: The bionic leaf includes a light harvesting system, which can be artificial or natural, to efficiently capture sunlight. In some designs, light-absorbing dyes or semiconductor materials are incorporated to enhance light absorption and conversion efficiency.

Catalysts: The bionic leaf incorporates catalysts, such as enzymes (Examples: Hydrogenase, Nitrogenase, etc.) or synthetic catalysts (Example: Rubisco (Ribulose-1,5- bisphosphate carboxylase/oxygenase)), to facilitate the chemical reactions involved in photosynthesis. These catalysts play a crucial role in splitting water molecules, generating electrons, and catalyzing the conversion of carbon dioxide into fuels or other chemical compounds.

Electron Transfer Pathway: An electron transfer pathway is an essential component of the bionic leaf system. It allows the generated electrons from water splitting to be efficiently transported to the catalysts involved in carbon dioxide reduction or other chemical reactions. This pathway ensures the flow of electrons necessary for fuel production or other desired chemical transformations.

Carbon Dioxide Source: To sustain the photosynthetic process, a bionic leaf requires a source of carbon dioxide. This can be obtained from various sources, including ambient air, industrial emissions, or concentrated carbon dioxide solutions.

Energy Storage or Conversion System: The bionic leaf includes an energy storage or conversion system to capture and store the chemical energy produced during photosynthesis. This can involve the production of hydrogen gas, liquid fuels, or other energy-rich compounds that can be stored and used as needed.

Control and Monitoring System: To optimize performance and ensure efficient operation, a bionic leaf typically incorporates a control and monitoring system. This system monitors various parameters such as light intensity, temperature, pH, and carbon dioxide levels, and allows for adjustments and optimization of the overall process.

Applications:

Renewable Energy Production: One of the primary applications of bionic leaf technology is in the production of renewable energy. Bionic leaf systems can harness solar energy and convert it into chemical energy in the form of hydrogen gas or other carbon-based fuels. These fuels can be used as clean energy sources for various applications, including transportation, electricity generation, and heating.

Agriculture and Food Production: Bionic leaf technology can have applications in agriculture and food production. By utilizing sunlight and carbon dioxide, bionic leaf systems can generate oxygen and energy-rich compounds that can enhance plant growth and improve crop yields. This technology can potentially contribute to sustainable agriculture practices and help address global food security challenges.

Environmental Remediation: Bionic leaf technology has the potential to aid in environmental remediation efforts. By utilizing the energy generated from sunlight, bionic leaf systems can power processes that remove pollutants or contaminants from air, water, or soil, contributing to the restoration and preservation of ecosystems.

BIRD FLYING:

Birds fly by flapping their wings and using their body weight and the movement of the air to stay aloft. They navigate using a combination of visual cues, the Earth's magnetic field, and celestial navigation. Aircraft, on the other hand, use engines to generate thrust and lift from the wings to stay in the air. They navigate using a combination of instruments and systems, including GPS (Global Positioning System), which uses satellite signals to determine the aircraft's position and help it navigate. Although birds and aircraft both fly, their mechanisms and methods of navigation are quite different.

Birds flying influenced the invention of aircraft in that early aviation pioneers, such as the Wright brothers, observed and studied the flight of birds to develop their flying machines. They noted how birds used their wings and body to achieve lift and control their flight, and used this knowledge to design and improve aircraft.

The development of GPS technology was not directly influenced by birds, but rather by the need for accurate and reliable navigation systems for various purposes, including aviation. GPS uses a network of satellites to provide location and time information, which is used by aircraft for navigation, communication, and safety purposes.

GPS(Global Positioning System):

GPS (Global Positioning System) is a technology that uses a network of satellites to provide location and time information to users. The technology works by measuring the time it takes for signals to travel from satellites to a receiver on the ground or in a vehicle, and using this information to calculate the user's position.

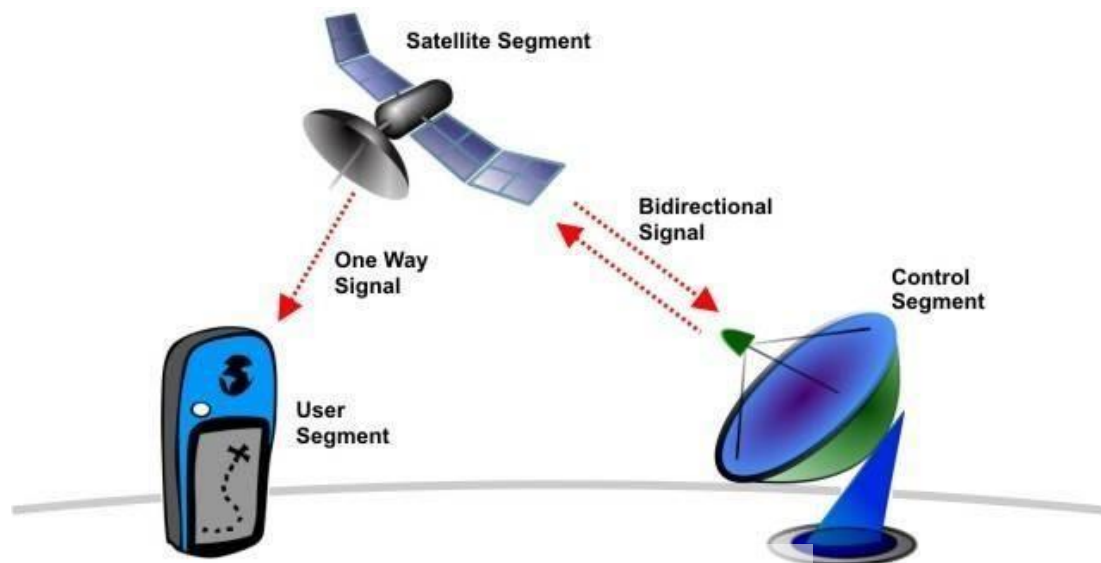
Here are some key components of GPS technology:

Satellites: The GPS satellite network consists of 24-32 satellites orbiting the Earth. These satellites continuously broadcast signals containing information about their location, time, and status.

Receivers: GPS receivers, which are typically integrated into devices such as smartphones, navigation systems, and aircraft, receive signals from GPS satellites and use the information to calculate the user's position.

Control segment: The control segment consists of ground-based monitoring stations that track the GPS satellites, check the accuracy of their signals, and make adjustments as needed.

User segment: The user segment consists of the GPS receivers used by individuals and organizations to obtain location and time information.



GPS technology has a wide range of applications, including navigation, mapping, surveying, search and rescue, and military operations. The accuracy and reliability of GPS have improved over time, and the technology continues to evolve with new developments in satellite and receiver technology, as well as the integration of GPS with other technologies such as augmented reality and artificial intelligence.

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Importance Of GPS Technology in Aircrafts



Representing GPS technology in aircrafts

Positioning and Navigation: GPS helps aircraft accurately determine their position and follow precise routes. Signals from satellites are received by GPS receivers onboard, allowing the system to calculate the aircraft's position.

Flight Planning: GPS assists pilots and planners in creating optimal flight plans, considering waypoints, altitudes, and current information on navigation aids, weather, and airspace restrictions.

Approach and Landing: GPS-based navigation systems provide precise guidance during approach and landing, even in low visibility. This enhances safety and reduces reliance on ground-based navigation aids.

Air Traffic Management: GPS is integrated into air traffic management systems, improving airspace efficiency, reducing congestion, optimizing routing, and enhancing aircraft tracking and situational awareness for controllers.

Collision Avoidance: GPS contributes to collision avoidance systems like TCAS and ADS-B. These systems use GPS data to track nearby aircraft, provide alerts, and ensure safe separation.

Flight Data Recording: GPS data is often recorded by flight data recording systems, aiding post-flight analysis, accident investigation, and overall flight safety improvements. GPS technology has revolutionized aircraft navigation and has become an integral part of modern aviation. It provides accurate positioning, enhances safety, improves operational efficiency, and contributes to the overall advancement of the aviation industry.

GPS AND BIRD FLIGHT:

Scientists have long known that birds navigate using the earth's magnetic field. Now, a new study has found subtle mechanics in the brain of pigeons that allow them to find their way.

A team at Baylor College of Medicine in the U.S. identified a group of 53 cells in a pigeon's brain that record detailed information on the Earth's magnetic field, a kind of internal global positioning system (GPS).

Experiment:

Prof. Dickman and his colleague Le-Qing-Wu set up an experiment in which pigeons were held in a dark room and used a 3D coil system to cancel out the planet's natural geomagnetic field and generate a tunable, artificial magnetic field inside the room. While they adjusted the elevation angles and magnitude of their, artificial magnetic field, they simultaneously recorded the activity of the 53 neurons in the pigeons' brains which had already been identified as candidates for such sensors.

Comparing Birds and Aircrafts with GPS Technology for Navigation

Criteria	Aircrafts	Birds
Mechanism	GPS technology in aircraft relies on signals received from satellites to determine precise position, velocity, and time.	Birds use a combination of visual cues, magnetic fields, landmarks, and celestial navigation to navigate and orient themselves during flight.

Accuracy	GPS technology provides highly accurate position information with a margin of error typically within a few meters.	Birds have remarkable navigational abilities but may not possess the same level of accuracy as GPS. However, birds can adjust their flight path based on real-time environmental cues, which allows for more dynamic and adaptable navigation
Sensory Input	GPS technology relies solely on receiving satellite signals	Birds integrate various sensory inputs for navigation. They can perceive and interpret visual cues, such as landmarks and the position of the sun or stars, and they may also have sensitivity to Earth's magnetic field, enabling them to navigate across vast distances.
Adaptability	GPS technology in aircraft provides consistent and reliable navigation regardless of the environmental conditions or time of day	Birds, on the other hand, demonstrate remarkable adaptability in their navigation abilities. They can adjust their flight paths based on changing weather conditions, wind patterns, and other factors, which allows for efficient long-distance migration and navigation through complex landscapes.
Evolutionary Aspect	GPS technology is a human-made innovation designed to enhance navigation and safety in aircraft.	Birds, however, have evolved over millions of years, developing specialized neural and physiological adaptations that enable them to navigate and fly efficiently in diverse habitats.

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AIRCRAFT:**MECHANISM:**

Lift, Drag, and Thrust: The fundamentals of bird flight are similar to those of aircraft, in which the aerodynamic forces sustain lift, drag, and thrust. Lift force is produced by the action of airflow on the wing, which is an airfoil. The airfoil is shaped such that the air provides a net upward force on the wing, while the movement of air is directed downward. The additional net lift may come from airflow around the bird's body in some species, especially during intermittent flight while the wings are folded or semi-folded (cf. tilting body).

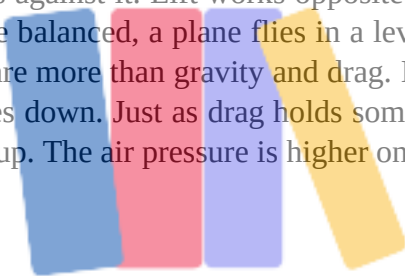
Aerodynamic drag is the force opposite to the direction of motion, and hence the source of energy loss in flight. The drag force can be separated into two portions, lift-induced drag, which is the inherent cost of the wing producing lift (this energy ends up primarily in the wingtip vortices), and parasitic drag, including skin friction drag from the friction of air and body surfaces and form drag from the bird's frontal area. The streamlining of the bird's body and

Wings reduces these forces. Unlike aircraft, which have engines to produce thrust, birds flap their wings with a given flapping amplitude and frequency to generate thrust.

Lift, Drag, and Thrust:

- **Thrust** is a force that moves an aircraft in the direction of the motion. It is created with a propeller, jet engine, or rocket. Air is pulled in and then pushed out in an opposite direction. One example is a household fan.
- **Drag** is the force that acts opposite to the direction of motion. It tends to slow an object. Drag is caused by friction and differences in air pressure. An example is putting your hand out of a moving car window and feeling it pull back.
- **Weight** is the force caused by gravity.
- **Lift** is the force that holds an airplane in the air. The wings create most of the lift used by airplanes.

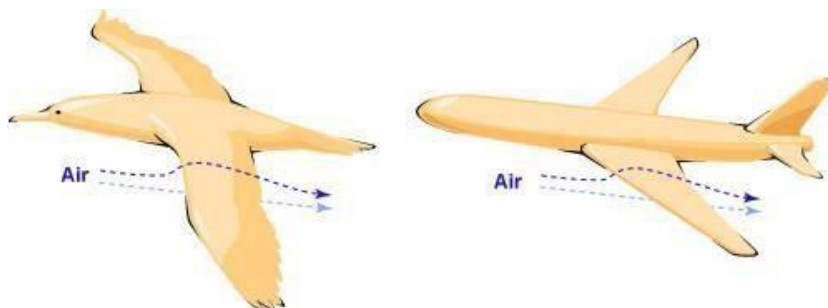
The way the four forces act on the airplane make the plane do different things. Each force has an opposite force that works against it. Lift works opposite of weight. Thrust works opposite of drag. When the forces are balanced, a plane flies in a level direction. The plane goes up if the forces of lift and thrust are more than gravity and drag. If gravity and drag are bigger than lift and thrust, the plane goes down. Just as drag holds something back as a response to wind flow, lift pushes something up. The air pressure is higher on the bottom side of a wing, so it is pushed upward.



Bio Mimicking Birds fly for Aircraft Technology

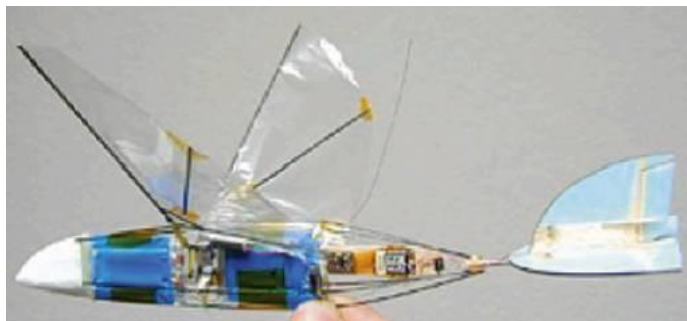
Biomimicry, or the practice of using designs and processes found in nature to solve human problems, has led to the development of various technologies inspired by birds' flight. Some examples include:

Wing design: The shape of bird wings has inspired the design of aircraft wings, which have evolved to be more aerodynamic and fuel-efficient as a result. The study of bird flight has also led to the development of winglets, small structures at the tip of wings that reduce drag and increase lift.



(Comparing the wing design of bird and aircraft)

Flapping-wing drones: Researchers have developed drones that use flapping wings to fly, mimicking the way birds and insects fly. These drones can be used for various applications, such as monitoring crops and wildlife, inspecting buildings and infrastructure, and search and rescue operations.



(Image of a flapping wind drone)

Soaring algorithms: Soaring refers to the flight technique used by birds and certain aircraft to stay aloft and travel long distances with minimal energy expenditure. It involves utilizing rising air currents, such as thermals, ridge lift, wind shear, or atmospheric waves, to gain altitude and maintain flight. Birds use thermals, or columns of rising warm air, to gain altitude and soar. Researchers have developed algorithms inspired by bird flight to help gliders and other aircraft use thermals more efficiently, leading to longer and more sustainable flights.

Landing gear: The legs and feet of birds have inspired the design of landing gear for aircraft, with shock-absorbing and retractable structures that help absorb impact upon landing.

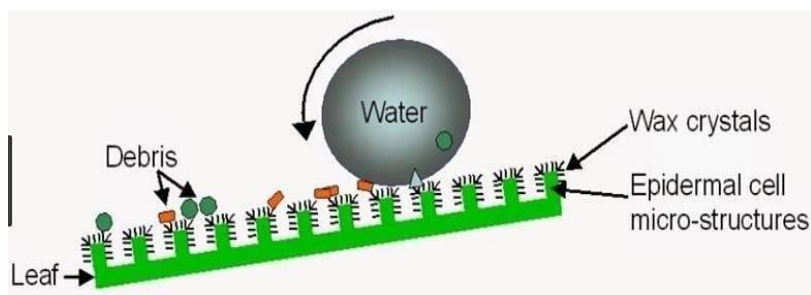
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LOTUS LEAF EFFECT easenotes.com

The lotus leaf is well-known for having a highly water repellent, or superhydrophobic, surface, thus giving the name to the lotus effect. Water repellency has received much attention in the development of self-cleaning materials, and has been studied in both natural and artificial systems.

The lotus leaf effect, also known as the "lotus effect," refers to the ability of lotus leaves to repel water and self-clean through their unique surface structure. This effect has inspired the development of super hydrophobic and self-cleaning surfaces, which have a wide range of applications in various industries.

The lotus leaf surface has a microscale and nanoscale structure that consists of numerous small bumps and wax-coated hairs. This structure creates a high contact angle between the water droplets and the surface, causing the droplets to roll off and carry away any dirt or debris. This self-cleaning property is due to the lotus leaf's ability to repel water and resist adhesion.



(Representing the surface of lotus leaf)

Super hydrophobic and self-cleaning surfaces have applications in industries such as aerospace, automotive, building materials, and medical devices. For example, self-cleaning coatings can be used on the exterior of buildings to reduce the need for cleaning and maintenance, while super hydrophobic coatings can be used to prevent icing on aircraft wings.

SUPERHYDROPHOBIC AND SELF-CLEANING SURFACES:

The self-cleaning function of superhydrophobic surfaces is conventionally attributed to the removal of contaminating particle by impacting or rolling water droplets, which

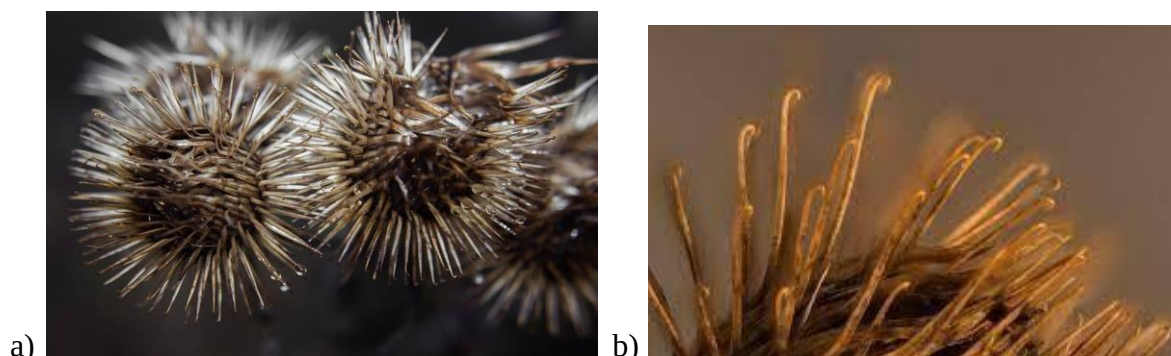
implies the action of external forces such as gravity. Here, we demonstrate a unique self-cleaning mechanism whereby the contaminated superhydrophobic surface is exposed to condensing water vapor, and the contaminants are autonomously removed by the self-propelled jumping motion of the resulting liquid condensate, which partially covers or fully encloses the contaminating particles. The jumping motion of the superhydrophobic surface is powered by the surface energy released upon the coalescence of the condensed water phase around the contaminants. The jumping-condensate mechanism is shown to spontaneously clean superhydrophobic cicada wings, where the contaminating particles cannot be removed by gravity, wing vibration, or wind flow. Our findings offer insights into the development of self-cleaning materials.

Mechanism:

An autonomous mechanism to achieve self-cleaning on superhydrophobic surfaces, where the contaminants are removed by self-propelled jumping condensate powered by surface energy. When exposed to condensing water vapor, the contaminating particles are either fully enclosed or partially covered with the resulting liquid condensate. Building upon our previous publications showing self-propelled jumping upon drop coalescence, we show particle removal by the merged condensate drop with a size comparable to or larger than that of the contaminating particle(s). Further, we report a distinct jumping mechanism upon particle aggregation, without a condensate drop of comparable size to that of the particles, where a group of particles exposed to water condensate clusters together by capillarity and self-propels away from the superhydrophobic surface.

PLANT BURRS and VELCRO:

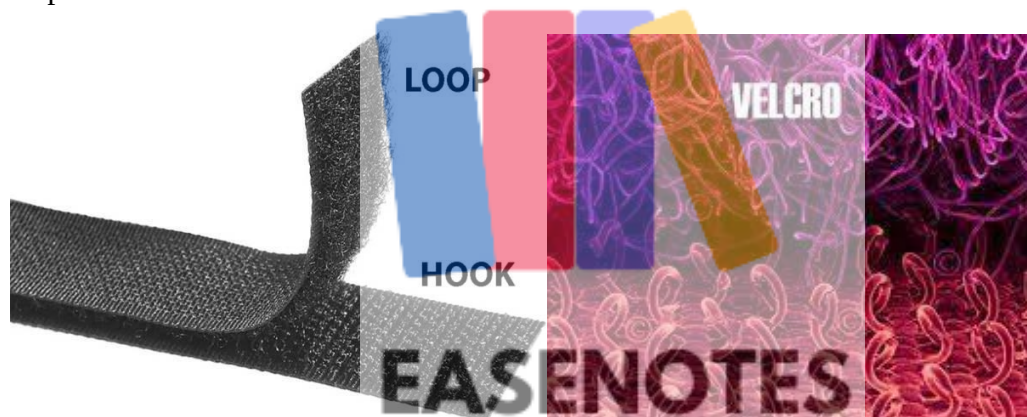
Plant burrs, such as those found on burdock, inspired the invention of Velcro, a popular hook-and-loop fastening system.



a) The globular flower heads of burdock

b) indicating the hook shape

The burrs have small hooks that can latch onto clothing, fur, or feathers, allowing them to disperse their seeds over a wider area.



a) hook and loops normal view of Velcro b) microscopic view of hooks and loops of velcro

Velcro was invented by Swiss engineer George De Mestral in 1941, after he became fascinated by the way burrs clung to his clothes and his dog's fur during a walk. He examined the burrs under a microscope and found that they had small hooks that could latch onto loops in fabric. De Mestral spent years experimenting with different materials before finally developing Velcro, which consists of two strips of nylon fabric, one with tiny hooks and the other with small loops. When pressed together, the hooks latch onto the loops, creating a strong bond that can be easily detached by pulling the two strips apart. Velcro has a wide range of applications, including in clothing, shoes, bags, and medical devices. It has become a popular alternative to traditional fasteners, such as buttons and zippers, due to its ease of use and versatility.

The name "Velcro" is actually a combination of the words "velvet" and "crochet," as the fabric strips resemble velvet and are hooked together like crochet. Velcro has since become a popular alternative to traditional fasteners, such as buttons and zippers, due to its ease of use and versatility.

Relevance to humans:

Burrs are best known as sources of irritation, injury to livestock, damage to clothing, punctures to tires and clogging equipment such as agricultural harvesting machinery. Furthermore, because of their ability to compete with crops over moisture and nutrition. Bur plants can be labels as weeds and therefore also be subject to removal. Methods of controlling the spread of bur plants include the use of herbicides, slashing and cultivation among others.

Some have however been used for such purposes as fabric fulling, for which fuller's teasel is a traditional resource. The bur of burdock was the inspiration for the hook and loop fastener also Known as Velcro.

Engineering Applications of Velcro Technology

Clothing and Footwear: Velcro is commonly used in clothing and footwear for closures and adjustable straps. It can be easily opened and closed, making it convenient for users with limited dexterity or mobility.

Medical devices: Velcro is used in medical devices such as braces, splints, and compression garments for its adjustable and secure fastening capabilities.

Aerospace equipment: Velcro is used in aerospace equipment, such as satellites and spacecraft, to secure components in place and prevent them from vibrating or shifting during launch or flight.

Automotive industry: Velcro is used in the automotive industry for a range of applications, such as securing carpets and headliners, and attaching door panels and seat cushions.

Packaging industry: Velcro is used in the packaging industry for resealable closures on bags, pouches, and other types of packaging.

Sports equipment: Velcro is used in sports equipment, such as helmets and gloves, for its ability to provide a secure and adjustable fit.

Shark skin and friction reducing Swin Suits

The denticles on shark skin have evolved over millions of years to reduce drag and increase swimming efficiency. These structures disrupt the flow of water around the shark's body, reducing turbulence and minimizing the formation of vortices. As a result, sharks can swim faster and with less effort compared to other fish.

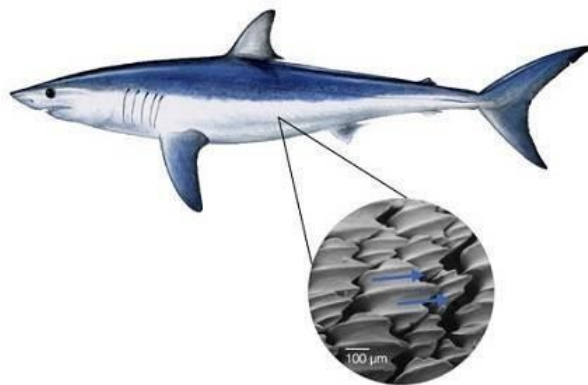


Figure: Indicating the denticles on shark skin

Denticles on shark skin are like tiny bumps or ridges. They disrupt the flow of water around the shark's body, making it smoother and reducing turbulence. This disruption reduces the resistance the shark experiences as it swim, allowing it to move faster and with less effort.

Turbulence in Water:

Turbulence is when a fluid, like water or air, becomes chaotic and unpredictable. Instead of flowing smoothly, it swirls and forms irregular patterns. This turbulence creates resistance or drag, which makes it harder for things to move through the fluid. In swimming, reducing turbulence is important because it helps to minimize resistance, allowing swimmers to move more easily and efficiently through the water.

Reducing Drag:

When a shark swims through the water, the water normally flows smoothly over its body. However, the denticles on the shark's skin disrupt this smooth flow. They create small disturbances in the water, which helps to break up turbulent currents that can slow the shark down. By reducing turbulence, the denticles make the flow of water around the shark's body smoother. This smoother flow reduces the resistance or drags the shark experiences as it moves through the water, allowing it to swim more efficiently.

Frictionless Swim Suits:

Shark skin has inspired the development of friction-reducing swim suits, which are designed to improve the performance of swimmers by reducing drag in the water.

Friction-reducing swim suits use a similar structure to that of shark skin to reduce drag and improve swimmer performance. These suits are made from high-tech materials that mimic the properties of shark skin, such as the shape and size of the denticles.

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Materials Used

The materials used to create friction-reducing swim suits inspired by shark skin include:

Polyurethane: A type of polymer that is commonly used in the production of swim suits, as it is durable and can be molded into a variety of shapes.

Lycra/Spandex: Lycra and spandex are made from the same synthetic fiber, which is technically called elastane. Elastane fibers are typically composed of a polymer called polyurethane which is then blended with other fibers like nylon, polyester, or cotton) that is known for its stretch and flexibility.

High-tech fabrics: A range of high-tech fabrics have been developed specifically for use in swim suits. These fabrics are designed to be lightweight, water-repellent, and hydrodynamic, and often incorporate materials such as silicone or Teflon to reduce drag.

Kingfisher Beak and Bullet Train



Figure: Indicating the shape similarities of kingfisher beak and design of the front of the bullet train

The kingfisher beak is an excellent example of nature's design for efficient diving and fishing. Its unique shape and structure enable the kingfisher to minimize the impact of water resistance and achieve a successful dive.

The Physics behind the Kingfisher Beak

Streamlining:

The beak of a kingfisher is long, slender, and sharply pointed, which helps reduce drag or air resistance as the bird dives into the water. The streamlined shape allows the kingfisher to smoothly cut through the air and minimize the energy required for the dive.

Surface Tension:

When the kingfisher hits the water, it encounters the resistance caused by surface tension. Surface tension is the cohesive force between water molecules that creates a "skin" on the water's surface. The sharp beak of the kingfisher helps to pierce through the water's surface, breaking the surface tension and reducing the force required to enter the water.

Minimizing Splash:

As the kingfisher dives, it needs to enter the water with minimal disturbance to avoid scaring away the fish it intends to catch. The shape of the beak helps to reduce the splash generated upon entry. The beak's narrow and pointed design helps create a smooth entry by minimizing the disturbance of the water surface, allowing the kingfisher to enter silently and effectively.



image of shinkasen bullet train of Japan

Technological Importance : The use of the kingfisher beak as a design inspiration for the front of the bullet train is an example of how nature-inspired engineering can lead to innovative solutions that improve the performance and efficiency of machines. Shinkansen bullet train of Japan is the best example which used the biomimicry of kingfisher's beak.

Aerodynamic Design: The front of the Shinkansen is meticulously shaped to reduce air resistance and improve aerodynamic performance. The streamlined design minimizes drag as the train travels at high speeds, allowing it to maintain stability and efficiency. The smooth, tapered shape reduces the pressure difference between the front and rear of the train, reducing noise and vibration.

Pressure Wave Reduction: When a high-speed train moves through a tunnel, it creates pressure waves that can cause noise and discomfort for passengers. The nose of the Shinkansen is designed to reduce these pressure waves by effectively managing airflow and minimizing the compression and expansion of air as the train enters and exits tunnels. This reduces the noise level and enhances passenger comfort.

HUMAN BLOOD SUBSTITUTES:

Shortages in blood supplies and concerns about the safety of donated blood have fueled the development of so-called blood substitutes. The two major types of blood substitutes are volume expanders, which include solutions such as saline that are used to replace lost plasma volume, and oxygen therapeutics, which are agents designed to replace oxygen normally carried by the haemoglobin in red blood cells. Of these two types of blood substitutes, the development of oxygen therapeutics has been the most challenging. One of the first groups of agents developed and tested were perfluorocarbons, which effectively transport and deliver oxygen to tissues but cause complex side effects, including flulike reactions and are not metabolized by the body.

Blood from the human umbilical cord has been studied for its potential as a substitute source of red blood cells for transfusion. Red blood cells can be extracted from cord blood via

sedimentation as the blood is cooled. Donated cord blood can be screened for infections organisms and other contaminants. Research concerning its potential use for transfusion is ongoing. Of particular concern for implementation are the establishment of safe, effective and ethical procedures for cord blood collection as well as the development of criteria that help to ensure safe transfusion and the preservation of cord blood quality.

Haemoglobin-based oxygen carriers (HBOCs) AND Perfluorocarbons(PFC):

Pharmaceutical companies attempted to develop HBOCs (also called oxygen therapeutics) and PFCs starting in the 1980 and at first, seemed to have some success. However, the results of most human clinical trials have been disappointing. A study published in 2008 in the journal of the American Medical Association summarized the results of 16 clinical trials on five different blood substitutes administered to 3,500 patients.

Those receiving blood substitutes had a threefold increase in the risk of heart attacks compared with the control group given human donor blood. However, a closer analysis of the results showed that some of the negative statistics were misleading. The artificial blood products reviewed in this study varied in their benefits and risks, and some blood substitutes had very few serious side effects. The findings suggest that some blood substitutes may be safer and more beneficial than scientists originally thought.

1) HBOCs:

Haemoglobin-based oxygen carriers (HBOCs) are “made of” natural haemoglobins that were originally developed as blood substitutes but have been extended to a variety of hypoxic clinical situations due to their ability to release oxygen. Compared with traditional preservation protocols, the addition of HBOCs to traditional preservation protocols provides more oxygen to organs to meet their energy metabolic needs prolongs preservation time, reduces ischemia-reperfusion injury to grafts, improves graft quality, and even increases the number of transplantable donors. The focus of the present study was to review the potential applications of HBOCs in solid organ preservation and provide new approaches to understanding the mechanism of promising strategies for organ preservation.

2)PFCs:

PFCs remain in the bloodstream for about 48 hours. Because of their oxygen-dissolving ability, PFCs were the first group of artificial blood products studied by scientists. They are first-generation blood substitutes. Unlike the red-coloured HBOCs, PFCs are usually white. However, since they do not mix with blood they must be emulsified before they can be given to patients. PFCs are such good oxygen carriers that researchers are now trying to find out if they can reduce swollen brain tissue in traumatic brain injury. PFC particles may cause flu-like-symptoms in some patients when they exhale these compounds.

Question bank Module 4

1. Explain Echolocation with its mechanics.
2. What is ultrasonography? Explain with example.
3. What is SONARs? Explain its Types.
4. Write a note on Photosynthesis process.
5. Explain Photovoltaic cells and write its applications.
6. Explain Bionic leaf with its mechanics and write applications.
7. Write a note on GPS system and Bird flying.
8. Explain Aircraft with its mechanism and also define Lotus leaf effect.
9. Discuss the superhydrophobic and self-cleaning surfaces with its mechanism.
10. Write a note on plant burrs.
11. Explain shark skin and how it is relevance to human.
12. How kingfisher beak looks and write its relationship with humans.
13. Write the strategy and potential in the beak that inspired a bullet train.
14. Briefly explain human blood substitutes with HBOs and PFCs.



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BIOLOGY FOR ENGINEERS

MODULE 5

TRENDS IN BIOENGINEERING (QUALITATIVE):

Muscular and Skeletal Systems as scaffolds (architecture, mechanisms, bioengineering solutions for muscular dystrophy and osteoporosis), scaffolds and tissue engineering, Bioprinting techniques and materials, 3D printing of ear, bone and skin. 3D printed foods. Electrical tongue and electrical nose in food science, DNA origami and Biocomputing, Bioimaging and Artificial Intelligence for disease diagnosis. Selfhealing Bioconcrete (based on bacillus spores, calcium lactate nutrients and biomineralization processes) and Bioremediation and Biomining via microbial surface adsorption (removal of heavy metals like Lead, Cadmium, Mercury, Arsenic).

MUSCULAR AND SKELETAL SYSTEM AS SCAFFOLDS:

Skeletal muscle architecture is one of the most important properties that determine a muscle's force and excursion capability. In the current review, basic architectural terms first are reviewed, and then specific examples relevant to upper extremity anatomy are presented. Specific examples of anatomic considerations required for surgical reconstruction after radial nerve palsy also are detailed. Together, these data show not only the wide variety of architectural designs in human muscles but the importance of considering architectural design when making surgical decisions. The relationship between structure and function in skeletal muscle has been described and probed for more than a century. A classic study has elucidated the microscopic and ultrastructural properties of skeletal muscle fibers, yielding great insights into their function. However, less attention has been given to excellent and insightful studies of the macroscopic properties of skeletal muscle tissues dating back to the 1600s. This macroscopic arrangement of muscle fibres is known as a muscle's architecture.

Architecture:

The musculoskeletal system (Locomotor system) is a human body system that provides our body with movement, stability, shape, and support. It is subdivided into two broad systems.

Muscular system, which includes all types of muscles in the body. Skeletal muscles, in particular, are the ones that act on the body joints to produce movements. Besides muscle, the muscular system contains the tendons which attach the muscles to the bones.

Skeletal system, whose main component is the bone. Bones articulate with each other and form the joints, providing our bodies with a hard-core, yet mobile, skeleton. The integrity and function of the bones and joints are supported by the accessory structures of the skeletal system. Articular, cartilage, ligaments, and bursae.



Besides its main function to provide the body with stability and mobility, the musculoskeletal system has many other functions. The skeletal part plays an important role in other homeostatic functions such as storage of minerals (eg: calcium) and hematopoiesis, while the muscular system stores most of the body's carbohydrates in the form of glycogen.

Mechanism:

The nerve system (body's command center) controls your voluntary muscle movements. Voluntary muscles are ones you control intentionally. Some involve large muscle groups to do activities like jumping. Others use smaller movements, like pushing a button.

Movements happen when:

Our nervous system (brain and nerves) sends a message to activate your skeletal (voluntary) muscles. Our muscle fibers contract (tense up) in response to the message. When the muscle activates or bunches up, it pulls on the tendon. Tendons attach muscles to bones. The tendon pulls the bone, making it move. To relax the muscle, your nervous system sends another message. It triggers the muscles to relax or deactivate. The relaxed muscle releases tension, moving the bone to a resting position. Hundreds of conditions can cause problems with the musculoskeletal system. They can affect the way you move, speak and interact with the world. Some of the most common causes of musculoskeletal pain and movement problems are:

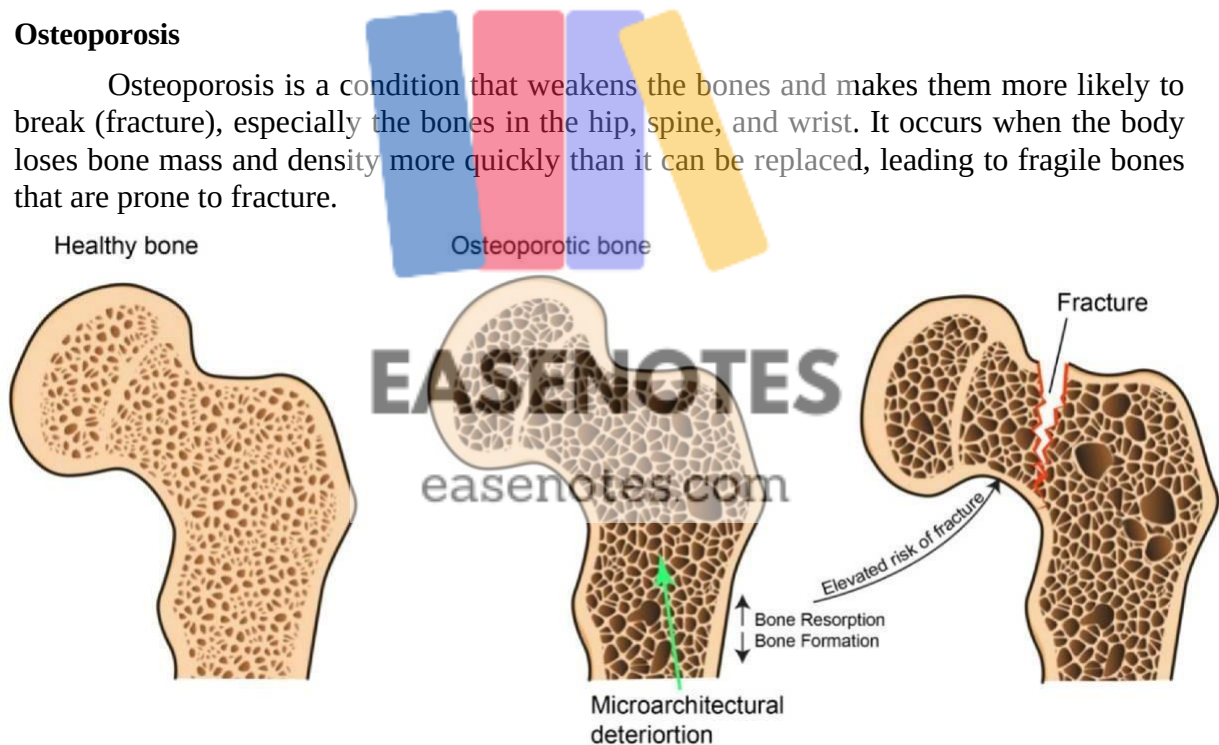
- Aging
- Arthritis
- Back problems
- Cancer
- Congenital abnormalities
- Injuries
- Osteoporosis
- Muscular dystrophy

Everyone has pain in their muscles and joints from time to time. One of the most common musculoskeletal conditions is Osteoporosis. More than 60% of people in the united states have Osteoporosis at some point in their lives. Arthritis is also very common. More than 54 million adults in the U.S have Muscular dystrophy. Most people recover from these disorders without long term health problems.

BIO ENGINEERING SOLUTIONS FOR MUSCULAR DYSTROPHY AND OSTEOPOROSIS:

Osteoporosis

Osteoporosis is a condition that weakens the bones and makes them more likely to break (fracture), especially the bones in the hip, spine, and wrist. It occurs when the body loses bone mass and density more quickly than it can be replaced, leading to fragile bones that are prone to fracture.



(Representing healthy bone and Osteoporosis bone).

Osteoporosis is a common condition, especially among older women, and it can increase the risk of falls and fractures, which can result in significant pain and disability. Risk factors for osteoporosis include being female, older age, having a family history of the condition, smoking, drinking excessive amounts of alcohol, being thin or having a small body frame, and having low calcium intake.

Treatment for osteoporosis aims to slow down bone loss, prevent fractures, and treat fractures if they occur. Some of the treatments include:

- Medications: Bisphosphonates, denosumab, and teriparatide are some of the medications that can slow down bone loss and reduce the risk of fractures.
- Calcium and Vitamin D supplementation: Calcium and Vitamin D are essential for healthy bones, and taking supplements can help maintain bone mass.
- Exercise: Weight-bearing and resistance exercises can help improve bone density and reduce the risk of fractures.
- Lifestyle changes: Quitting smoking, reducing alcohol consumption, and eating a healthy diet that includes enough calcium and Vitamin D can help maintain healthy bones.

It's important to work closely with a healthcare provider to develop a comprehensive treatment plan for osteoporosis, as the right approach may vary depending on the individual's specific needs and medical history.

Bioengineering solutions for osteoporosis

Bioengineering solutions for osteoporosis aim to improve bone health and prevent fractures. Some of the approaches being explored include:

- Tissue engineering: This involves using scaffolds and growth factors to stimulate the growth of new bone tissue and promote the repair of damaged bones. The goal is to create functional bone tissue that can replace the lost bone mass and density in individuals with osteoporosis.
- Stem cell therapy: This involves using stem cells to replace the damaged bone cells and promote the repair and regeneration of bone tissue. Stem cells can be taken from the patient's own body (autologous stem cells) or from a donor (allogenic stem cells).
- Biomaterials: This involves using synthetic or natural materials to replace or augment damaged bone tissue. Biomaterials can be designed to mimic the properties of natural bone and promote the growth of new bone tissue.
- Gene therapy: This involves delivering a functional copy of a gene involved in bone growth and repair to the affected bone cells. The goal is to restore the production of the missing protein and improve bone health.

These approaches are still in the early stages of development, but hold promise for the future treatment of osteoporosis. Clinical trials and further research are needed to determine the safety and efficacy of these therapies.

In addition, traditional treatments for osteoporosis, such as medication, exercise, and lifestyle changes, will likely continue to play an important role in preventing fractures and maintaining healthy bones in individuals with osteoporosis.



Bioprinting Techniques and Materials

Bioprinting is a rapidly growing field that uses various techniques to produce three- dimensional (3D) structures and functional biological tissues for medical and scientific applications. The main objective of bioprinting is to mimic the structure and function of human tissues and organs, leading to the development of replacement parts for damaged or diseased organs.

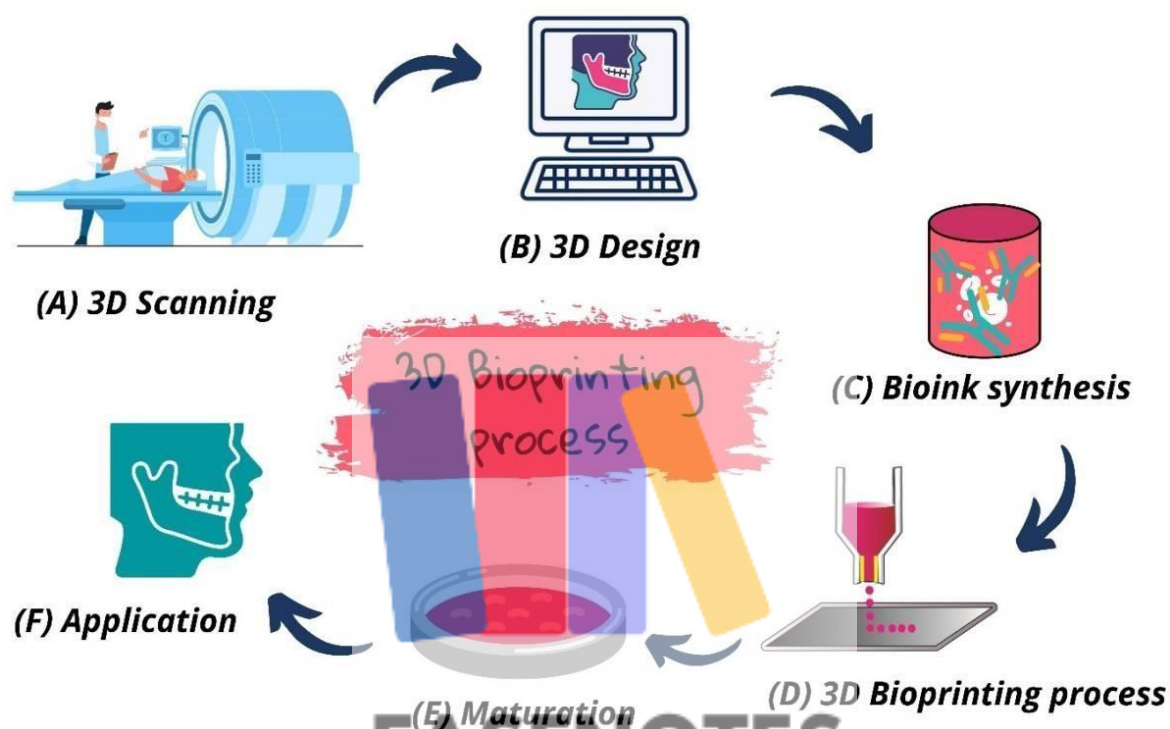
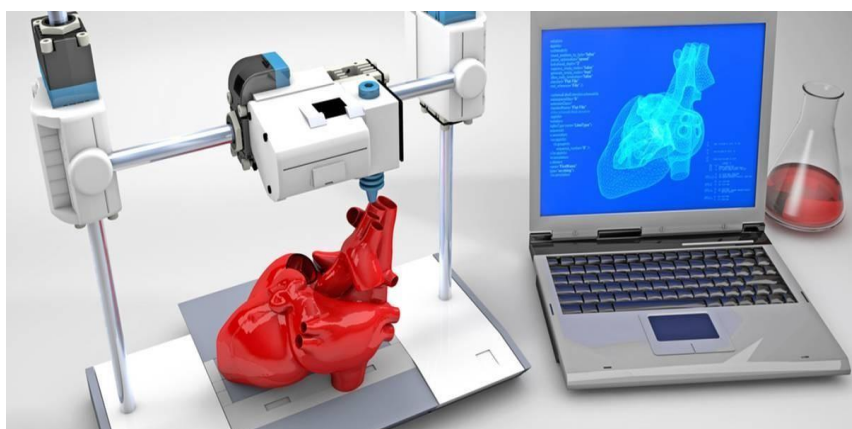


Figure: Schematic representation of bioprinting process

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Schematic representation of 3D bioprinting concept

Comparison between 3D printer and bioprinter

The table provide a concise overview of the comparisons, advantages, and limitations of 3D printers and bioprinters.

Aspect	3D Printers	Bioprinters
Printing Purpose	General-purpose printing of objects	Fabrication of living tissues and organs
Materials	Plastics, metals, ceramics, resins, etc.	Bioinks (hydrogels, extracellular matrices, cell aggregates, etc.)
Applications	Manufacturing, engineering, product design, architecture, etc.	Regenerative medicine, tissue engineering, drug development, etc.
Printing Process	Additive manufacturing, layer-by-layer deposition	Precise deposition of bioinks layer-by-layer
Cell Compatibility	NA	Bioinks must support cell viability and function
Challenges	NA.	Development of suitable bioinks, cell viability, vascularization, scaling up, etc.
Advantages	Versatile, wide range of applications Enables rapid prototyping Cost-effective for non-biological objects	Potential for tissue and organ transplantation Enables tissue engineering and regenerative medicine Can create tissue models for studying diseases Potential for personalized medicine and drug testing.
Limitations	Limited ability to create functional living tissues, limited choices of materials for certain applications, lack of cell compatibility and tissue functionality.	Complex and rapidly evolving technology, challenges in developing suitable bioinks and scaling up vascularization and long term functionality of printed tissues.

Bioprinting materials

Bioprinting materials, also known as bioinks, are specifically designed to be compatible with living cells and provide a supportive environment for their growth and organization. Here are some examples of commonly used bioprinting materials:

Hydrogels:

Hydrogels are water-based polymer networks that closely mimic the extracellular matrix (ECM) found in living tissues. They offer excellent biocompatibility, mechanical support, and can be formulated to have similar physical properties to native tissues. Examples of hydrogels used as bioinks include:

- Gelatin-based hydrogels
- Alginate hydrogels
- Fibrin-based hydrogels
- Collagen-based hydrogels

Cell-laden Aggregates:

In some cases, cells are first aggregated into biomolecules and biomaterials (or microtissues) before being incorporated into the bioink. These aggregates provide a more physiological environment for the cells and enhance their viability and functionality.

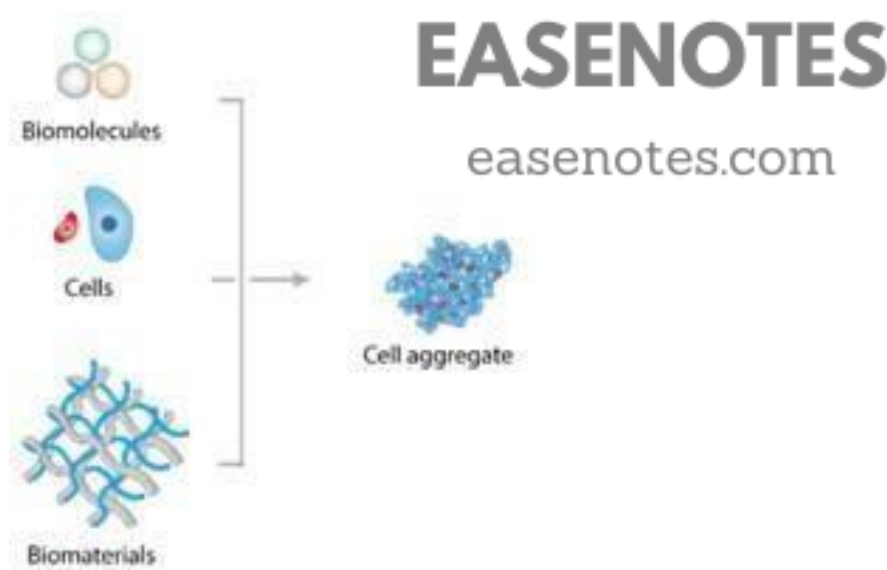


Fig:Schematic representation of formation of cell aggregates

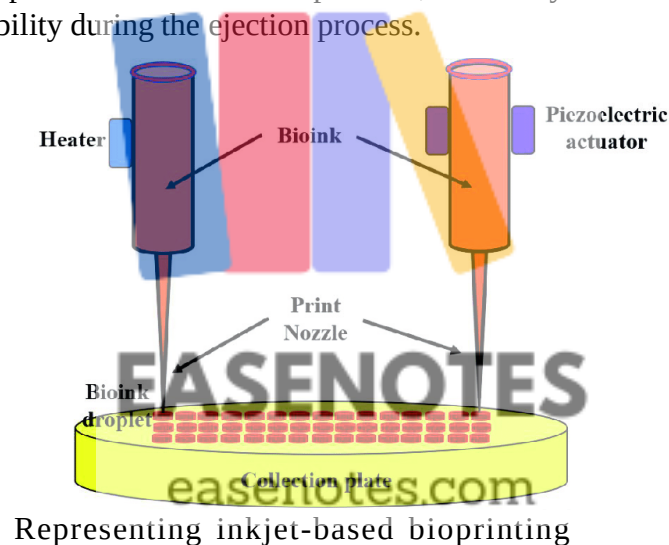


Most Commonly used Bioprinting Techniques

Bioprinting techniques involve the precise deposition of bioinks to create three-dimensional structures with living cells. Several techniques have been developed to accomplish this, each with its own advantages and limitations. Here are some of the most commonly used bioprinting techniques:

Inkjet-based Bioprinting:

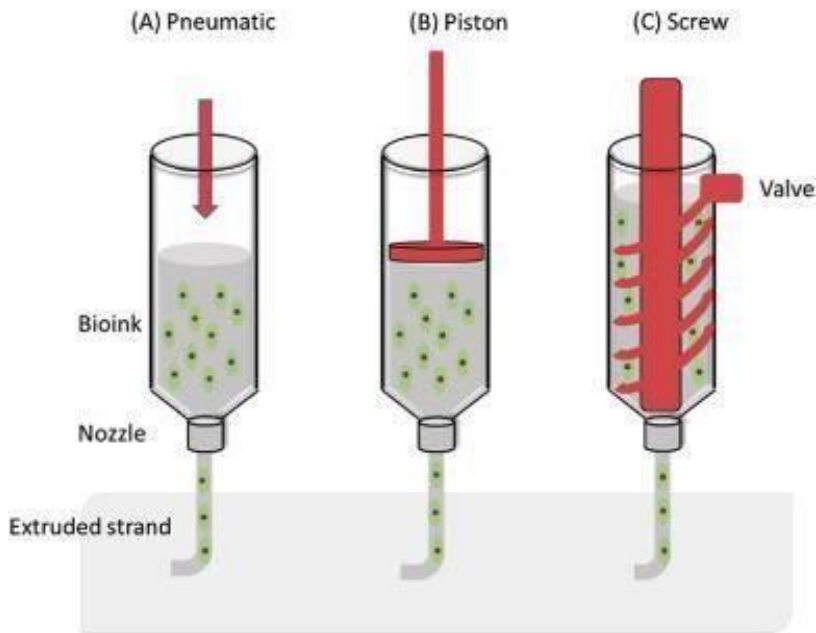
Inkjet bioprinting works similarly to standard inkjet printing. The bioink is loaded into cartridges, and droplets of the bioink are ejected through fine nozzles onto a substrate. The droplets form layers, and the structure is built by depositing subsequent layers. Inkjet bioprinting allows for high-resolution printing and precise control over droplet size, but it may be limited by the viscosity of the bioink and cell viability during the ejection process.



Extrusion-based Bioprinting:

Extrusion-based bioprinting uses a syringe or a similar mechanism to extrude the bioink through a nozzle. The bioink is deposited layer-by-layer to create the desired structure. This technique is versatile and can handle a wide range of bioinks with varying viscosities, including those with living cells or cell aggregates. It allows for high cell viability and can produce structures with controlled porosity. However, it may have limitations in achieving high resolution and complex geometries.





Representing extrusion based bioprinting

Laser-assisted Bioprinting:

Laser-assisted bioprinting utilizes laser energy to precisely deposit bioinks onto a substrate. The bioink is placed on an energy-absorbing layer, and the laser creates a pressure wave that propels the bioink onto the substrate in a controlled manner. This technique offers high resolution, precision, and the ability to print complex structures. It can be used with delicate bioinks and allows for cell viability. However, laser-assisted bioprinting can be relatively slow and may have limitations in terms of bioink viscosity.

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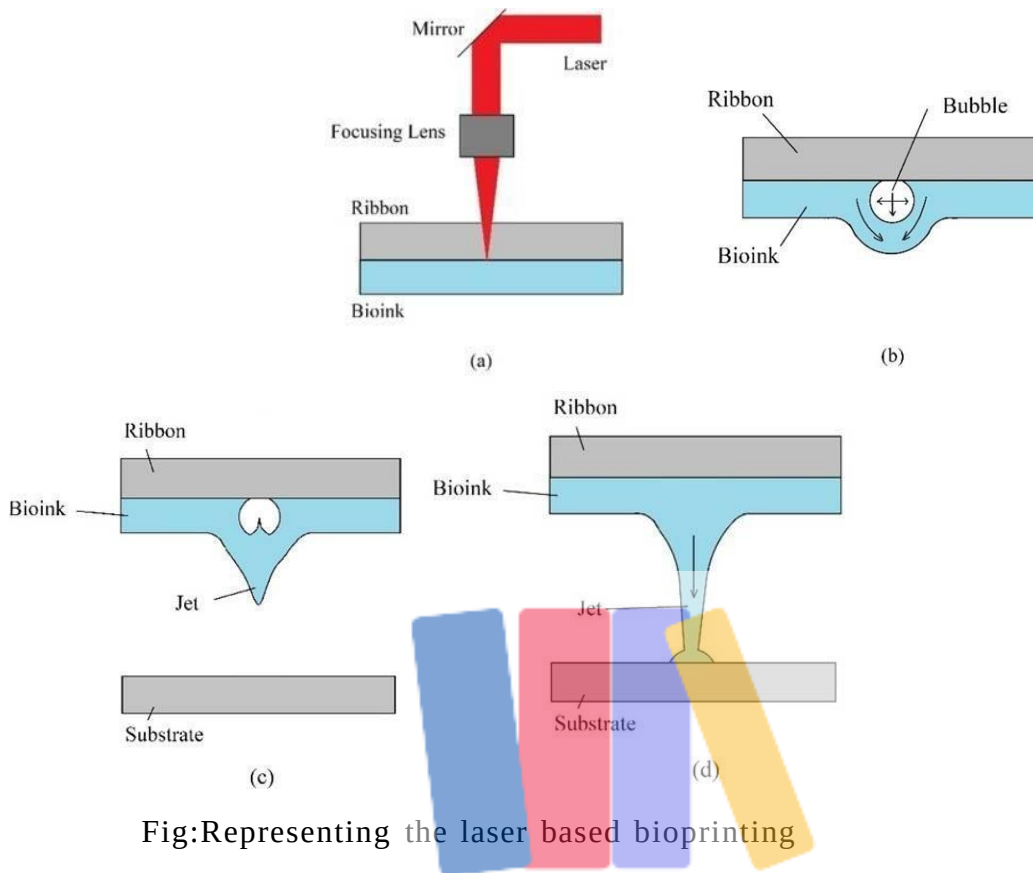


Fig:Representing the laser based bioprinting

3D printing of Ear

3D printing has revolutionized the field of medicine, and one of its applications is the 3D printing of human ears. This process involves using a 3D printer to create an ear-shaped structure using a special material, such as a biocompatible polymer or a hydrogel, as the "ink." The printed ear structure is then seeded with human cartilage cells, which grow and develop into functional ear tissue over time.

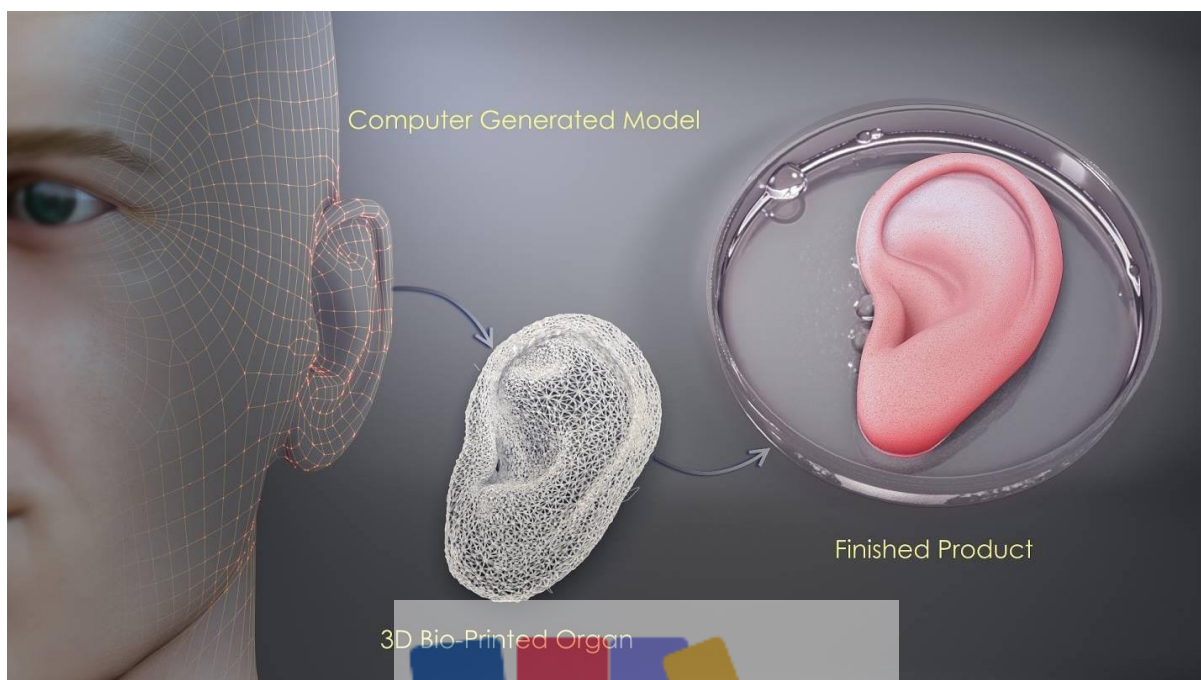


Fig:Representing 3D printing human ear

The main advantage of 3D printing an ear is that it allows for the creation of an ear that is custom-fitted to an individual patient, based on their specific ear shape and size. This can be especially useful for children with congenital ear deformities or individuals who have suffered ear injuries or losses.

Additionally, 3D printing can also be used to create ears that are anatomically and functionally similar to a patient's normal ear, reducing the risk of complications associated with traditional surgical methods.

Materials Used for 3D Printing of Human Ear

The material used for 3D printing of human ears can vary, depending on the specific technique used and the desired outcome. Some of the most commonly used materials for 3D printing of ears include:

Hydrogels:

Hydrogels are soft, gel-like materials that are commonly used in bioprinting due to their ability to mimic the mechanical properties of human tissues. They can be used as the "ink" in 3D printing, providing a supportive structure for the cells to grow and develop into functional tissue. Examples of hydrogels used in 3D printing of ears include alginate, gelatin, and collagen. They have been used in the 3D printing of ear structures due to their ability to mimic the mechanical properties of human ear tissue.

Biocompatible polymers:

Biocompatible polymers are synthetic materials that are compatible with human tissues and do not cause adverse reactions. They are commonly used as the "ink" in 3D printing of human ears because they provide a stable structure for the cells to grow and develop into functional tissue. Polylactide (PLA): Polylactide is a biocompatible polymer that has been used in 3D printing of ear structures. This material is favored for its biocompatibility and ability to support cell growth.

Scaffolds:

Scaffolds are structures that provide a supportive framework for the cells to grow and develop. In the case of 3D printing of ears, scaffolds can be used to create a specific shape or structure for the ear tissue to grow around.

Cell-embedded materials:

Cell-embedded materials are materials that contain living cells, which can be used to seed the 3D printed structure. The cells then grow and develop into functional ear tissue over time.

Ceramics:

Ceramics, such as hydroxyapatite, can be used in 3D printing of ear structures. This material is a natural component of human bones and has been shown to be biocompatible and effective in 3D printing of bones and other tissues.

Technological Importance of 3D Printing of Human Ear

Personalized ear prosthesis:

3D printing allows for the creation of customized ear prostheses that match the unique anatomy of each patient.

Faster production and lower costs:

Traditional methods of ear prosthesis fabrication can be time-consuming and expensive.

3D printing can reduce the production time and cost of ear prosthesis.

Biocompatibility:

3D printing can use biocompatible materials for the production of ear prostheses, reducing the risk of adverse reactions and improving patient outcomes.

Medical education:

3D printing of human ears can be used to educate medical students and health care professionals on the anatomy and treatment of ear defects and injuries



3D printing Of Bone

3D printing has revolutionized the field of medicine, and one of its applications is the 3D printing of bones. This process involves using a 3D printer to create a bone-shaped structure using a special material, such as a biocompatible polymer or a ceramic material, as the "ink." The printed bone structure can then be implanted into a patient to replace missing or damaged bone tissue.

There are two main approaches to 3D printing of bones: additive manufacturing and scaffold-based techniques. Additive manufacturing involves building up the bone structure layer by layer, whereas scaffold-based techniques involve creating a porous structure that provides a framework for bone cells to grow and develop.

Additive manufacturing in 3D Printing of Bone

Additive manufacturing involves building up the bone structure layer by layer using biocompatible materials. The layer-by-layer deposition of material enables the creation of complex three-dimensional structures that mimic the natural bone tissue. The process of additive manufacturing in 3D printing of bone involves several key steps.

Steps involved in additive manufacturing of 3D Printed Bone

Patient Imaging:

The process begins with obtaining accurate imaging data of the patient's bone defect or the area requiring reconstruction. This is typically done using techniques like CT scans or MRI scans.



Digital Model Generation:

Using specialized software, the acquired imaging data is processed to create a three-dimensional digital model of the patient's bone structure. This digital model serves as the basis for designing the customized bone scaffold.



Scaffold Design:

With the digital model in place, the next step is to design the scaffold or implant. This involves determining the appropriate shape, size, and internal structure of the scaffold to match the patient's anatomy and specific requirements. Software tools are used to create the design, ensuring proper support, porosity, and structural integrity.



Material Selection:

Biocompatible materials suitable for bone tissue engineering are chosen for the 3D printing process. These materials should be capable of supporting cell attachment, growth, and eventual



bone regeneration. Common materials include biocompatible polymers, ceramic composites, or biodegradable materials.



3D Printing Process:

Once the scaffold design and material selection are finalized, the actual 3D printing process takes place. The chosen technique, is used to build the scaffold layer by layer. The 3D printer precisely deposits or fuses the chosen material, following the digital model's specifications.



Post-processing:

After the 3D printing is complete, post-processing steps may be required. This can include removing support structures, cleaning the scaffold, and performing any necessary surface treatments to enhance biocompatibility and optimize the scaffold's properties.



Sterilization:

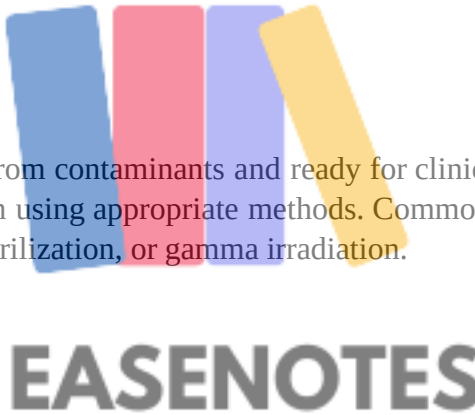
To ensure the implant is free from contaminants and ready for clinical use, the 3D printed bone scaffold undergoes sterilization using appropriate methods. Common techniques include autoclaving, ethylene oxide sterilization, or gamma irradiation.



Surgical Implantation:

The final step involves the surgical implantation of the 3D printed bone scaffold into the patient.

Surgeons carefully position the scaffold in the intended area, ensuring proper alignment and stability. Over time, the scaffold provides support for bone regeneration and integrates with the surrounding tissue.



Materials Used for 3D Printing of Bone

Materials used for 3D printing of bones can vary, depending on the specific 3D printing technique used and the desired outcome. Some of the most commonly used materials for 3D printing of bones include:

Biocompatible polymers:

Biocompatible polymers are synthetic materials that are compatible with human tissues and do not cause adverse reactions. They can be used as the "ink" in 3D printing, providing a supportive structure for the cells to grow and develop into functional bone tissue. Examples: polyethylene, polycaprolactone, polylactide, and polyvinyl alcohol

Ceramics:

Ceramics, such as hydroxyapatite, are natural components of human bones and can be used as the "ink" in 3D printing. Hydroxyapatite is a biocompatible material that has been shown to be an effective material for 3D printing of bones. Examples: Hydroxyapatite, Calcium phosphate, Tricalcium phosphate.

Scaffolds:

Scaffolds are structures that provide a supportive framework for the cells to grow and develop. In the case of 3D printing of bones, scaffolds can be used to create a specific shape or structure for the bone tissue to grow around. Examples: Polyglycolic acid (PGA), Poly-L-lactic acid (PLLA), Polyethylene terephthalate (PET).

Cell-embedded materials:

Cell-embedded materials are materials that contain living cells, which can be used to seed the 3D printed structure. The cells then grow and develop into functional bone tissue over time. Examples: Gelatine methacryloyl, Alginate.

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3D printing of Skin

3D printing of skin refers to the process of creating three-dimensional human skin tissue using a 3D printer. The goal of 3D printing skin is to create functional, living tissue that can be used for a variety of purposes, such as cosmetic testing, wound healing, and drug development. The process involves the use of bioprinting technology, where a bioink made from living cells and growth factors is printed in a specific pattern to create the desired tissue structure.



Figure: 3D image of skin

The Process of 3D Printing of Skin

The process of 3D printing skin typically involves the following steps:

Preparation of the bioink:

A bioink is made by mixing human skin cells, such as fibroblasts and keratinocytes, with a hydrogel matrix that provides a supportive environment for cell growth.

Design of the tissue structure:

The tissue structure to be printed is designed using computer-aided design (CAD) software, which is then used to control the dispensing of the bioink.

↓

Printing:



The bioink is printed layer by layer using a 3D printer to create the desired tissue structure.



Incubation:

After printing, the tissue is incubated in a controlled environment, such as a cell culture incubator, to promote cell growth and tissue formation.



Assessment:

The printed tissue is assessed for its functional properties, such as cell viability, tissue structure, and tissue function.

Materials used for 3D printing of Skin

Hydrogels:

Hydrogels, such as alginate and collagen, are hydrophilic materials that can be used to create 3D structures for cell growth. These materials have been used in the 3D printing of skin due to their ability to mimic the mechanical properties and water-retaining capacity of human skin.

Polymers:

Biocompatible polymers, such as polyethylene glycol and polycaprolactone, can be used in 3D printing of skin. These materials are synthetic and biocompatible, making them suitable for use in the creation of 3D printed skin structures.

Cell-laden hydrogels:

Cell-laden hydrogels are materials that contain living cells and can be used to create 3D printed skin structures. The cells within the hydrogel will grow and develop into functional skin tissue over time.

Scaffolds:

Scaffolds are structures that provide a supportive framework for cells to grow and develop. In the case of 3D printing of skin, scaffolds can be used to create a specific shape or structure for the skin tissue to grow around.

These materials can be used alone or in combination with other materials to create the desired structure and properties for 3D printing of skin. The choice of material will depend on several factors, including the specific 3D printing technique used, the desired outcome, and the intended use of the 3D printed skin.

3D printed Foods



3D printed food refers to food items that are created using 3D printing technology. This technology allows for the creation of food items with intricate shapes and designs, which can be customized based on individual preferences and dietary needs. The process of 3D printing food involves the use of edible materials, such as pastes, gels, and powders, which are combined and printed layer by layer to create the final product.

The use of 3D printing in the food industry has the potential to revolutionize the way food is produced, as it allows for the precise control of portion sizes and ingredients, which can be beneficial for individuals with specific dietary needs or restrictions. Additionally, 3D printing technology can be used to create unique and customized food items that would be difficult to achieve using traditional cooking methods.



Fig:A sample image of 3D printed food

Materials used for 3D Printing of Food

Edible pastes:

Edible pastes, such as pureed fruit, chocolate, and cream cheese, can be used in 3D printing of food. These materials are easily printable and can be used to create intricate shapes and designs.

Edible gels:



Edible gels, such as agar and gelatin, can be used in 3D printing of food. These materials are flexible and can be used to create 3D structures that are both aesthetically pleasing and functional.

Edible powders:

Edible powders, such as flour and sugar, can be used in 3D printing of food. These materials can be combined with liquids to form a printable mixture that can be used to create 3D structures.

Examples of 3D Printed Food

Sweet and savory snacks:

3D printed snacks, such as crackers, cookies, and chips, can be customized to include intricate shapes and designs.

Pastries:

3D printing technology can be used to create intricate and aesthetically pleasing pastries, such as cakes and cupcakes.

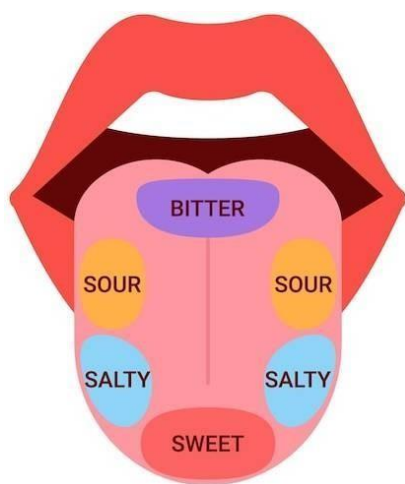
Decorative garnishes:

3D printing technology can be used to create unique and attractive garnishes for dishes, such as cheese and fruit designs.



Electrical Tongue in food Science

The Human Tongue



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Fig: Map of human tongue with taste buds section



The human tongue plays a crucial role in the sense of taste, allowing us to perceive and distinguish various tastes. Here's an overview of how the human tongue functions in sensing tastes:

Taste Buds:

The surface of the tongue is covered with tiny structures called taste buds. Taste buds contain specialized cells called taste receptor cells, which are responsible for detecting different taste qualities.

Taste Receptor Cells:

There are five primary taste qualities recognized by taste receptor cells: sweet, salty, sour, bitter, and umami (savory). Each taste receptor cell is sensitive to specific taste compounds associated with these qualities.

Taste Pores:

Taste receptor cells have small openings called taste pores that are in direct contact with the oral cavity. Through these pores, taste compounds dissolved in saliva come into contact with the taste receptor cells.

Binding of Taste Compounds:

When taste compounds enter the taste pores and come into contact with the taste receptor cells, they bind to specific receptors on the surface of the cells. Each taste receptor cell is specialized to detect a particular taste quality.

Neural Signals:

The binding of taste compounds to the taste receptor cells triggers an electrical signal in the form of action potentials. These signals are then transmitted to the brain via the cranial nerves, specifically the facial nerve, glossopharyngeal nerve, and vagus nerve.

Taste Processing in the Brain:

The neural signals from taste receptor cells reach the brain, specifically the gustatory cortex, where the signals are processed and interpreted. The brain combines the information from different taste receptor cells to create the perception of taste.

Taste Perception:

The brain's interpretation of the signals from taste receptor cells allows us to perceive and differentiate various tastes. The combination and intensity of signals from different taste qualities give rise to the complex flavors we experience when we eat or drink.

The Electrical Tongue



The electrical tongue is a device used in food science to analyze the taste and flavor of food and beverages. It works by measuring the electrical conductivity, impedance, and capacitance of a food or beverage sample, which are related to the concentration of ions in the sample and the texture of the sample.

This technology allows for the rapid and non-invasive analysis of food and beverages, as it does not require human taste testers. Instead, the electrical tongue provides a numerical representation of the taste and flavor of the sample, which can be used to compare and analyze different food and beverage products.

Materials Used in Electrical Tongue Technology

Examples of biomaterials used in Electrical Tongue technology include:

Polymers: Polymers, such as polyvinyl alcohol (PVA) and polyethylene oxide (PEO), are often used as the substrate or matrix material in electrical tongue sensors, as they have high sensitivity to changes in ion concentration and are flexible.

Metal Oxides: Metal oxides, such as tin dioxide (SnO_2) and zinc oxide (ZnO), are commonly used in electrical tongue sensors because of their high sensitivity to changes in ion concentration and ability to undergo changes in electrical conductivity in response to different tastes.

Carbon Nanotubes: Carbon nanotubes are small tubes made of carbon atoms that have high electrical conductivity and sensitivity to changes in ion concentration, making them an attractive material for use in electrical tongue sensors.

Dendrimers: Dendrimers are synthetic, branched nanostructures that can be functionalized with specific receptors or enzymes to target specific tastes. They are being explored as potential materials for use in electrical tongue sensors.

Microfluidic Devices: Microfluidic devices, which are small devices that can manipulate small volumes of fluid, are being used in the development of electrical tongue sensors. These devices can be made from a variety of materials, including silicon, glass, and polymers, and can be functionalized with specific receptors or enzymes to target specific tastes.

Comparison of Functioning of Human Tongue and Electronic Tongue

Aspect	Human Tongue	Electronic Tongue
Sensing Mechanism	Taste buds on the tongue detect taste compounds	Electronic sensors detect chemical properties or patterns
Taste Perception	Humans perceive basic taste qualities: sweet, salty, sour, bitter, umami	The electronic tongue can be programmed to detect various taste qualities, but it may not perceive tastes in the same way humans do
Sensitivity	Human taste buds are sensitive to low concentrations of taste compounds	Electronic sensors can have high sensitivity to detect minute differences in chemical properties

Subjectivity	Perception of taste is subjective and can vary among individuals	Electronic tongue provides objective and standardized measurements
Limitations	Human taste perception can be influenced by factors like smell, temperature, texture, and personal preferences	Electronic tongue may not fully capture the complexity and nuances of human taste perception
Throughput	Human tasting is a relatively slow process	Electronic tongue can analyze multiple samples simultaneously, providing fast and high-throughput analysis
Maintenance and Calibration	No maintenance or calibration required for the human tongue	Electronic tongue requires calibration to ensure accuracy and consistency of sensor responses
Application	Human taste testing is commonly used in food and beverage industries for sensory evaluation and quality control	Electronic tongue is used in various applications, including food and beverage analysis, quality control, and flavor profiling

Electrical Nose in Food Science

The human Nose in Food Science

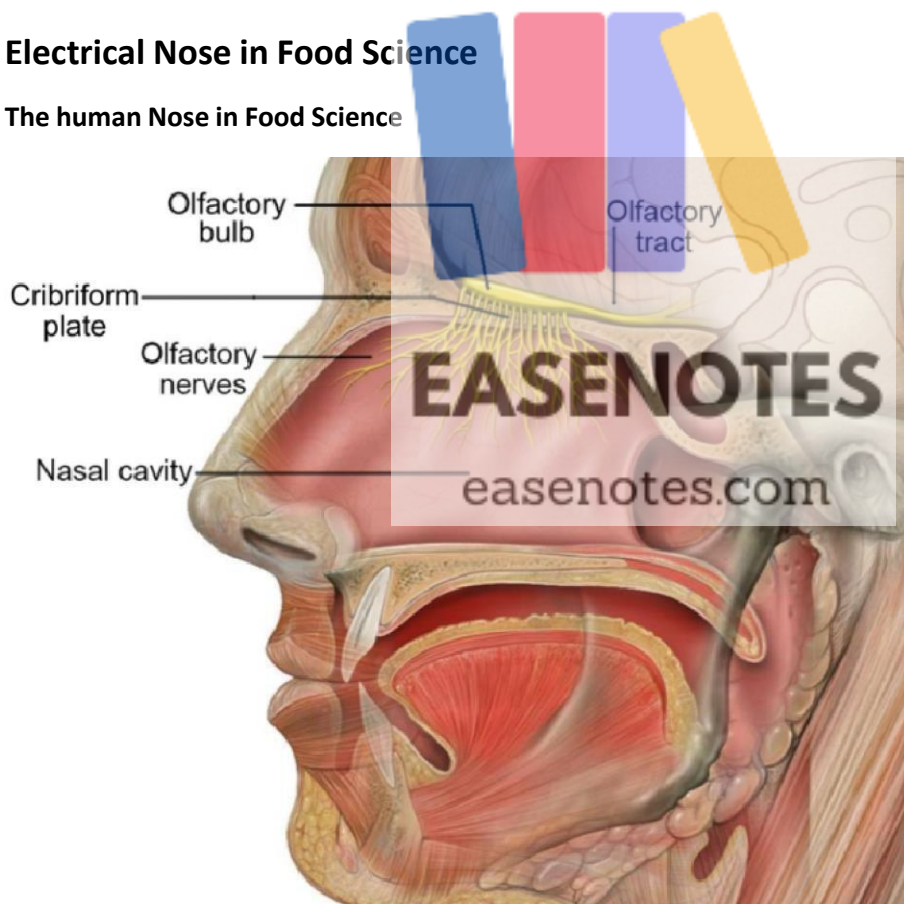


Figure: Representing the olfactory system

The Electronic Nose



The electrical nose, also known as an electronic nose, is a technology used in food science for the analysis and characterization of food and beverage aromas and flavors. The electrical nose typically consists of a sensor array that is capable of detecting and quantifying volatile organic compounds (VOCs) in food and beverage samples.

Technology behind the Electronic Nose

The sensors in the electrical nose work by measuring the changes in electrical resistance or capacitance that occur when the sensors are exposed to volatile organic compounds. Each sensor in the array is designed to respond to a specific range of volatile organic compounds, and the combination of signals from all of the sensors allows for the analysis of the overall aroma and flavor profile of a sample.

Materials Used in Electrical Nose Technology

Examples of biomaterials used in Electrical Nose technology include:

Polymers: Polymers, such as polyvinyl alcohol (PVA), are often used as the matrix or substrate material in electrical nose sensors, as they are flexible and have a high sensitivity to volatile organic compounds.

Carbon Nanotubes: Carbon nanotubes are small tubes made of carbon atoms that have high electrical conductivity and sensitivity to volatile organic compounds, making them an attractive material for use in electrical nose sensors.

Metal Oxides: Metal oxides, such as tin oxide (SnO_2) or zinc oxide (ZnO), are commonly used in electrical nose sensors because of their high sensitivity to volatile organic compounds and ability to undergo changes in electrical conductivity in response to different aroma compounds.

Dendrimers: Dendrimers are synthetic, branched nanostructures that can be functionalized with specific receptors or enzymes to target specific aroma compounds. They are being explored as potential materials for use in electrical nose sensors.

Microfluidic Devices: Microfluidic devices, which are small devices that can manipulate small volumes of fluid, are being used in the development of electrical nose sensors. These devices can be made from a variety of materials, including silicon, glass, and polymers, and can be functionalized with specific receptors or enzymes to target specific aroma compounds.

Comparing the functioning of human nose and electronic nose

Aspect	Human Nose	Electronic Nose
Sensing Mechanism	Olfactory receptor cells in the nasal cavity detect odor molecules	Electronic sensors detect and analyze chemical properties of odor molecules



Odor Perception	Humans can perceive a wide range of distinct odors	Electronic nose can identify and differentiate various odors, but may not perceive them in the same way as humans
Sensitivity	Human sense of smell is highly sensitive to trace amounts of odor molecules	Electronic sensors can have high sensitivity to detect and quantify odor compounds
Subjectivity	Perception of odors can vary among individuals due to personal preferences and experiences	Electronic nose provides objective measurements, eliminating subjective variations
Limitations	Human perception of odors can be influenced by factors like adaptation, context, and individual differences	Electronic nose may not fully capture the complexity and nuances of human olfaction
Throughput	Human olfaction is relatively slow and limited in throughput	Electronic nose can analyze multiple samples simultaneously, providing fast and high-throughput analysis
Maintenance and Calibration	No maintenance or calibration required for the human nose	Electronic nose requires periodic maintenance and calibration to ensure accurate and consistent results
Application	Human olfaction is used in various industries, including fragrance, food and beverage, and environmental monitoring	Electronic nose is used in diverse applications, such as quality control, environmental monitoring, and product development

DNA Origami

DNA Origami is a technique in nanotechnology that involves folding DNA molecules into specific shapes. The process involves using a long, single strand of DNA, called the scaffold, to guide the folding of short, complementary DNA strands, called staples, into a desired shape.

The first DNA origami structures were developed in the mid-2000s and since then, the technique has been widely used in a variety of applications, including the creation of nanoscale structures, the study of molecular interactions, and the development of new drug delivery systems.

Technological Importance of DNA Origami

The technological importance of DNA origami lies in its potential to be used in a wide range of applications, including nanotechnology, materials science, and biomedicine. Some of the key ways in which DNA origami can impact technology include:

Nanoscale manufacturing: DNA origami can be used as a template for the precise assembly of nanoscale structures, which have applications in areas such as electronics, photonics, and materials science.

Drug delivery: DNA origami can be used to develop new approaches for drug delivery, as it can be designed to carry therapeutic agents directly to specific cells or tissues.

Biosensors: DNA origami can be used to develop new biosensors that can detect specific biological molecules and signals in real-time.

Biomedical imaging: DNA origami can be used as a tool for biomedical imaging, as it can be designed to target specific cells or tissues and provide high-resolution images.

Gene therapy: DNA origami can be used as a delivery vehicle for gene therapy, as it can be programmed to target specific cells and deliver therapeutic genes to those cells.

Biocatalysis: DNA origami can be used to develop new approaches for biocatalysis, as it can be designed to perform specific chemical reactions and act as a catalyst.

Nanopatterning: DNA origami can be used as a tool for nanopatterning, as it can be programmed to arrange and position nanoscale structures with precise control.

Bio Computing

Bio-computing refers to the use of biological systems, such as cells, enzymes, and DNA, for computing and information processing. This field combines the principles of computer science, biology, and engineering to create novel systems for computing and data storage.

Technological Importance

The technological importance of bio-computing lies in its potential to provide new and innovative solutions for computing and information processing. Here are some of the key ways in which bio-computing can impact technology:

- **Computational power:** Bio-computing systems have the potential to provide new levels of computational power, as they can perform complex tasks and calculations using biological processes.
- **Data storage:** Bio-computing systems can be used to store and process large amounts of data, as DNA has a high information density [consider that a single gram of DNA can theoretically store up to 215 petabytes (1 petabyte = 1 million gigabytes) of data] and can be easily synthesized and amplified.
- **Medical applications:** Bio-computing systems can be used to develop new diagnostic and therapeutic approaches in medicine, such as biosensors and gene therapies.
- **Environmental monitoring:** Bio-computing systems can be used to monitor and track environmental conditions, such as air and water quality, in real-time.
- **Energy efficiency:** Bio-computing systems are energy-efficient, which is becoming increasingly important as we face the challenge of climate change and the need to reduce our energy consumption.
- **Robustness:** Bio-computing systems are highly robust, as they are less susceptible to errors and failures compared to traditional electronic systems.
- **Versatility:** Bio-computing systems can be programmed and reprogrammed to perform different tasks, which makes them highly versatile and adaptable.

Advantages of Bio-computing:



- **Biocompatibility:** Bio-computing systems are made from biological components, which are biocompatible and less likely to cause an immune response compared to traditional electronic devices.
- **Energy efficiency:** Bio-computing systems use significantly less energy than traditional electronic computers, as they rely on biological processes that occur naturally and do not require external power.
- **Scalability:** Bio-computing systems can be easily scaled up or down, as they are based on biological processes that can be repeated and multiplied.
- **Robustness:** Bio-computing systems are often more robust and reliable than traditional electronic systems, as they are less susceptible to errors and failures.
- **Flexibility:** Bio-computing systems can be programmed and reprogrammed to perform different tasks, which makes them highly flexible and adaptable.

Limitations of Biocomputing:

- **Speed:** Bio-computing systems are generally slower than traditional electronic computers, as they rely on biological processes that occur over time.
- **Complexity:** Bio-computing systems can be complex and challenging to design and build, requiring specialized knowledge and expertise.
- **Reliability:** Bio-computing systems can be unreliable, as they are subject to the fluctuations and errors inherent in biological systems.
- **Cost:** Bio-computing systems can be expensive to produce, as they require specialized materials and equipment.

Bioimaging for Disease Diagnosis

Bio-imaging is the use of imaging technologies to visualize biological processes and structures in living organisms. It plays a crucial role in disease diagnosis by providing detailed

images of the body's internal structures and functions, and can help healthcare professionals to identify and diagnose a wide range of diseases and conditions.

Examples of Bioimaging Techniques

Some examples of bioimaging techniques used for disease diagnosis include X-rays, CT scans, MRI, PET scans, ultrasound, and optical imaging. These technologies can be used to visualize a wide range of structures and functions, including bones, tissues, organs, blood vessels, and more.

Table: Comparing the analyses performed by few important techniques

Imaging Technique	Analyzed Structures/Conditions	Advantages	Limitations
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X-rays	Bones, fractures, lung conditions, etc.	Quick, widely available, relatively low cost	Limited soft tissue detail, exposure to radiation
CT scans (computed tomography scans)	Organs, bones, blood vessels, tumors	Detailed images, good for trauma cases	Exposure to radiation, not suitable for some patients
MRI (Magnetic Resonance Imaging)	Soft tissues, organs, brain, tumors	Excellent soft tissue contrast	Long scan times, restricted for some patients
PET (Positron Emission Tomography) scans	Metabolic activity, cancer, brain	Detects diseases at cellular level	Limited anatomical detail, requires radioactive tracer
Ultrasound	Organs, fetus, blood flow	Real-time imaging, no radiation exposure	Limited penetration, operator-dependent
Optical Imaging	Cellular and molecular processes	Non-invasive, high-resolution imaging	Limited depth penetration, restricted to surface

Artificial intelligence for Disease diagnosis

Artificial Intelligence (AI) has the potential to revolutionize the field of disease diagnosis by providing healthcare professionals with more accurate and efficient tools for identifying and treating various conditions.

Advantages

Some of the key ways in which AI is being used in disease diagnosis include:

- **Image analysis:** AI algorithms can analyze medical images, such as X-rays, CT scans, and MRIs, to detect signs of diseases. For example, AI algorithms can identify patterns in medical images that may indicate the presence of a particular condition, such as a tumor or an injury. This type of image analysis is known as computer-aided diagnosis (CAD).
- **Data analysis:** AI algorithms can analyze large amounts of patient data, such as electronic health records, to identify patterns and trends that may indicate a disease. This type of data analysis is known as predictive analytics.
- **Diagnosis:** AI algorithms can be used to diagnose diseases by evaluating symptoms, test results, and other patient information. AI algorithms can help healthcare professionals make faster and more accurate diagnoses, reducing the risk of misdiagnosis.
- **Personalized medicine:** AI algorithms can be used to create personalized treatment plans for patients based on their specific medical histories, lifestyles, and other factors. For example, AI algorithms can analyze a patient's medical history, lifestyle habits, and genetic information to recommend the best course of treatment for their condition.
- **Clinical decision support:** AI algorithms can be integrated into electronic health records to provide healthcare professionals with real-time decision-making support. For example, AI algorithms can provide physicians with information about the best

diagnostic tests to order, the most effective treatments to consider, and the best ways to manage patient care.

Limitations

In addition to these advantages, there are also some limitations to the use of AI in diseasediagnosis. Some of these limitations include:

- **Lack of understanding of the underlying algorithms:** AI algorithms can be complex and difficult to understand, making it difficult for healthcare professionals to interpret the results. This can lead to confusion and mistrust of AI-based tools, particularly among healthcare professionals who are not familiar with AI technology.
- **Bias:** AI algorithms may be biased, leading to inaccurate or unfair diagnoses. For example, if an AI algorithm is trained on data from a predominantly male population, it may not accurately diagnose conditions that affect women differently.
- **Regulation:** The use of AI in healthcare is heavily regulated, and it can be challenging to get approval for new AI technologies. In many countries, AI algorithms must undergo a rigorous evaluation process before they can be used in healthcare.
- **Cost:** The development and implementation of AI algorithms can be expensive, which may limit access to these technologies for some patients and healthcare facilities. This is particularly true in low- and middle-income countries, where access to healthcare is already limited.

Despite these limitations, AI has the potential to revolutionize the field of disease diagnosis, providing healthcare professionals with new and more accurate tools for identifying and treating a wide range of conditions.

Self-Healing Bio- Concrete

Self-healing bio-concrete is a type of concrete that incorporates microorganisms, such as Bacillus fragments, into the mixture, along with calcium lactate as a nutrient source. The microorganisms are activated when the concrete cracks, and they produce calcium carbonate, which fills in the cracks and repairs the concrete. This process is known as bio-mineralization.

The benefits of self-healing bio-concrete include increased durability, reduced maintenance costs, and improved sustainability, as the concrete is able to repair itself without the need for human intervention. Additionally, because the microorganisms used in the concrete are naturally occurring and non-toxic, self-healing bio-concrete is considered to be environmentally friendly.

Self-healing bio-concrete is still a relatively new technology and is currently in the research and development phase. However, initial studies have shown promising results and have demonstrated the potential for self-healing bio-concrete to be a viable alternative to traditional concrete in certain applications.

Self-healing Process

Process Flow Chart



Mix Bacillus bacteria and calcium lactate with concrete



Bacteria remain dormant within the concrete



Concrete cracks



Water and oxygen enter the crack



Bacteria become activated



Activated bacteria produce calcium carbonate



Calcium carbonate fills in the cracks



Concrete is repaired and structural integrity is restored

Self-healing bio-concrete works by incorporating Bacillus bacteria into the concrete mixture, along with calcium lactate as a nutrient source. The bacteria are dormant within the concrete and do not become active until the concrete cracks.

When the concrete cracks, water and oxygen enter the crack and activate the Bacillus bacteria. The bacteria then produce calcium carbonate, which is a type of mineral that is commonly found in natural stone. The calcium carbonate acts as a binder and fills in the cracks, repairing the concrete and restoring its structural integrity. This process is known as biomineralization.

The Bacillus bacteria used in self-healing bioconcrete are naturally occurring and non-toxic, so they are considered to be environmentally friendly. They are also able to survive in a wide range of temperatures and pH levels, making them well-suited for use in concrete.

In addition to repairing cracks, self-healing bioconcrete also has the potential to improve the overall durability of concrete by reducing the amount of water that is able to penetrate the surface. This can help to prevent the development of further cracks and increase the longevity of the concrete.

Technological Importance of Self-Healing Bioconcrete

Self-healing bioconcrete has several important technological advancements that make it a promising alternative to traditional concrete:

- Increased durability: Self-healing bioconcrete has the ability to repair itself, which can help to increase its overall durability and reduce the need for maintenance.
- Improved sustainability: By using naturally occurring and non-toxic microorganisms, self-healing bioconcrete is considered to be a more environmentally friendly



alternative to traditional concrete.

- **Reduced maintenance costs:** Because self-healing bioconcrete is able to repair itself, it has the potential to reduce the need for costly maintenance and repairs over time.
- **Increased longevity:** By repairing cracks and reducing the amount of water that is able to penetrate the surface, self-healing bioconcrete can help to extend the lifespan of concrete structures.
- **New applications:** The ability of self-healing bioconcrete to repair itself may open up new applications for concrete that were not possible with traditional concrete.
- **Reduced carbon footprint:** The biomineralization process used in self-healing bioconcrete has the potential to reduce the carbon footprint associated with concrete production, as it eliminates the need for concrete to be transported and replaced when it becomes damaged.

Bioremediation and Biomining via Microbial Surface Adsorption

Bioremediation and biomining are two related but distinct processes that utilize living organisms to clean up contaminated environments or extract valuable minerals, respectively.

Bioremediation refers to the use of microorganisms, plants, or animals to clean up contaminated environments, such as soil, water, or air. This process occurs naturally over time, but can also be accelerated through the addition of specific microorganisms or other biotic agents. The goal of bioremediation is to remove contaminants from the environment and restore it to a healthy state.

Biomining, on the other hand, refers to the use of microorganisms to extract valuable minerals from ore deposits. This process involves the use of microorganisms to dissolve minerals from ore, creating a solution that can be separated and purified to obtain the valuable minerals. Biomining is often used in the extraction of metals such as copper, gold, and nickel, and has several advantages over traditional mining methods, including lower energy costs, reduced waste, and increased metal recovery.

Table: Comparing bioremediation via microbial surface adsorption and biomining via microbial surface adsorption

Aspect	Bioremediation via Microbial Surface Adsorption	Biomining via Microbial Surface Adsorption
Objective	To remove or neutralize pollutants/contaminants from the environment	To extract valuable metals or minerals from ores
Process	Microorganisms adsorb and degrade pollutants/contaminants	Microorganisms adsorb and extract metals from ores
Targeted Contaminants/Metals	Focuses on organic pollutants or contaminants	Focuses on desired metals or minerals
Microorganisms	Diverse range of microbial strains with pollutant-degrading capabilities	Specific microbial strains with metal adsorption capabilities

Surface Adsorption Mechanism	Microorganisms attach to pollutant surfaces	Microorganisms attach to metal surfaces
Environmental Impact	Can restore ecosystems and improve environmental quality	Can potentially cause some environmental disturbances
Timeframe for Results	Can take months to years for significant remediation	Quicker results for metal extraction in controlled conditions
Waste Generation and Disposal Considerations	May generate waste that requires proper disposal	Waste generation and disposal considerations in mining operations
Applications	Soil, water, and air pollution remediation	Mining operations for metal extraction

Bioremediation and biomining via microbial surface adsorption is a process that utilizes microorganisms to remove heavy metals like lead, cadmium, mercury, and arsenic from contaminated environments or ore deposits, respectively.



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The process of removing polluting heavy metals using bioremediation or biomining via microbial surface adsorption

Identification of heavy metal-contaminated site:

Identify the site or area contaminated with heavy metals, such as soil, water, or industrial wastesites.



Isolation and characterization of metal-resistant microbial strains:

Select and isolate microbial strains that have demonstrated resistance to heavy metals. These can include bacteria, fungi, or archaea.



Culturing and enrichment of microbial strains:

Culture and propagate the selected microbial strains in a suitable growth medium under laboratory conditions. This step aims to obtain a sufficient quantity of active microbial biomass for subsequent applications.



Preparation of microbial suspension:

Harvest the microbial biomass and prepare a suspension by suspending the biomass in a carrier solution, such as water or a nutrient broth. This suspension will serve as the delivery system for the microbes during application.



Application of microbial suspension to the contaminated site:

Apply the microbial suspension to the heavy metal-contaminated area. This can be done through spraying, injection, or soil/water mixing, depending on the specific site conditions.



Microbial adsorption and sequestration of metal:

The applied microbial strains adsorb to the surfaces of metal particles or form biofilms. Through their metabolic activity, the microbes produce extracellular compounds such as organic acids or biofilm matrix components that have an affinity for binding metal ions.



Separation or removal of metals from the contaminated site can be achieved through different methods

Methods used for the Separation or Removal of Metals

After the steps of microbial adsorption and sequestration of heavy metals, the subsequent separation or removal of metals from the contaminated site can be achieved through different



methods. Here are a few common approaches:

Phytoremediation:

In this method, plants are used to remove heavy metals from the soil or water. The metal-accumulating ability of certain plant species, called hyperaccumulators, allows them to take up metals from the environment and store them in their tissues. After the plants have absorbed the metals, they can be harvested and disposed of properly, effectively removing the metals from the site.

Chemical extraction:

Chemical agents can be applied to the contaminated area to facilitate the release of heavy metals from the microbial biomass or the surrounding matrix. Chelating agents, such as ethylenediaminetetraacetic acid (EDTA) or citric acid, can be used to form complexes with the metals, increasing their solubility and facilitating their removal.

Biosorption:

In this method, the metal-loaded microbial biomass or biofilms can be harvested and separated from the site. The biomass can then be processed to recover the metals through techniques such as acid leaching or thermal treatment. The metals can be further purified or recycled for various industrial applications.

Physical removal:

In some cases, physical methods such as sedimentation, filtration, or membrane separation can be employed to separate the metal-loaded microbial biomass or biofilms from the surrounding environment. These techniques rely on the physical properties of the biomass or biofilms, such as size, density, or adsorption capacity, to separate them from the water or soil.

Electrochemical methods:

Electrochemical techniques, such as electrokinetic remediation or electrocoagulation, can be utilized to remove heavy metals from the contaminated site. These methods involve the application of an electric field or the generation of metal precipitates through electrochemical reactions, resulting in the migration or precipitation of metal ions, which can then be collected and removed.

Advantages of Bioremediation and Biomining

- **Environmentally friendly:** The use of microorganisms to remove heavy metals from contaminated environments or ore deposits is an environmentally friendly alternative to traditional methods such as chemical leaching, which can produce toxic waste products.
- **Cost-effective:** Bioremediation and biomining using microbial surface adsorption is often less expensive than traditional methods for removing heavy metals, as it does not require the use of costly chemicals or equipment.
- **Selective:** Microorganisms can be selected based on their ability to remove specific heavy metals, which allows for the removal of specific contaminants in a targeted manner.
- **Effective:** Microorganisms can effectively remove high levels of heavy metals from



contaminated environments or ore deposits, making this process a useful tool for environmental remediation and mining.

- **Sustainability:** The microorganisms used in bioremediation and biomining can be cultured and reused, making the process sustainable over the long term.

Limitations of Bioremediation and Biomining

- **Slow process:** The process of removing heavy metals via microbial surface adsorption can be slow, as it may take several months or even years for the microorganisms to adsorb the heavy metals.
- **Incomplete removal:** While microbial surface adsorption is effective in removing high levels of heavy metals, it may not be able to remove all of the contaminants, leaving some heavy metals behind.
- **Microbial inhibition:** Some environmental conditions, such as high levels of other heavy metals or low pH, can inhibit the growth and activity of the microorganisms, reducing their ability to remove heavy metals.
- **Difficulty in harvesting:** Harvesting the microorganisms that have adsorbed the heavy metals can be difficult, as the microorganisms may form dense biofilms or be difficult to separate from the contaminated environment or ore deposit.
- **Limited application:** The effectiveness of microbial surface adsorption for removing heavy metals is limited by the ability of the microorganisms to adsorb specific heavy metals. Some heavy metals, such as mercury, may not be effectively removed using this process.

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QUESTION BANK MODULE 5

1. Give a comparison between 3D printer and bioprinter
2. Write a note on bioprinting materials and explain different types of bioprinting techniques.
3. What are different types of material used for 3D prints of human ear and write technological importance of 3D printing of human ear.
4. Write steps involved in additive manufacturing of 3D printed bone.
5. What are the different types of material used for 3D printing of Bone.
6. Explain the process of 3D printing of skin and write different material used for 3D printing of skin.
7. What are different types of material used for 3D printing food and write examples of 3D printed foods.
8. Explain human tongue with its different taste buds section.
9. What are different types of material used in electrical tongue technology and write the comparison between human tongue and electric tongue.
10. Explain the technology behind the electronic nose and different materials used in electrical nose technology.
11. Compare the functioning of human nose and electric nose.
12. Briefly explain DNA origami with its technological importance.
13. Explain the technological importance and advantages of bio computing.
14. Explain Bioimaging and its importance.
15. Write applications and limitations of artificial intelligence works in disease diagnosis.
16. Write a brief note on Self-healing bio concrete.
17. Write a note on Bioremediation and Biomining via Microbial Surface Adsorption
18. Compare the Bioremediation and Biomining.

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